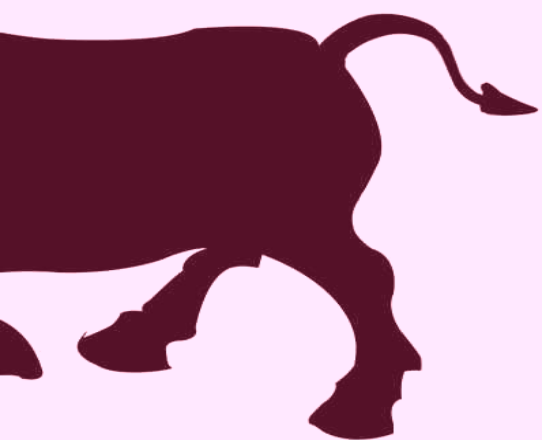


CHRONICLE OF INDIA'S HIGH-TECH STOCK MARKET
SCANDAL & THE CABAL THAT WENT SCOT-FREE.

THE MARKET



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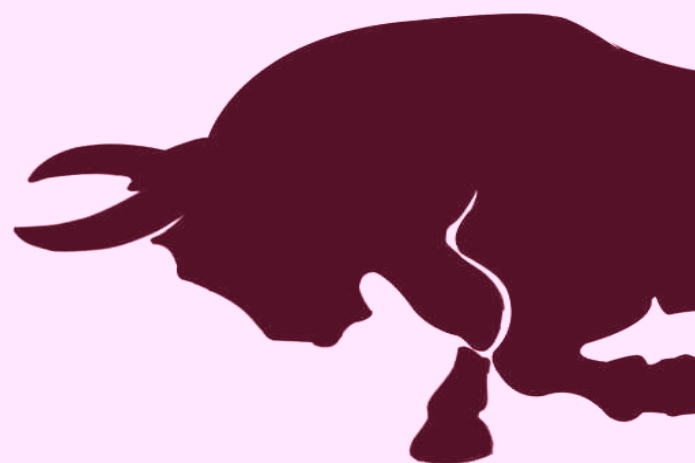
SHAH

Foreword by:

ANAND SWAMY



MARKET



MARKET

PALAK

Foreword by:

DR. SUBRAMANIAM

THE MARKET MAFIA

CHRONICLE OF INDIA'S HIGH-TECH STOCK MARKET
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Foreword by:

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AUTHOR'S NOTE

"PUBLISH AND BE DAMNED"

When I started writing *The Market Mafia*, an ex-bureaucrat cautioned me that I'll be 'damned' for publishing the story. For a journalist, to publish and be damned and publish again is routine.

I want to be absolutely candid about the fact that: "The book is a work of nonfiction. No names were changed, no characters were invented or events fabricated here." Let me also submit: "It is not my intention in any way to defame any person, entity or institution(s) in this book. My endeavour has been to bring out the 'truth' about the rampant stock market manipulation in India that stretches far beyond Harshad Mehta and Ketan Parekh scandals."

The work is a culmination of months and years of painstaking follow-up of developments in India's stock market and the regulatory regime. Facts unravelled in the book are also emanating from articles published as part of my job. Thus, I stand by every single word, right up to the last full stop.

I humbly challenge anyone to write a counter-narrative to my story and dismantle the facts that I have dug-up through my own investigation and those of the regulators.

My close friends and colleagues also cautioned me about possible defamation. Well, to be honest, the only legal action I fear is likely from my wife, who may sue me for spending months writing the book and almost

ignoring her.

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I will always be indebted to late Mr. Pradeep Shinde, Mumbai's best-known crime reporter, who introduced me to journalism and mentored a cub. An editor par excellence, Mr. Shinde epitomised the best of investigative journalism and English literature, in an era long gone.

The book would not have been possible without the help of many people and my trusted sources. I would like to express my deep gratitude to all those who supported me, shared crucial information, talked things over, offered their invaluable comments and insights and assistance in editing, proofreading and designing.

FOREWORD

Mr Palak Shah's book *The Market Mafia* is an exposé of the inner rot in India's stock exchange ecosystem. The Securities and Exchange Board of India (SEBI), whose role is to represent the shareholders, has performed much like the exploiters.

The author's account of the Algo trading scandal, data theft at the National Stock Exchange (NSE) and the Multi Commodities Exchange (MCX) is a sordid narrative of how corruption was institutionalised under the Congress-led United Progressive Alliance rule for 10 years.

When I first raised concerns over the use of Participatory Notes (P-Notes) in the stock market between 2005 and 2007, there was no doubt to me that the 'Black Money' instruments were flourishing under the lustful eyes of then Finance Minister Mr P Chidambaram. For the first time, a journalist has explained in very simple terms Chidambaram's craze for P-Notes, his backing of controversial SEBI chief Mr C B Bhavé, and planting of dubious bureaucrats like Mr K P Krishnan on the regulators' Board.

The author reveals how Mr Chidambaram prevailed over PM Manmohan Singh on P-Notes and Mr Bhavé's appointment, which laid the foundation for a hijack of the SEBI and the NSE. Could Mr Chidambaram have accomplished this without the backing of Ms Sonia Gandhi? Stock markets had become their 'cash cow' and the book aptly reveals the plot without beating around the bush.

The author has done a praiseworthy job of highlighting the meticulous details and ramifications of the fraudulent activities of prominent economists such as Mr Ajay Shah and Ms Susan Thomas and raised some relevant questions. Yet, SEBI with its deliberately shoddy investigations have attempted to bury the many facets of the scam, some of which are coming out in the book for the first time. SEBI's diagnosis is just a farce, albeit a deliberate one when you see various other angles to the scam that the author talks about. The culprits have surely got away with no punishment at all.

In my view, SEBI is not a competent authority to probe the scandal as it is in conflict with itself and a suspect. Up to the last sentence, the book makes some big revelations, which require a thorough probe either from the Central Bureau of Investigation or from a Special Investigation Team that should comprise even financial market and forensic experts. Mr. Shah's exposé should go a long way in ensuring that SEBI is taken to task, sooner rather than later.

Dr. Subramanian Swamy,
Economist & Statistician,
Member of Parliament, Rajya Sabha

HOUSE OF CARDS

The Blindfolded Cop

As the man at the helm of affairs of the premier financial market regulatory agency, he was expected to be more vigilant. On the contrary, Chandrashekhar Bhasker Bhave was conveniently turning a blind eye to the subterranean changes that the Indian stock markets were undergoing around 2008. High-Frequency Trading (HFT) driven Algorithms (Algos) – the computer-assisted tech tools that deftly buy and sell stocks – were altering the high-octane equity trading game.

Securities and Exchange Board of India (SEBI), the ‘*super-cop*’ responsible for policing it, was at a nadir in following its *Dharma* of ensuring ‘*fair market access*’ to all participants when HFT-guided Algos came into play. Bhave, the then SEBI chairman, was apparently busy fighting his own battles.

As a sacred rule, no trader can have preferential access to a stock exchange platform. And the ethics demand that no tool, order type or service can be obtained exclusively by a few from the exchange by whatever means, even as the rest remain forbidden from it. But technology changed everything; it turned the stock markets into an ‘*Animal Farm*’. The elite few with their tech tools became ‘*more equal*’ than the rest.

When the cat's away, the mice will play. Without supervision, people will do as they please, especially in disregarding laws or breaking rules, and that's exactly what happened. Bhavé's laxity opened the field to manipulators when the new-age trading technology was coming up at the National Stock Exchange (NSE), India's powerhouse for equity derivatives.

Ajay Shah and Susan Thomas, the two Mumbai-based professors who were celebrities in the realm of capital market research, had spun a flourishing family enterprise at the NSE. Among other perks, their chicanery as exchange insiders exposed them to the real gold mine – Data, which is most vital in coding of Algo trading software.

“You have to swear everyone to silence about the fact that the data we are getting out from the NSE is going into Algorithmic trading work. It would be a severe problem if this fact comes to light, since NSE has not given this data to anyone else,” Ajay wrote in an email to Susan's sister Sunita Thomas, who was a dealer of Algo software. HFT driven Algos were just starting to gain supremacy in the Indian stock markets at the beginning of 2008.

Sunita conducted her business through Infotech Financials Private Limited (IFPL), a company that emerged as a mere front for the professors. Coincidentally, she was also the wife of Suprabhat Lala, a top rung NSE executive. Data, research and Algo software all became an in-house production job for Ajay and Susan.

Algos are computer coded trading strategies that can react to events – every price tick is an event. HFT machines are electronic trading bots that give these monster brains the speed of execution as they spit automated orders faster than the latency of light. Before the human mind can assess and decide whether it wants to partake in a trade, the HFT Algos have already executed it and made a killing.

Speed of trading and advanced information is a *Mantra* for winners in the stock market – HFT, Algos and data its form. Technology required that brokers put up their bots as close to an exchange trade engine as possible for

them to be fast at the play. Since every bot trader wanted to get in close proximity of NSE, they started locating near its headquarters in Mumbai's plush Bandra-Kurla Complex (BKC), ironically a stone's throw away from SEBI's headquarters.

In 2010, the exchange set up a high-tech data hub within its sprawling office building, and for the first time, brokers were invited to trade from inside its premises by NSE. The exchange offered them space in the data centre that also housed the master order matching engine in the vicinity. Armed with tech tools and an unquenched greed for profit, it was the closest a broker could get to NSE's engine that tirelessly matched buy-and-sell orders in a jiffy.

This was an incredible development, since those who were searching to locate closely around NSE, could now actually set up their bots inside the exchange. The heat had turned up a notch higher. NSE's heart was now the broker's server farm, termed as Co-location (COLO), a highly disruptive force. Algo bots became lethal when the machines that drove them were placed cheek-by-jaw with NSE's master engine.

There was little realisation among wider market participants about what the monolithic change meant, the regulator it seems was also wearing blinders. SEBI's foremost *Dharma* was to inspect the new tech changes of NSE and grant its certificate of approval or disapproval after weighing the pros and cons of a radical new change in market operations. But Bhav displayed sheer apathy and NSE began COLO operations without any formal approval from SEBI.¹ The COLO rack space inside an exchange premise required top dollars but was made truly worth the investment with rich data-feeds called tick-by-tick (TBT) price quotes provided by NSE to these brokers.

This was the real gold mine. It gave advanced knowledge of buy-sell orders. TBT alerts COLO machines of the coming order and the bots can hit and run a trade in '*milliseconds*' or '*microseconds*'. It is less time than a photo

flash takes to '*disappear*'. Since speed is yielding profits, it spurred an arms race. Nano-second, pico-second became the new street lingo. Interestingly, the high-tech trading grid at NSE had escaped the regulator's radar even as a privileged few were reaping its benefits. Ensconced at his comfortable office just across the road from NSE, Bhavé was watching the changes at the bourse, albeit, rather silently. The marshal was aloof and blatantly ignoring his fundamental duty.

By the constitutional powers vested in it, the idea of SEBI approvals has been core to the working of stock exchanges in India, so much so that the regulator can even reject appointments of top officials, veto strategic deals, set limits for compensation packages or can if it feels deemed fit, stall or allow an exchange from extending trading hours. Thus it was surprising how NSE could move ahead with COLO and TBT data sharing without SEBI approval.² But then nothing that NSE did during Bhavé as SEBI chairman called for any scrutiny from the regulator, which otherwise had its say even in minute things like, categorisation of stocks into groups that limited their daily rise and fall through circuit breakers.

The traditional ritual played out in the markets when SEBI or the stock exchanges came up with any new proposal – the same happened prior to NSE's launch of COLO operations. Committees were set up, discussions were held, white-papers put-out and public opinion sought before implementing the changes. NSE's rival, the Bombay Stock Exchange (BSE), often complained that several of its ideas, proposals and even deals got stuck without SEBI approvals for months and years.

Ironically, Bhavé, who was much benevolent with NSE, was kind of giving the stepmother treatment to BSE. He had thwarted BSE's bid to pick-up a majority stake in the mutual funds database company CAMS. The ground was that the exchange with its trading business would give rise to a conflict of interest if it also owned a company that collected data on clients of mutual funds who traded on the BSE and its rival NSE. Interestingly, the

stake in CAMS was later picked up by NSE. Similarly, when equity trading through mobile phones was proposed and BSE seemed enthusiastic, the NSE opposed that move. Bhavé ensured that an expert committee was set-up to '*study*' the new proposal and form '*rules*'. In the process, the implementation of mobile trading was delayed for nearly three years.

With COLO and HFT, technology had created a vacuum for wider market participants. They had no iota of clue about how the trading game they had played for decades was about to change with the advent of the tech tools at NSE.

Only a few brokers who used NSE's COLO grid knew the nuts and bolts of the technology that was driving the markets, while the outside world remained in pitch-darkness on the insides of the space-age like tech platform. But that invoked no conflict from Bhavé's point of view. He allowed the most disruptive changes including HFT driven Algos and COLO operations to seep into the stock market trading microstructure by NSE without a deeper scrutiny. He turned into a recluse when someone told him that NSE was sharing TBT data exclusively.

Bhavé was appointed as Chairman of SEBI in February 2008 for three years, but he was not the first choice of the then Prime Minister (PM) Manmohan Singh to lead the regulatory body for Indian capital markets. He got selected due to the powerful hand that was vying for him.

Before Bhavé took the corner office at SEBI, he was the MD and CEO of National Securities and Depository Ltd. (NSDL). He had resigned as an IAS (Indian Administrative Service) bureaucrat in 1996 to take up a more lucrative corporate job and with due credit to him, he built NSDL from scratch. Although the fact remains that his task was made easy with the monopoly business that India's first share depository enjoyed.

NSDL dematerialised (demat) physical share certificates into electronic form and its business grew rapidly under Bhavé. Later on, he extended NSDL's operations into tax information services and other lucrative areas,

however, the company's primary income came from its account holders who bought and sold shares on stock exchanges but stored them in digital form at NSDL.

NSE was a platform to buy and sell stocks but the digital transfer of shares from one account to another took place inside NSDL. These transfers within NSDL attracted a minuscule charge, besides which the depository also recovered an annual maintenance cost from account holders.

As the buy and sell volumes soared at NSE, so did NSDL's business, and along with it, Bhavé's stature. His stock was surely hitting the peaks, but it would not be for long. An Income Tax investigation shattered NSDL and Bhavé's reputation as the pioneer of corporate governance and compliance measurement systems. The drama that followed shocked even Bhavé's most ardent admirers. NSDL came under a cloud for fake demat accounts scam.

Earlier in 2006 when Income Tax raided a broker, Purushottam Budhwani, they accidentally made a startling discovery that he was controlling over 5,000 demat accounts. Being the market regulator, tax officials passed on the information gathered from their investigation to SEBI, which then started deep scrutiny of several thousand suspicious demat accounts at NSDL.

The SEBI probe found that stock market operators were able to open multiple dubious digital share accounts by incomplete documentation and flaunting of know-your-customer (KYC) norms with ease. Nearly 100,000 of them were unearthed. Around 50,000 were in the single name of one Rupalben Panchal and associates.³ These demat accounts were used to corner shares reserved for retail investors in initial public offers (IPOs) that hit the market between 2003 to 2005. Since the portion reserved for retail investors in IPOs was small, multiple applications in the category via such fake demat accounts created artificial demand. The share price rose on the listing of the IPO and illegally cornered shares through fake demat accounts were then transferred to its beneficiary who dumped them in the market.

IDFC, Yes Bank, Suzlon, IL&FS, Indiabulls... the list of IPOs where the operators' cornered shares meant for retail players was long. The party went on until G Ananthraman, then SEBI member and head of investigations cell, unearthed all the facts and the scam came to light.

After the investigations, SEBI adjudicated the matter. In his order against NSDL, Ananthraman said, "Initial findings demonstrated contributory negligence on the part of depositories and their management."

This IPO scam was pegged at more than Rs. 500 crore and a systemic lapse due to inadequate '*checks*' by the management of NSDL and CDSL (Central Depository Services Limited), the only other rival to Bhavé's company in the segment, were blamed for it. NSDL had a lion's share of the dubious accounts as it was the market leader, with nearly 80 per cent of demat business at that time.

A deep dive into the history of stock market scams will show that the faces or the front in any mega fraud are the boorish bunch of operators who fall victim to their greed, not knowing that the tie-suit executives are their silent handlers exploiting the systemic weaknesses they created themselves, knowingly or unknowingly.

Could NSDL fake demat scam be the same kind of game? The then SEBI chairman, Meleveetil Damodaran, a no-nonsense former IAS officer, decided to take a swan dive into it. He roped in Indian Police Service (IPS) officer, Sanjay Pande, an Indian Institute of Technology (IIT) Kharagpur alumnus and a security technology expert, to conduct a thorough independent probe against NSDL.

Pande, an upright officer, meticulously detailed the procedural and technology lapse at the depository. His report, which was crucial evidence of NSDL's hollow and mediocre standards, was so incriminating against the depository that it got suppressed with the brutal force of politico-bureaucratic nexus. A few privy to the investigation could see that he was also stonewalled by those in the upper echelon of power.

Pande reportedly told Damodaran: “There was absolute lack of cooperation by NSDL’s management during the probe.” The media meanwhile pitted Damodaran against Bhavé and created a spectacle of professional rivalry.

In Ananthraman’s reckoning, there were grave management lapses, which led to a stark situation at NSDL. SEBI asked the promoters of the depository to take action including a swift management revamp. There was a stoic silence on Ananthraman’s order and nobody from among NSDL’s promoter entities raised even a finger against Bhavé. Incidentally, NSE was the largest promoter of NSDL.

When Bhavé became the SEBI chairman, he was just returning the favour.

In fact, the camaraderie between Bhavé and NSE’s core team had started long back. When NSE was being set up, it was a routine affair for the senior executives of the exchange to call upon Bhavé, who was then working as an executive director with SEBI (between 1992 to 1996), for necessary approvals. When NSE was setting up NSDL, Bhavé came in as a senior executive.

Under his watch between 2008 and 2011, SEBI went blindfolded to the happenings at NSE.

Bhavé’s appointment as the chairman of the regulatory body in 2008 thus laid the foundation of massive data heist by Ajay and Susan and the COLO scam by brokers at NSE – the lid of which was blown in 2015. Both Bhavé and Ajay too shared a deep bonding. Ajay was the member of NSDL’s Executive Committee between 1996–2008.

Shadow Boxing

Bhavé’s second stint at SEBI, this time as its head, became possible as the then Finance Minister (FM) Palaniappan Chidambaram rallied his weight

behind him.⁴ The *Mundu* clad politician, who was also a lawyer and a graduate from Harvard Business School, was actually a darling of the Indian stock markets and Foreign Institutional Investors (FIIs). His cheeky, off the cuff statements often moved Sensex and Nifty, so much so that the two benchmark share indices of BSE and NSE had come to be known as the barometer of Chidambaram's announcements.

He was the most vociferous FM India has seen as he spoke plenty during market hours between 9 AM to 3:30 PM and traders loved it. On the floor of the Parliament during the start of one of his annual budget speeches, Chidambaram said, "I'll keep it short." That was the longest budget speeches he had delivered in his stint as FM and market volatility shot-up sharply that day.

On the other hand, Chidambaram's boss, PM Singh was a stark contrast, who rarely uttered a word in public, unless it was the customary occasion of Independence Day where he addressed the nation. It had earned him the sobriquet as '*Maun*' Mohan Singh or '*Mauni Baba*', the '*Silent Hermit*'.

In 2008, Singh had been advised by his senior colleagues Montek Singh Ahluwalia, Deputy Chairman of the Planning Commission and C Rangarajan, Chairman of PM's Economic Advisory Council to extend Damodaran's term as the SEBI chairman.

The two advisors told the PM that Damodaran had, "High reputation among the investor community." They further informed Singh that, "SEBI's overall performance under Damodaran was good and there was nothing amiss with the regulator under his tenure."

Singh looked inclined to take the good counsel, however, Chidambaram differed on that advice and confronted the PM.

"There are serious concerns about the ability of Damodaran to pull along as a team player, which would impose great stress on the system," the FM told PM Singh.

Chidambaram also provided Singh with a series of instances where he felt SEBI could have performed better.

“In the last year, regrettably, communications between the government and SEBI have virtually broken down,” (iv) Chidambaram tried to justify his point in a letter to Singh. It appeared that he was vehemently opposing Damodaran who otherwise was his department subordinate.

So why did the communication between Chidambaram and Damodaran break down? Who was threatened by whom? The secret was known to one and all. Damodaran’s move to go behind Participatory Note (P-Note) investments in India made him Chidambaram’s nemesis.

P-Notes are Off-Shore Derivative Instruments (ODIs), the trail of which could mainly lead to entities based out of Mauritius and Cayman Islands. For decades, the two tiny islands have been known to the world for disguising laundered cash and rerouting it to desired destinations with the ease of a click of a button. Funds or corporations registered in these lax tax jurisdictions issued promissory notes against the cash they received from undistinguished individuals and companies.

P-Note issuing fronts were mainly SEBI-registered big guys, FIIs like Goldman Sachs, Deutsche Bank, Citi Bank, JP Morgan and others, who themselves or through their step-down entities had a base in tax havens. But the ultimate P-Note beneficiaries had no face. When FIIs bought and sold stocks or cut a derivative deal, their names were reflected in India’s market database and never of those holding the P-Note. It was a mirage and gullible market investors believed that large FIIs were bearish or bullish on markets even though P-Note holders were driving the sentiments from the shadows.

A huge proportion of FII money flowed into India through P-Notes and who could be using them? It is common knowledge that politicians, policymakers and company promoters in possession of crucial data, information on listed companies and the economy used P-Notes to indulge in derivative trading. When FIIs issued P-Notes against cash, the money

came into their kitty and remained parked until the beneficiary decided to take a market bet. It could all be done at a short notice and P-Notes could be sliced and diced into various interest and non-interest bearing instruments. Voila – a perfect alibi.

Imagine heads of political parties in India, ministers, stock exchange officials, company promoters and even the SEBI chief, all of whom had inside information on market moves, using P-Notes to trade.

There was another serious ramification of this, which also compromised national security. The instruments also came handy for underworld don, terrorists or enemies of the country who wished economic disruption in India. P-Note misuse had no limits.

India's home-grown experts have several times blamed P-Notes for the sudden market spike and crash. Data often revealed huge FII derivative positions ahead of crucial events and government policy announcements. The union budget presentation being one such favourite occasion for P-Note play. Some experts even pointed towards P-Notes for insider trading.

But all the emotional calls to ban P-Notes instantly fell on the deaf ears of Chidambaram. Like a damsel with billions of dollars, P-Notes had romanticised the FM, and he seemed to be loving it.

Data shows that under Chidambaram's watch, P-Note investments in India rose sharply. Their holding by FIIs in stocks and derivatives combined jumped to a colossal Rs. 3,53,484 crores (51.6 per cent of FII assets under custody) by August 2007. In March 2004, when Chidambaram had just become the FM, the same P-Note holding of FIIs stood at Rs. 31,875 crores (20 per cent of assets under custody).

In 2004, before P-Notes gained supremacy, SEBI had first asked FIIs to wind-up their positions through these instruments in five years. Even the Reserve Bank of India (RBI) did not hide its unhappiness with P-Note-laden FII flows. In 2005, relations between Chidambaram and then RBI governor,

Dr. Yaga Venugopal Reddy, had reached a flashpoint on the P-Note issue. On January 12, 2005, a call from Reddy's office late in the evening to news reporters suggested he would make an important statement, barely two hours after a thought-provoking speech on taxing increased foreign flows. It was a historic high point for any news reporters in their career.

A few minutes after journalists assembled, Reddy walked in with a piece of paper and read out an amendment to the speech he made earlier in the day and walked back to his suite without even looking straight up in the eyes, which had dozens of questions. Usually jovial, Reddy was grim, thanks to a direction from Chidambaram to retract the reference in his speech to Tobin Tax (also known as the financial transactions tax), a levy intended to temper foreign exchange inflows.⁵ The bone of contention obviously was P-Note. Even though he retreated that day, Reddy continued with his efforts to convince others in the government on curbing P-Note inflows.

In 2006, when PM Singh visited RBI headquarters in Mumbai, Reddy alerted him about the malaise of P-Notes, and its long-term ramification on the financial system of the country

“Fight against black money was incomplete without P-Note curbs. If a bank account needs PAN (permanent account number), so should million-dollar P-Note investments,” Reddy told the PM in that meeting. Being an economist, a hardcore finance guy, besides a former governor of the RBI himself, Singh got the message loud and clear.

He proposed a high-level committee to suggest a way forward on FII investments. A group led by RBI's former deputy governor SS Tarapore was constituted. The Tarapore Committee made strong recommendations against FII's use of P-Notes in India.⁶ A July 2006 committee report said: “Existing P-Note-holders may be provided an exit route and the instruments should be phased out completely within one year.”

Along with Reddy, Damodaran was very vocal on the issue of P-Notes.

He constantly spoke about it in various forums. But when Damodaran launched his offensive on P-Note by proposing curbs, it made Chidambaram furious. The SEBI chief had dared to open Pandora's box.

At its peak in October 2007, when the marauding P-Note investments in Indian stock markets stood at near Rs. 4.5 lakh crore (USD 82 billion considering rupee at 55), Damodaran invited public comments on the proposal to immediately ban these hot money instruments in the '*derivative*' segment and to impose a limit on their fresh issuances for stocks. (vi) The official announcement on this from SEBI was made after the markets closed for trading on October 16, 2007.

Chidambaram and his ministry officials were in a state of shock as the SEBI chairman neither consulted nor communicated his decision in advance to the *Big Daddy* of financial markets then.

SEBI's announcement had sent jitters to the already nervous traders and investors, and it led to a sudden crash when the markets opened on the following day (October 17, 2007), within a minute of opening for trade, the BSE SENSEX crashed by 1,744 points or about 9% of its value (the biggest intraday fall in Indian stock markets in absolute terms). This led to an automatic suspension of trading for one hour. The markets recovered somewhat after opening. But in three trading sessions since the P-Note discussion paper was launched, Sensex plunged 1,500 points.

Along with several others, Anil Ambani, the Reliance family scion, who was then waiting to launch his multi-billion-dollar dream IPO of Reliance Power was left sulking. He had to wait for a few more weeks for markets to stabilise before he could launch the company's mega IPO roadshows.

Chidambaram, on the other hand, could not rest in peace. As he tried to salvage the situation, a series of contradictory statements followed from the FM and from the regulator SEBI.

Chidambaram said the idea was not to ban P-Notes but only restrict the

flow of money through it, while Damodaran asked FIIs to shun the back door entry as SEBI was going to ease the registration norms for them. This P-Note controversy sparked a deeply agonising animosity between Chidambaram and Damodaran. They never saw eye to eye after that ever again.

In 2008, when the SEBI chief's appointment came up for renewal, Chidambaram obviously, overwhelmingly battled for a change and suggested Bhavé's name to the PM for the top job at the regulatory body. Since the SEBI chairman's appointment required the PM's signature, the FM harped upon Singh about Bhavé's '*impeccable record*' and how the NSDL chief fulfilled all requirements to become the SEBI chief. (iv)

The question was, what would Singh do in this situation? Especially since he knew that Damodaran had made and proved his point in an efficient and adroit manner. But it seemed that Chidambaram would have had his way. To stick to his point, the grapevine had it that Chidambaram threatened to resign if Damodaran was granted an extension.

It was no secret that Chidambaram, a wily politician that he was, had the ears of the then Indian National Congress (INC) party President Sonia Gandhi on matters regarding the appointment of senior officials in the financial sector and regulatory bodies. Be it the chief of large government banks, insurance companies or RBI and SEBI, all had to have the FM's blessings, and he had the blessing of the high command.

Nudged by Mrs. Gandhi and accustomed to his tradition, PM Singh surrendered. Bhavé, who was neither interviewed nor shortlisted by the committee responsible for the selection of SEBI chairman, was given the top job at the regulatory body.

But what about SEBI's pending case and order against NSDL that had even sought Bhavé's removal for severe '*lapse*' in duty?

Chidambaram and his protégé, Dr. KP Krishnan, the then Joint Secretary

(Capital Markets) at the Ministry of Finance worked out the idea of '*ring-fencing*' him, i.e., keeping Bhave away from the NSDL matter when SEBI presided over it.

Krishnan at that time was also the board member of SEBI and had policymakers and experts in the financial market at his beck and call, as it was well known that the bureaucrat enjoyed the full support of Chidambaram. Notes put up by Krishnan at the Finance Ministry were seldom rejected by his boss. A Tamilian from Delhi, Krishnan was allotted the Karnataka cadre, but he never left the capital city until Chidambaram was in power.

After Damodaran's ouster from SEBI in 2008, a committee was constituted by the Finance Ministry to suggest a '*roadmap*' for P-Note investments. Meanwhile, Chidambaram was shifted to the Home Ministry to drive the United Progressive Alliance (UPA) government's public relations machinery after the November 26, 2008 terror attacks in Mumbai but the bureaucrats he had appointed continued with their work.

The P-Note committee of 2009 had Krishnan, his colleague CKG Nair, NSE's MD, CEO Ravi Narain and Ajay – all the Chidambaram blue-eyed boys. His agenda was safeguarded with the loyal '*bunch*'. After he quit the capital market division in the Ministry of Finance, Nair was made a judge at Securities Appellate Tribunal (SAT) while his daughter had worked with the NSE since 2013.

After a few meetings, those in the P-Note committee that were not from the '*bunch*' were left disillusioned. They could clearly see that the group's suggestions were a myth. It was not at all an overall collective job. Key members would keep the minutes of the meeting and even copies of draft reports ready in advance to be approved by everybody. It is this reason that the attendance sheet of the committee gatherings would show as to how some members, who disapproved of the procedure, largely remained absent from the meetings.

***'Ring-Fenced'* Yet Exposed**

After Bhavé's appointment as SEBI chief, the NSE had its way on COLO operations and TBT data sharing as the regulator conveniently looked the other way. But where exactly was SEBI focused on?

Sometime in February 2010, SEBI board got together and disposed of the Show-Cause Notice (SCN) that the regulator had issued to NSDL under Damodaran for the illegal demat accounts scam. The move could not have come at a more opportune time for the NSE, as it kicked-up a storm and never let media or market draw its attention towards COLO operations that NSE said it had commenced in January that year.

Chidambaram and Krishnan kept insisting that Bhavé remained *'ring-fenced'* from the proceedings, in which the SEBI board had disposed-off orders against NSDL without seeking any action against the depository. But Bhavé's conflicted role and the shenanigans of Chidambaram and Krishnan could not be suppressed for long.

Dr. Mohan Gopal, head of India's National Judicial Academy, who was also the SEBI board member during Bhavé and Krishnan's time, laid bare the skulduggery that was deployed in giving NSDL a clean chit and attempting to wipe the traces of the IPO scam. He said the *'fence'* that guarded SEBI in NSDL matter from Bhavé had collapsed badly.⁷

As part of the *'ring-fencing'* measures in 2008 to appoint Bhavé as SEBI chief, a two-member committee including Gopal and V Leeladhar, the former deputy governor was constituted into the NSDL matter. Both of them held Bhavé guilty of glaring failure in duty of supervising, investigating, monitoring data in the demat account scam and directed NSDL to conduct an independent inquiry to establish individual responsibility.

As soon as Gopal submitted his order to the SEBI board in December 2008, some 10 months after Bhavé's appointment as the chief of the regulatory body, Krishnan swung into action. A board meeting of SEBI was

called and all of them vetoed the Gopal-Leeladhar order as '*non-est*', simply non-existent in legal parlance.

In that meeting, although Bhavé remained absent as he was supposedly '*ring-fenced*' from the matter, the senior SEBI officials who reported to him and even a corporate honcho, Mohandas Pai (whose company Infosys Ltd was regulated by SEBI), were present. Gopal, who was also a member in that meeting, of course, gave a dissenting note against such an action of the board that he believed was deeply laced in conflict with every board members' fiduciary duty. But the SEBI board under Krishnan's dominance said that they had taken legal opinion for exonerating the depository. The opinion was that the committee, which had passed three orders against NSDL and its top management, had gone beyond its scope of work and its orders were outside the confines of delegation and without the authority of law.

In essence, Gopal's counter-argument dismembered the then SEBI board's argument in holding his order void, which he said was to give an escape route to NSDL and Bhavé.

Gopal carried a reputation of a law reformer, whose opinion was seldom disregarded even by the Supreme Court (SC) of India. This time he wrote a long letter to PM Singh highlighting the cesspool of policymaking that SEBI had become under the influence of Bhavé and Krishnan. His letter told a tale of raw abuse of power by the '*bureaucratic brotherhood*' that ran deep in the vein of India's administrative system. Gopal educated Singh about SEBI's system and processes being vitiated to protect Bhavé and whitewash his and NSDL's role in the IPO scam.

“‘SEBI-under-Bhavé’ judged and exonerated ‘NSDL-under-Bhavé’ through a process that was thinly disguised as independent, but was, in fact, deeply vitiated and subverted,” Gopal told the PM. (vii)

He did not mince words or hide his anguish in describing the happenings inside SEBI. Gopal said there was a subversion of the action against NSDL to protect Bhavé and the coterie that was defending him was highly conflicted.

“An informal clique of current and serving bureaucrats, SEBI officials, lawyers and corporate interests orchestrated this subversion of the due process of law. They illegally interfered with independent SEBI adjudication, manipulated legal opinions, suppressed and misrepresented facts and misled the SEBI Board and Government officials about the legality of the Orders. Law, regulations and established precedent were violated. NSDL was given undue special treatment. NSDL was relieved of a fine of crores of rupees, and Securities Appellate Tribunal (SAT) decisions adverse to SEBI but favouring NSDL, were not appealed to the SC as they should have been,” Gopal quoted.

A legal expert-par-excellence, he further noted:

“One of the most shocking and unprecedented actions taken by SEBI to exculpate NSDL was the board—for the first time in SEBI’s history – setting aside quasi-judicial orders which are, under the law, subject only to judicial review.”

SEBI is a quasi-judicial authority vested with powers to issue legally binding orders like a court of law. These SEBI orders can only be appealed in SAT or a higher court. But orders passed by Gopal, as a SEBI board member, were disposed-off by other SEBI officials, which not only was unconventional but a clear cause of conflict.

Complimenting what Gopal said, J S Verma, another eminent jurist and former Chief Justice of India, too came in his support and said, “SEBI’s action violated established legal and constitutional principles.”

To act as a *‘Devil’s Advocate’*, even if one accepted Bhavé’s contention that he had no role in obtaining this special treatment from SEBI – orders against NSDL being suppressed, then reviewed, then reversed and finally substituted by the board – was something that was never done for any other company in SEBI’s history.

“Who then was behind this? Perhaps a judicial inquiry should be

conducted into this affair to determine who was subverting the normal procedures of SEBI, and for what motive and for what consideration – inside and outside SEBI, including the counsel. This is a serious matter because the underlying issue involved is the IPO scam and we can't rule out the hand of those responsible for the great securities fraud perpetrated on retail investors. The board relied on the legal opinion of a lawyer, whose firm was defending the prime accused in the IPO scam," Gopal averred.

He also highlighted four structural fault lines in SEBI's legal framework that had made this power abuse possible. These were: "Inadequate Transparency, (lack of) Public accountability; and Parliamentary Oversight."

On the '*Lack of Protection against Conflict of Interest*', Gopal wrote, "A Code of Conduct for the SEBI board was evolved at his instance, but the mechanism was violated and then dismantled in the context of the NSDL matter."

He said SEBI had perverse practices compared to the Securities Exchange Commission (SEC) in the US.

"The SEC held open public meetings and the US Senate exercised close scrutiny over its workings. There is nothing comparable in India." Gopal did not stop here but further pointed to the PM that he was subjected to, "Retaliation and attack without any protection."

On Pai and Infosys, Gopal said, "SEBI board generously excused the conflict of interest arising out of the business relationship between Infosys and NSDL. It was perhaps for the first time in Indian history that judicial power was exercised by a serving private sector corporate official."

As a tech company Infosys was engaged by NSDL and Bhavé had a say in fixing the remuneration.

Gopal also drew his anger towards SEBI's ineffective framework for law enforcement.

“The structure for law enforcement in SEBI is seriously flawed. There are overlapping enforcement and punitive provisions in the Act, which needs to be rationalised. This subjects a regulated entity to multiple proceedings without a clear distinction between them. Major violations established through investigation are excused without punitive action through opaque consent orders and faulty adjudicator orders favouring wrongdoers—in such cases review by SAT would never be sought because neither SEBI nor the wrongdoer want it. A company guilty of ‘*criminal*’ market manipulation was let off by a whole-time member (WTM) asking it to be more careful in future. SEBI does not have adequate focus and priority on law enforcement, with the result that it bent backwards and violated the law to protect a favoured regulated entity rather than pursue it to enforce the law.”

Gopal then directly accused Krishnan for, in his words, “Exercising undue influence in the functioning of an independent regulator through informal back channels, through which SEBI officials were funnelling information and documents to him, which he legally should not have access to.”

It was the most serious charge levelled against Krishnan by an eminent legal eagle and laid bare the bureaucrats’ true fangs and covert conduct.

On the point of reforms, Gopal had the following suggestion, “SEBI Act badly needs to be redesigned. It contains too many explicit and implicit levers of bureaucratic and political control of the regulator on one hand and too little public oversight, transparency and public accountability on the other hand. SEBI in effect is run by an informal caucus of serving or former civil servants rather than domain experts.”

The five-page letter ended with asking the PM to order a high-level inquiry into SEBI’s decisions in relation to NSDL during Bhavé’s tenure and to look into the structural issues raised by him. He said that the SEBI board lacked relevant expertise because it (was) dominated by *Babus* – serving and ex-bureaucrats – pointing towards Krishnan and Bhavé.

SEBI under the influence of Bhavé and Krishnan was waiting to explode. Gopal's revelations had just torched it.

"The government's interaction with the regulator would be over the counter and not below the table," Gopal cautioned PM Singh in as many words.

The letter was so explosive and blistering it blew the lid off on the theatrics playing at SEBI. Gopal's indictment of Bhavé and Krishnan was stunning. However, *'Caesar's wife must be above suspicion.'* When fingers were pointed at her, *Julius Caesar divorced Pompeia.*

Under any other government, immediate action would have been taken, heads would have rolled, and the due process of law would have been fast tracked. But not under the *'Accidental PM'*. Singh just forwarded the letter to Chidambaram, who returned as the FM in 2012. Gopal was later dumped from SEBI's board.

This was yet another instance where history will judge Singh as the weakest PM of India.

Nevertheless, Gopal's letter had by then damaged the prospects of Bhavé who did not win another extension as SEBI chief, since his tenure came to end when Chidambaram was still the Home Minister and Pranab Mukharjee the FM.

In his latter revelations, Gopal has asserted how high were the stakes involved in SEBI's coveted post. Apart from the fact that he said he was threatened and pressurised, he made some explosive revelations.

"Yes, pressured and threatened," he told the media in an interview on November 17, 2011.⁸

So who was so powerful to pressurise the head of India's National Judicial Academy?

News reporters were abuzz with gossip about a powerful figure in SEBI

having reached out to Gopal asking him to reverse his order.

Those who could issue threats to a person with the stature of Gopal were '*The Market Mafia*'. The power they derived from their position was the harsh, repressive force that wielded like a '*Rampuri*' (Indian gravity knife) to cut people who came in their way. Threats to Gopal were splashed in leading newspapers but those at whom fingers were being pointed at, remained silent – an act more deadly than their violence.

Before Bhavé's ouster as SEBI chief, the high-tech trading game was all set at NSE with all advanced tech tools fully in operation. Gopal's letter was a stark picture of why SEBI under Bhavé allowed NSE to do what it wanted to. As NSE, the largest promoter of NSDL, had let Bhavé do what he wanted to at the depository. NSE had raised no finger at Bhavé even when the SEBI probe under Damodaran highlighted the then NSDL chief's laxity in the fake demat accounts scam.

Bhavé's detachment from his *Dharma* as the SEBI chairman opened the door for data theft at India's largest asset pool, NSE. It also led to the installation of defective COLO trading infrastructure at the exchange, by an unofficial decree, which gave select few brokers preferential access to equity derivatives trading. Fast-moving trading bots, deep-rooted nexus between enterprising PhD scholars, economists, top dollar earning executives, politicians, bureaucrats and salivating brokers were at play.

LIFE IN THE SPEED LANE

A Mystery of High Degree

Trading in stock derivatives for many is a passion as strong as gambling, and it's known to be addictive as well. The use of Algos for a wager was made popular at the world's largest horse racing circuit in Hong Kong. Sometime in November 2001, two American gamblers, Bill Benter and his associate, Paul Coladonato, cracked the horse racing code by writing an Algo code that couldn't lose at the track. That year they won the '*Triple-Trio*', the biggest jackpot at Hong Kong's *Happy Valley Racecourse*, where at least a million people placed their bet. The wager was a trifecta of trifectas where gamblers predicted the top three horses. 35 of Benter and Coladonato's bets correctly predicted the finishers, giving them USD 16 million in windfall. Years later, it came to fore that smart Algos had kept the duo ahead of phenomena called the '*Gamblers Ruins*'.

Benter and Coladonato hired anyone including coders, academics, journalists – who could improve their Algos. In horse racing, front-runners break a leg, jockeys fall, champion thoroughbreds decide, for no apparent reason, that they're simply not in the mood. But Algo software code that Benter and Coladonato wrote captured every aspect of the unforeseen circumstance and predicted probable winners. The duo constantly kept

updating their Algo codes with newer probabilities and data.

Akin to Derby, stock derivatives are all about predicting near-term winners. A pure stock market investor may buy company shares and hold them for long to see its value rise. But derivative traders look for shares that are ripe for sharp moves on various probabilities. They are an agile breed of traders who want to get-in and get-out of bets, fast. The spreads in a derivative wager could be tiny but leverage, very high. Such stock derivative bets are known as Futures and Option (F&O) trading. The wager could be as brief as a fraction of a few seconds or stretch up to three months, the furthest that F&O contracts between buyers and sellers can stay alive.

Gamblers are attracted to the excitement of F&O trading as it requires only tiny margin money to be paid up-front. This, since nobody takes delivery of shares in F&O but only speculate the price move. Hence, a sum that can cover the anticipated risk of loss to traders is collected as a security from each party by the exchange in advance.

The prize money is paid out every day on the settlement of the bet and also a call is made for a shortfall in the security margin, which is called '*Marking to Market*'.

The greed of fast buck in F&O ensures that business remains brisk for the exchanges as they impose a small cost on each transaction for parties on both sides of the trade. NSE with its monopoly in the F&O segment has often been teased by brokers for making supernormal profits, something which has risen to upwards of Rs. 1,000 crore annually post the taxes. It was the monthly expiry F&O contracts that generated a high churn, setting higher volume and turnover records regularly. The F&O turnover touched over Rs. 15 lakh crore in a '*single day*' in September 2017 (approximately Rs. 200 billion at an exchange rate of Rs. 75 against the USD). In 2020 the volume churn record was more than double that amount. A study of margin money required for such colossal volume churn could throw up a measly number of a few thousand crores, which can demonstrate how mind-

boggling was the use of leverage at play in F&O.

Like the millions at the Happy Valley Racecourse, all stock market traders would like to think that they could win, not knowing their wager is against Algos similar to Benter and Coladonato, which wouldn't lose. The parallel between horse racing and equity derivative ends at the finishing line. In derby, there could be multiple winners who could have predicted the same horse to win. But in stock derivatives, each improvement in tech tools had turned the game into the fastest finger first. Human hands have no contest against HFT-driven Algos as the stock price often moved by the time a physically punched trade by the dealer reached the master order matching engine. This depleted the profits for non-machine traders. Still, the only edge that could have humans dominate the scenario was information. That too was killed when NSE started giving TBT data and invited brokers inside its premises to set up the COLO farm.

Ajay and Susan were captivated by Algo trading. As PhD scholars from the University of Southern California (USC), they behaved more like modern-day Utopians, idealistic reformers in shaping the policy narrative for NSE. Both were hardened advocates of derivatives trading and wrote papers, blogs and newspaper columns churning out pompous theories to showcase its traits. NSE zealously guarded its higher volume churn in the F&O from rivals like the BSE.

Ajay and Susan had theorised that Algos got more liquidity to markets and widened the pool. What they left unsaid was that comp bots were also reminiscent of jobless professional day-traders who fell behind in the velocity and latency race. Any probability a day trader could think of to punch a trade, Algos had it inbuilt in their memory and they struck with precision and alacrity that was impossible to match for a dealer's hand.

Ajay and Susan's association with NSE goes back to the mid-1990s when Nifty-50, the key index and most successful product of the exchange, was conceived. In 1996, Ashishkumar Chauhan, Vice President (VP), NSE was

in-charge of setting up the derivatives market. His job also involved creating a new index that could be traded in the derivatives. NSE's crowd puller was the F&O contract of the Nifty-50 index, which allowed traders to take a position in top 50-stocks in a single trade. Nifty index derivatives allowed a trader to wager on its near-term moves, which was akin to betting on the whole market rather than a single stock counter as the index covered nearly two-third of NSE's total capitalisation. Since betting on the whole market to move up or down required a measly margin, it attracted many starry-eyed gamblers. The index was also the delight of P-Note insiders.

Chauhan, who is presently the MD, CEO of BSE, had already developed the NSE-100 index in 1994, which was used by the exchange to showcase top 100 companies on its platform in the initial days. That index could not be traded. But as the head of equity derivatives, Chauhan figured that NSE needed an index that could be traded to gain participation.

An '*index committee*' was formed and the modalities of Nifty worked out. Ajay and Susan, who had just returned to India with claims of being armed with a PhD in 1995 and were trying to understand the core working of exchange and market, have often accepted full credit for the creation of Nifty. Whatever work was done by the index committee got lost in their narrative.

A book titled *The Future of Fund Management in India: Challenges and Opportunities* published by Tata-McGraw Hill in 1997 and edited by Tushar Waghmare details the formation of the index committee, Chauhan's role and how Ajay and Susan were roped in later after the idea was already floated.

"We decided to constitute an index committee consisting of imminent market participants and academicians to ensure the decision governing grey areas are made in a sensible way," Chauhan told Waghmare as quoted in the book.

Being part of such a high profile committee, Ajay and Susan along with others gave their suggestions on aspects including how large the index should

be, the kind of securities to be included and on matters involving its calculation. But since both published their research papers on Nifty-50 index and derivatives in the following years, a narrative was being fed that the duo was solely responsible for the index creation. Media reports also suggest that they were paid annual royalty for higher churn than F&O trading in Nifty index generated. But nobody including the NSE has ever confirmed or denied it.

In interviews, Ajay told the press that he and Susan wrote the code for simulating the management of NSE, the evolution of the index at computing the diversification gains of the size of the index and the measurement of the index cost. It took them three months overall but they wrote all of the codes.

Very pompously, he was quoted as saying, “It was herculean task and we worked like dogs. There was no staff or team or support. Every little secretarial job was done by us.”⁹

According to NSE officials, it would not be wrong to say that Ajay and Susan did a lot of hard work on the Nifty index. They are also known to have assisted NSE in generating favourable opinions on every major policy initiative that included the listing of Nifty-50 index for trading in Singapore.

The Nifty index was listed on the Singapore Stock Exchange (SGX) sometime in the year 2000 and only FIIs could take a position there. All trades were dollar-denominated. Like P-Notes, this presented yet another avenue for disguised traders, who did not want the Indian Government to know that they were betting on the country’s market.

Although theories were floated that SGX Nifty played an important role in bringing incremental volume churn for NSE in Mumbai, the fact remains that Singapore trading has so far only benefited FIIs. Noticeably, on several occasions, SGX Nifty pulled the volumes away from NSE in Mumbai as the market share of the off-shore trading was seen higher. NSE earned listing fees from SGX but traders in India watched FIIs decide India’s market

moves in Singapore. The most glaring instance of such a play was seen in 2007 when Damodaran proposed a ban on P-Notes. From the next day, FII open interest and volumes on SGX Nifty swelled as P-Note trading shifted there.

SGX Nifty listing has pained India and the government has made unsuccessful attempts either to delist the index or somehow bring the off-shore trading volumes back home by offering tax breaks. But disguised traders have their game set and do not bother for tax breaks as long as the source of their money is not questioned by authorities in an off-shore market. Curiously, even SGX may not have data on the ultimate beneficiary of funds operating on its platform as it has allowed ‘Omnibus account structures.’

Omnibus accounts refer to those that hold more than one item (*omni-meaning ‘many’ and -bus meaning ‘business’*). Such accounts are normally overseen by a futures manager, who complete trades on behalf of participating individual investors. So the source of money of each client is unavailable with the exchange. The alibi of SGX is that it allows funds flow only from globally approved and tax complainant jurisdictions.

Singapore is a large asset pool of global funds as the tax outgo for money managers in that country starts at zero and is capped at a higher percentage point depending on citizenship factors. Most fund managers have acquired Singapore citizenship now but have their funds pooling in assets from tax havens across the globe. Also, India’s double tax avoidance treaty with Singapore ensured virtually zero tax on capital gains generated through equity trading. Betting on SGX Nifty was a real sweetheart deal for FIIs.

“Even though Susan and Ajay had no formal role in the NSE projects, they had the luxury of being treated as insiders and were invited for discussions.” (ix)

Among other things, their work gave the researchers unprecedented access to NSE’s goldmine in plain sight – data.

SEBI's investigation that followed a whistleblower letter in 2015 about Algo trading scam at NSE revealed how Ajay got crucial data from the exchange for years without any formality. What and how much data did NSE share with Ajay and Susan? This is simply not traceable. SEBI too is clueless. Later, it came to light that the couple with their family enterprise had devised a lucrative business model of Algo trading software and even sold it to clients.

How did Ajay and Susan achieve this unusual distinction of being both an entity and action right from the beginning of their careers?

Couple's Early Days

Economics and much less the inner workings of India's financial markets was never their first calling. Ajay was more interested in building aeroplanes and Susan in dams and bridges. When they went to study at the IIT Bombay in the early 1980s, Ajay came out as a computer programmer and a BTech in Aeronautical Engineering and Susan graduated as a Civil Engineer. Though the two came from different cultural backgrounds and geographies, they bonded over their common love for hiking and eventually came close to each other. After graduating from IIT, they both moved to the US in the early 1990s. There, they enrolled at USC and on their return to India a few years later, started their professional career as PhD scholars.

Princeton Review, an academic journal, described USC as a school best known for its festive fraternity parties and sports teams until the early 1990s. As a matter of fact, it was once teasingly called the University of Spoiled Children. As per LaMag (Los Angeles Magazine), with the help of strong leaders and their knack for fundraising and public relations (PR), the once lethargic institution was turned into a research and academic behemoth that began to shoulder its way among more established East Coast rivals in the US. The sobriquet of USC known as the most scandal-plagued campus in America is still talked about in the United States.¹⁰

Along with academic studies, Ajay and Susan also worked with RAND Corporation, a shadowy policy think-tank at Santa Monica. RAND, which stands for research and development, has a self-imposed mission to *improve global decision making through research and analysis*. A deeper look at its history reveals that RAND, which rose from the ashes of World War II, virtually tries to manipulate and influence public policy discourse in the Americas. With the success of the Manhattan Project – a USD 2 billion initiative in the US to create the first atomic bomb – a five-star Air Force General, Henry Hap Arnold concluded that America needed a team of great minds to keep the country's technology ahead of the world.

In 1946, he put together a small group of scientists, with USD 10 million in funding and started RAND with the help of a family friend and aircraft magnate Donald Douglas in his factory in Santa Monica. It is said that the internet, the first human satellite to have orbited around the earth or IBM's first modern computer, all saw their breakthrough at RAND. It is here that both Ajay and Susan cut their teeth into ways of influencing public policy and thinking, which would go a long way in devising strategies to seep into India's policymaking *Juggernaut*.

But more than their stint at RAND, it is the PhD thesis they wrote beckons as it lays a foundation for them as researchers.

Ajay wrote a thesis on '*A Life Cycle Model of Fertility Choice*', and submitted a dissertation paper to Andrew Weiss in 1993, the records of which and his PhD certificate are available on the website of USC.¹¹

The thesis is an expansion of study on '*fertility literature*', which was pioneered by Gary S Becker, who was awarded a Nobel in 1992 for broadening the range of problems considered by economists for bringing it closer to sociology. In 1989, Robert Barro and Becker had published a paper called '*Fertility Choice in a Model of Economic Growth*'. Becker continued his study on '*fertility*' churning out crucial papers.

Ajay's thesis has an empirical implementation on birth probabilities, death of children, death of parents, old-age support, etc., which fall under the categorisation of general economics as per USC and carries a PhD certificate accorded to him by the university.

However, when it comes to credits of Susan completing her PhD and being awarded a degree certificate in the early 1990s, the records are nowhere available on the USC website, which otherwise displays scores of papers and PhD certificates. Correspondence with USC to dig for the PhD degree certificate of Susan too did not yield results. "Therefore, her claims that she was a PhD for two decades are shrouded in mystery."

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INFORMATION YOU PROVIDED

Subject Name:	SUSAN	THOMAS
	First Name	Last Name
School Name:	UNIVERSITY OF SOUTHERN CALIFORNIA	
Degree Award Year:	1995	
Attempt To:	Verify a degree	

UNABLE TO CONFIRM

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Interestingly, the USC displays a dissertation paper submitted by Susan as recently as June 20, 2017, which according to the university was in '*partial fulfilment of requirement for degree of Doctor of Philosophy*'.¹² Like USC displays a PhD certificate for dissertation paper submitted by Ajay, no such

evidence is available for Susan even for her paper submitted in 2017 that could prove her credentials as a PhD scholar beyond doubt.

Below is the degree certificate of Ajay

UNIVERSITY OF SOUTHERN CALIFORNIA
THE GRADUATE SCHOOL
UNIVERSITY PARK
LOS ANGELES, CALIFORNIA 90007

This dissertation, written by
..... Ajay Shah
under the direction of h.i.s..... Dissertation
Committee, and approved by all its members,
has been presented to and accepted by The
Graduate School, in partial fulfillment of re-
quirements for the degree of
DOCTOR OF PHILOSOPHY
Barbara Solomon
.....
Dean of Graduate Studies

Date July 19, 1993

DISSERTATION COMMITTEE
Andrew A. Weiss
..... Chairperson
LA Leland
.....
Quinn Morris
.....

Ph.D.
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In the absence of such crucial piece of evidence, it can be fairly concluded that there are no records to prove that Susan, one of India's leading researcher on stock markets and economy since the 1990s and a senior professor at Indira Gandhi Institute of Developmental Research (IGIDR), had a PhD degree for over "two decades" as claimed by her.

If indeed she acquired a PhD degree after due research and submission of dissertation in 2017, what did Susan base her paper on? As per records, she

submitted a paper titled '*Empirical Characterisation of The Bombay Stock Exchange*', which analyses the behaviour of Indian markets using the returns on an index calculated for the period from '*April 1979 to March 1995*'. (xii)

It was 1995 when Susan had returned to India, post, what she claimed, acquiring a PhD. In that case, how was a dissertation paper submitted to '*Aizenman, Joshua*' at USC in 2017, as a partial fulfilment of requirement for degree? Even for the 2017 paper of Susan, the USC does not display any PhD certificate.

Astoundingly, the same paper was also submitted by Susan in 1995 to '*Andrew Weiss*', who was also a guide to Ajay. It was titled '*An Empirical Characterization of the Bombay Stock Exchange*'.

On July 1, 1995, Susan had even published a paper that included a chapter titled '*Heteroskedasticity Models on the Bombay Stock Exchange*', from her thesis, then submitted just 10 days ago to the USE. The paper and the chapter are available on the internet¹³ and even gives a reference to her so claimed 1995 thesis: '*An Empirical Characterization of the Bombay Stock Exchange*'.

The USC website now says that Susan defended her thesis on June 20, 2017. For USC, it could be a fallacy of research if a doctorate could be awarded by the university to someone who based her study in 2017 on data analysed between 1979 to 1995 and tries to find a pattern or conclude market behaviour.

Much water has passed under the bridge for the stock markets in India as well as globally between 1995 to 2017, in terms of trading regulations and technology that could determine the market behaviour.

What could one do in 2017 by studying stock market patterns between 1979 to 1995? What did Susan aim for in the research paper?

"The aim of this thesis is thus to document the features of the BSE with a view to measuring the degree of efficiency of the market, using '*newly*

constructed data’ from original sources. Research will be done both from the point of view of individual companies as well as the BSE Sensex. The time period for the data-set is five years of daily data from 1990 to 1994 for individual company returns. Most of the individual company research was done using a set of 92 company returns data. Daily BSE Sensex returns data is available from April 1979 to April 1995,” she said in her paper. (xii)

She concluded her paper by stating that, “Both the higher rates of return as well as the lack of correlation of Indian stock markets with other, global equity markets would make investments into India attractive from the viewpoint of a well-diversified portfolio controlled in New York. A rigorous empirical characterisation of the market behaviour would be useful in portfolio optimisation involving Indian stocks.”

Is India’s market not correlated to the global equity scenario now? Multiple studies before 2017 have proven they are. Can the features of BSE Sensex be documented in 2017 by studying data up to 1995? Almost all the companies that formed constituents of Sensex in 1995 have changed and it is futile to any cause, so much for a PhD to write such a paper. Is she a PhD at all? These are very perplexing questions, which have no logical answers.

The onus to clear this mystery over the PhD degree lay both on Ajay and Susan right from the beginning of their careers. Despite not holding a valid degree, Susan received sitting fees for attending committee meetings, professional fees for conducting training, workshops and research projects, among others from the NSE and its affiliate companies. How could the exchange not check the educational qualifications of Susan, unless some patronage from higher up was at work?

It was through a criminal act of deception that the couple gained influential positions as consultants, researchers, PhD guides and most importantly – access to NSE data. The seeds of the data scandal at NSE were sown around 1995 when the exchange started sharing crucial information with the couple on the basis of falsehood. It was only the climax of the

scandal that hit the exchange some two decades later in 2015 – when the COLO scam came to light.

Ajay is made a key accused by SEBI for the data heist but Susan's name has conspicuously escaped. How could SEBI not ask questions about the educational qualification of Susan during its entire course of COLO investigation and seek reasons on how she and Ajay were chosen for projects by NSE despite not holding any official position?

Not surprising. Such questions could have been raised only if there was any genuine intent to probe. To the level of its inefficiency, SEBI has not even '*named*' Susan in the NSE COLO scam report. After Susan escaped SEBI lens in 2015, she targeted another large Indian bourse – Multi Commodity Exchange (MCX) – for data theft.

The Facade

Dr. Narottam Shah, Ajay's father, was one of India's most influential economists during the 1970s-1980s. In 1976, he established an independent economic think-tank – Centre for Monitoring Indian Economy (CMIE) – which sounded like just another government department but actually was a private limited company. Today the company claims crores of rupees in just general reserves out of its activities of providing information solutions in the form of databases and research reports.

At a time when *dhoti*-clad, *paan*-chewing stock brokers dominated equity trading in a circular building at BSE, rightly named *Rotunda* and access to much information was restricted to a closed circle of traders, CMIE's products came as a great tool for their business. Its claim of dissemination of streamlined financial market and poverty-related economy data apart from exhaustive information of most of the listed companies was enough to make it a most coveted product. Such data is the oxygen that drives the market.

For his time, Shah was a very influential man. His clout can be gauged

from the fact that several leading economists of that era were vying to partner with his enterprise. No wonder people with high stature like that of DT Lakdawala, noted economist and deputy chairman of the Planning Commission, was also a director on CMIE board.

The Planning Commission was the *Holy Grail* for all socio-economic planning in India. It has been India's crucial government think-tank that dominates discussions around policymaking and is heard by every key institution in the country. However, it could not function without key economic data. For decades, India's bureaucrats, are known to have relied on data and other inputs from CMIE.

Since in the 1970s and '80s data related to private unlisted companies was never easily available in the public domain and there were no means to search through records at the hard copy library, nor there was internet, CMIE's business was a virtual monopoly.

Aiding to the fact was that Shah had named his enterprise in such a way that many institutions and even academics perceived CMIE as some government agency. Presence of people like Lakdawala and others of his stature on its board¹⁴ and the name of the company unlike any private limited firm, only gave those perceptions much legitimacy.

Even the financial analysts at brokerage houses, research firms and journalists loved CMIE, since it offered data related to shareholding patterns and other filings of any private company on a platter, which was otherwise a tedious job during that time.

When Shah passed away in 1984, CMIE was run by Lakadawala and later by Shah's son-in-law Mahesh Vyas. Under Vyas, the popular perception about CMIE gained even more weight when Surendra Ambalal Dave, the first chairman of SEBI (between 1988 and 1990), was appointed as the chairman of CMIE, right after his retirement.

Dave brought in more recognition and revenues to CMIE, as he worked

on several government committees while being the chairman of CMIE and used data extensively from his '*own company*' for official projects of these committees. At the end of committee reports, farcical, Dave even issued thank you letters and acknowledgements to his own company CMIE.

While still in his teenage, Ajay lost his father but the hold of CMIE in the financial sector and blessings of personalities like Lakdawala and Dave gave him a launchpad much early in his career. To understand the importance of Dave and how it eventually acted as the biggest boon for Ajay, one should pay attention towards the fact that friendship between the former SEBI chief and the founder chairman of NSE, Dr. Ramchandra H Patil, went back to the days when they worked together at IDBI Bank. Patil was also much younger in age to Ajay's father Shah.

Dave was named as the first SEBI chairman in April 1988 after PM Rajiv Gandhi announced in his 1987-88 budget, his government's intention to form a separate board for the regulation and orderly functioning of the stock exchange and the securities market.

In the initial days of SEBI's formation, Dave operated from IDBI's head office in Mumbai's Cuffe Parade. As part of his core team, he got young, bright officials from IDBI Bank to write the securities law. These included Ravi Narain, Chitra Ramkrishna, G V Nageswara Rao and Pratip Kar – all of whom later gained key positions in NSE or its group companies and rose to the ranks to reach the top position. Chauhan, however, moved out of the NSE due to a power struggle among the high ranking executives and joined Dhirubhai Ambani.

The entire top brass of NSE was junior to Dave at IDBI Bank, the institution which was the key founder and the largest promoter of the exchange. Suresh Shankar Nadkarni who became chairman of SEBI after 1992 also came from IDBI Bank. Damodaran, who headed SEBI between 2005 and 2008 was a former IDBI Bank chairman after he quit IAS. Even Bhawe, who had worked as an executive director at SEBI in the early 1990s

was obliged to Dave, but Dave was in turn obliged by data disseminating company CMIE, promoted by Ajay and his family. No wonder Dave was offered CMIE's chairmanship soon after his retirement from SEBI.

On his return to India with the PhD, Ajay now the owner of a majority stake in CMIE, was at the helm of affairs at the company as its President. Susan, *'who could not complete her doctorate'* too joined Ajay at CMIE. Undoubtedly, the network of his father among government officials, leading economists and the edge of CMIE among key institutions and SEBI were important assets for them. It made them important personalities in the financial market at the very beginning of their careers.

Along with their stint at the family concern, and despite having no PhD certificate, Susan along with Ajay were also inducted as professors with an independent private research and educational institution based out of Bombay, the IGIDR. Kirit Parikh, the Founder Director of IGIDR was an economist and a close friend of Shah, Lakdawala and Dave. IGIDR further gave Ajay and Susan the legitimate recognition as researchers. Although a private institution, IGIDR had managed to get funding by the apex bank of the country RBI, which could be termed as rare if not an unusual fate.

IGIDR was registered as an autonomous society in 1986 and as a public trust in 1987. Its sprawling campus based on government allotted land in Mumbai suburbs was inaugurated by former PM Rajiv Gandhi and subsequently, the institute was even recognised as a deemed university.

With the facade of CMIE, IGIDR and the backing of Dave, Lakdawala gave Ajay and Susan an advantage to mask the fact that Susan, who has also been a guide to a SEBI official in doctorate studies, did not herself hold a PhD degree. Blessings of Dr. Patil, NSE's founder boss and a father figure to the staff at the exchange but much junior to Dave in IDBI, also helped Ajay gain a foothold at the exchange. Dave remained the chairman of CMIE and Ajay, who was never officially employed by NSE in any capacity, except board seats on group companies, operated like a perfect insider in the

exchange.

Since the narrative as a creator of Nifty index had stuck, Ajay was engaged at NSDL by Bhavé, who was tutored by Dave when he was a junior official in SEBI during his first stint. Ajay was made a board member of NSE promoted Clearing Corporation of India for over a decade between 2001 and 2012 and National Commodity Derivative Exchange (NCDEX) between 2003 and 2005. Ajay and Susan were even made part of SEBI committees, one such also had Bhavé and Ajay working together in 2005.

Ajay's bio gives his illustrious connections in the government and financial markets as he was associated with many committees, some of which were very crucial ones during Chidambaram's tenure as the FM. He was a virtual member of the Planning Commission Committee on Financial Sector Reforms, chaired by Raghuram Rajan in 2009.

Prior to that, Ajay and Rajan had written a paper together titled '*New Directions in Indian Financial Sector Policy*' way back in 2005. Then, Rajan was not known much in India as he was on a teaching job in the United States. Ajay even wrote a paper with Urijit Patel on '*Pension Funds and Social Security*'. Both Rajan and Patel became governors of RBI successively. So clearly, Ajay had been hobnobbing with the who's who of the Indian financial sector and he leveraged on it very successfully.

NSE was inaugurated by Manmohan Singh when he was the FM in 1994 and the economist-turned-politician often felt that the exchange was his baby. NSE's Narain was known to be close to Singh's blue-eyed boy, the prominent economist and fellow Punjabi, Montek Singh Ahluwalia. *Narain* is a Jat surname of a strong ethnic group, who have their origins in Haryana. Several of the Jat tribesmen had migrated to south-western Punjab. Notably, Narain's father, Hari Narain, too, was a well-known economist and a contemporary of Singh. Narain and Ahluwalia shared an old bonding, as both are Cambridge University alumnus.

When Singh became the PM, he made Ahluwalia his key advisor. Ajay

who, with the help of NSE had already established a foothold in New Delhi, became a consultant to the Ministry of Finance between 2001 and 2005 and was known to be close to Chidambaram when the latter became the FM in May 2004. It was a well-known thing among bureaucrats in the Finance Ministry in Delhi that Ajay also played a role in writing Chidambaram's budget speech.

While Susan still carried on with most of the work from Mumbai, Ajay remained in Delhi and got employed with the National Institute of Public Finances and Policy (NIPFP) in 2007 as a research professor. The organisation fell directly under the preview of India's Finance Ministry and the RBI. NIPFP is a think-tank that provides economic inputs to the government on planning and is a go-to organisation for the Finance Ministry.

The governing body of NIPFP comprises three representatives of the Finance Ministry, who then reported directly to Chidambaram when Ajay was employed there. Interestingly, NIPFP board also had a representative of IGIDR, where Ajay and Susan were already engaged. A few members also included executives from corporate houses.

The chairman of NIPFP until March 2020 was one of India's celebrated former economists and bureaucrat Vijay Kelkar. He was the non-executive chairman at NSE between 2010 and 2013 when the COLO scam was at its peak. Kelkar got into NSE when Bhavé was the SEBI chairman and his appointment at the exchange gave rise to a conflict of interest debate. Kelkar joined the exchanges' board soon after laying down the office as the chairman of the 13th Finance Commission. Ideally, when bureaucrats retire and join private companies, they should follow some cooling-off period¹⁵ but Bhavé did not pay any heed to this and did not object to Kelkar's appointment at NSE. SEBI holds the power to approve or disapprove key appointments at exchanges. At NSE, even though Kelkar was a non-executive chairman, his pay could easily be a matter of envy for many top corporate executives. A flat in an up-market locality and a chauffeur driven

car, are the few luxuries that bureaucrats can't make-do without. NSE, under Narain, never deprived Kelkar of that. Similarly, during Kelkar's tenure at NSE, the exchange raised Narain's annual pay package to around Rs 8 crore.

Like Kelkar, a few more NSE board members, too, have not bothered to follow such asceticism of cooling-off period after retiring from government jobs and RBI, which were directly regulating NSE. After the COLO scam broke, Kelkar wrote letters to all the top bureaucrats in support of Ajay explaining his innocence. It was a closely-knit group thriving on the high patronage of the chain of the *'bureaucratic brotherhood'*, the protection of political *'Godfather'* and corporate interests.^{16 17}

As on date, both Bhavé and Kelkar are together ensconced on the board of The Indian Institute for Human Settlements (IIHS), a private limited company that promotes research.

Kelkar has been on the IIHS board since 2010 and Bhavé joined him in 2013. On February 6, 2020, IIHS arranged a talk on Kelkar and Ajay's book in Bangalore, where key speakers also included Bhavé and Krishnan. Together, Kelkar and Ajay are on the board of National Bulk Handling Corporation.¹⁸ The company is 100 per cent promoted by a private equity firm, True North, which was originally incorporated as India Value Fund (IVF) in 1999 and invested in private companies. Ajay's association with IVF can be traced back to the early 2000s. The fund's promoter company True North Trusteeship Private Ltd now has Dave and Susan as its board of directors. It has huge money flowing in from Mauritius and Cayman.

Even more than being an academician, Ajay was NSE's chief lobbyist in Delhi and kept promoting derivatives trading. In turn, the exchange offered him unhindered access to the data of Mumbai markets. The data he was being given on a platter from NSE was used by Ajay and entities linked to him to create a high-tech trading software. This software was also sold to brokers for derivative trading at NSE COLO and the deal was 40 per cent sharing of the trading profits. There was no agreement on loss sharing – perhaps the

software developers were self-assured of its prowess.

Despite there being evidence of such arrangements, which was uncovered during investigations that followed the 2015 whistleblower exposé on NSE, SEBI fell woefully short in investigating the software deal. This cuts the most crucial piece in the trail that could lead to the beneficiaries – albeit by design.

Moreover, the full money trail of companies that have Ajay, Susan and Dave on its board or as advisors, has also been left untouched by the regulator. SEBI has itself said that Ajay exploited crucial NSE data for commercial gains, therefore, it was binding and imperative on the regulator to investigate the extent of benefits he derived from data and his position at NSE. But officials in SEBI who are accustomed to deviating from the path of their *Dharma*, have also ignored this very conveniently.

Money flow in some of the companies, where Ajay and his family members are linked, is laced in an intricate web in tax paradise. The family also shares links with big Mumbai brokers that cater to large FIIs. Susan was also a board of director of Geojit Securities, one of South India's leading stock broking firms, between 1999 and 2003. It was a clear case of conflict then too, as she was an NSE insider and got data without any agreement.

Susan was also a member of SEBI's secondary market advisory committee between 2009-2017. During the same time, Susan's sister, Sunita, who owned IFPL along with her partner, Krishna Dagli, provided Algo trading software to many brokers to trade at NSE's COLO. Ajay's brother-in-law, Vyas, who is a shareholding partner in CMIE, is still on the board of Geojit group company. He was appointed in 2003 soon after Susan quit. Interestingly, Susan's brother-in-law, Lala, a top rung NSE official and a suspect in the COLO scam, earlier worked with Geojit before joining NSE.

NSE on Top

Over 20 million people trade on the NSE and BSE as per the number of

demat records found with NSDL and CDSL. Nearly 50 per cent of retail trading volumes come from tier-II and tier-III cities. But there is little understanding among these traders of how high-tech the systems have grown over the years. Mobile app-driven trading with direct access to the exchanges is still in its nascent stage in India when compared to countries, let's say like Korea. In such a scenario, when the NSE allowed HFT-driven Algos and COLO-based trading between 2008 and 2010, retail traders knew nothing about those monsters. None of the large domestic institutions too participated in COLO-based trading or used bots before that.

Mutual funds and insurance business giants including the Life Insurance Corporation of India (LIC) that dominate the buy-side trading in Indian markets are considered counterbalance to the might of FIIs. But the political and regulatory environment has moulded fund managers at these home-grown institutions in such a way that they seldom question the sudden change of rules of the game. It is what happened when new trading tools were transforming Indian markets.

Through its COLO facility, the NSE had divided traders into two classes. Those not present in COLO or use that service were undoubtedly '*Second Class*' traders and casualty of their turtle-like speed of trading. On the other hand, there was the category of '*First Class*' traders, a handful of those who could always execute their orders from NSE's COLO. Those who got into COLO were also those who were using bots to trade. Rich proprietary brokers, who trade note for clients but for themselves and few FIIs first filled up all the COLO server space at NSE. '*Fair market access*' was now just a cliché.

Machine trading is a fascinating tale but it cannot be understood well without knowing the market operations before these tech tools took over.

In the late 1990s, NSE broke the backbone of BSE's open-out-cry system. It replaced a vociferous vegetable market like a trading ring at BSE's Rotunda building with snappy computer screens spread across the country.

In the open-out-cry system, brokers in a BSE ring inside the exchange premises shouted on top of their voice for hours to buy or sell shares. Its big flaw was that only those in the ring had the precise knowledge of all the data of stock wise positions and the faces behind big buy and sell orders. It was just like the P-Notes where the faces behind the buy and sell were only known to the FIIs.

NSE's online system gave the traders a '*breath of fresh air*' as they could now trade with distant participants on the click of a button with a reasonable amount of transparency.

NSE became a runaway hit after Dhirubhai Ambani decided to list the shares of Reliance Industries Ltd on the exchange in 1995. Then, Reliance was shunned by the brokers club at the BSE where it was trading since 1977. A bear cartel led by Manu Manek had sided with the Wadia Family of Bombay Dyeing, rivals of Dhirubhai, and together they often launched a selling sortie on Reliance shares.

In 1997, BSE also went online with the introduction of BSE On-Line Trading (BOLT) system.

For a decade after the computer-based trading was introduced at NSE, things went well until, in 2001, when BSE erupted as it came to light that the exchange officials were sharing stock position information for a price. Anand Rathi, the then BSE President, had to resign and the exchange lost investor confidence. The trust deficit in BSE gave way to more confidence in the brand NSE and the exchange with thousands of screens to display stock prices started scaling new highs.

BSE had other vagaries too. Trading at the exchange would come to an abrupt halt if a broker was raided by the Income Tax and it remained shut for days. The exchange even celebrated the birthday bash of big brokers with a trading holiday. Manek's birthday was much awaited.

For the public at large, NSE's rise was gratifying. When Dhirubhai

approached NSE for listing, both the exchange and SEBI gave him a red carpet welcome. Then, Dhirubhai was considered to be a close friend of the Gandhi family that controlled the INC party. It is well known that Dhirubhai later convinced the government to ask all the public sector undertakings to trade on NSE.

After the NSE launched derivatives trading on its platform sometime around 1999, it has maintained a near 100 per cent concentration of F&O volumes so far. Due to NSE's monopoly, FIIs largely aligned their trading terminals only to a single exchange and BSE remained parched for business.

By 2012, the NSE was crowned as the largest venue for stock trading globally by the World Federation of Exchanges. In derivative trading, NSE was ranked ahead of New York Stock Exchange (NYSE) Euronext, Nasdaq and OMX in the US, KSE of South Korea and China's Shanghai exchange. Trade data from January to June that year showed that more than 735 million derivative contracts were traded on NSE.

As per a 2012 interview given by NSE's then chief technology officer Ravi Apte to me in Business Standard,¹⁹ NSE ran the largest electronic speed-highway connecting India's 2000 towns and cities to its trading engine at BKC, Mumbai.

Main brokers at NSE had various sub-broker tie-ups in every nook and corner of the country to cater to small retail investors. The master trading engine at NSE headquarters formed the ring where order matching happened for trades coming from across India.

So, how did the NSE connect these towns and cities to its BKC headquarters?

Initially, the exchange had a wide network of VSAT (Very Small Aperture Terminals) that connected brokers with the exchange via satellite link. Investors anywhere could place trade orders over the phone and brokers would key-in the trade on the computer screen. VSAT is nothing but dish

antennas that transmute narrowband or broadband data. For a long time, NSE touted VSATs as last-mile connectivity until they were replaced by the internet.

The BSE slowly wriggled out of the brokers' control and along with NSE, it too upgraded its technology. Both combined have brokers operating between two lakh to four lakh screens across the country. But the mode of order placement is manual and dealers are required to key-in trade orders, which takes several minutes. What Apte did not say in his interview to me was that such order dictation over the phone and manual punching was no match to the HFT and Algo trading at NSE.

India's 'Flash Boys'

In just a year after Bhavneet took charge at SEBI amidst the raging controversy over P-Notes and his own NSDL fiasco, India had its tryst with the '*Flash Boys*' in 2009. It was before the world woke up to the speed link fibre optic cable that was established between Chicago and New York. The flipside of HFT and Algos was witnessed by the world, first during the US flash crash of May 2010.

But a similar Flash Trade incident in India, a year before the US HFT-driven crash, passed without much ado. Unlike the US, what India witnessed was a sharp vertical leap of main market indices that negated the panic.

On May 18, 2009 NSE Sensex and NSE Nifty hit 10 per cent upper circuit limit immediately on opening. It's a technical term called '*index-based market-wide circuit-breaker system*'. As per norms, market trading was shut for two hours to let the emotions cool down. At 11:55 AM, traders were stuck with deja-vu. Both the index hit another circuit up on opening again. Bots are emotionless and they existed even before COLO came into play in India. So what was this euphoria all about?

Buoyancy was in the air as Sonia Gandhi-led INC was returning to power

for the second consecutive term that year. Lok Sabha election results announced on Sunday May 17th, showed that INC and its alliance partners had won 262 seats. They were just 10 short of the majority mark but certain to seize power, nobody doubted Gandhi's manoeuvring. Singh was to be the second PM in India's political history, since independence from British rule in 1947, to have been re-elected for a consecutive five-year term. He enjoyed the full support of the Gandhi family and hence even the INC and the parties backing the government, everyone knew that his sycophancy had paid off.

The story in the stock markets was not Singh's second term as PM but the two consecutive upper circuit breakers hit immediately on opening were – historic.

Did the stupendous rise in the indices justify the euphoria? Data suggested otherwise. Trading volumes on the NSE and BSE were abysmally low. Roughly, Rs. 127 crores (around USD 28.22 million, considering rupee to dollar conversion rate at Rs. 45) worth stocks could only be traded on the BSE in a few seconds or minutes when markets remained open. On NSE, the cash segment volumes stood at around Rs. 170 crores (nearly USD 37.77 million). The derivative turnover, mainly concentrated in the index segment, stood at around Rs. 1,000 crores (USD 222.22 million).

In total, trade turnover worth less than USD 300 million had shut down a stock market with a total market cap of around trillion dollars even before much trading could happen.

Clearly, the index gains were not quite in the manner of serendipity but like events matching stupidity. Incidentally, many large institutions, otherwise proactive in the market, were just bystanders that particular day.

So, how can measly volumes shut equity trading in India, and that too so fast?

Country's markets were witness to similar or much bigger events in the past but they were never shut for hitting upper circuit limit consecutively

prior to May 18, 2009. If at all, the circuits were hit, it was often on the down and backed by large sell orders that got executed at constant lower prices. Be it the Harshad Mehta crash in the 1990s, the Dot-com bubble in 2001, 2004's election crash when the Bharatiya Janata Party (BJP) lost, P-Note crash of 2007 or the financial crisis of 2008, the markets during these events fell under the weight of heavy sell orders.

During the financial crises fall, the Sensex and Nifty witnessed a gap-down opening but it took a long enough trading session to shut down the markets. Both the indices even managed to recover sharply post the cool-off session, since hitting 10 per cent circuit breaker. Most important, the volumes traded on BSE and NSE were in excess of Rs. 6,800 crores (USD 1.51 billion) on each exchange.

Considering these factors, the vital question arises, did '*Flash Boys*' orders play a role in propelling Sensex and Nifty to hit upper circuit and shutdown down markets?

The answer to this can be found in the book titled Indian Financial Market: An Insider's Guide to How the Markets Work, which has studied the resilience of Indian markets prior to the advent of HFT-driven Algos. The book is primarily written by Ajay Shah and Susan Thomas with inputs from Michael Gorham, a professor at the Stuart School of Business in the Illinois Institute of Technology. Gorham also served on the board of National Commodity and Derivatives Exchange of India.

Ajay and Susan, have confirmed in their research that a market shut down in India is highly impossible without the distortion caused by 'Flash Boys' orders.

The book presents a similar scenario like the 2009 general elections. Here is an abstract from the book:

“General elections took place in 2004 with an unexpected outcome (BJP-led by Atal Bihari Vajpayee was the favourite to win, but lost). In response to

this negative news, the Nifty dropped by an unprecedented 21 percent in the first 2-hours of trading session on the day. At the end of the trading day (after staging a recovery of 9 percent), the market closed with a largest-ever one-day drop of 12 percent. Despite such a large drop, the equity market institutions did not collapse. Instead, markets stayed open for business. What is more, the liquidity in the market remained sound.”

Takeaway: Despite the massive crash, markets remained open and liquidity was sound.

During the 2019 general elections, the victory for Narendra Modi-led BJP was far bigger and unexpected than the election results of 2004 and 2009 had accorded to the incumbent INC. But stock trading never reached a feverish pitch to have shut markets. On March 13, 2020, when Sensex and Nifty hit the lower circuit, the reason was global market rout and shut down of the US market, the largest pool of equity assets in the world due to COVID-19 spread, a ‘*Black Swan*’ event with no parallel.

What makes the 2009 ‘*upper circuit*’ in Sensex and Nifty and market shut down so intriguing? Was it the use of HFT Algos by the ‘*Flash Boys*’ that pushed the markets to a grinding halt? Investigations that followed the 2015 revelations by an anonymous whistleblower about a scandal brewing in the bowels of NSE, also concealed an answer to the 2009 giant market leap.

The 2009 market melt-up on thin volumes had an alibi?

A vital clue to this surfaced when SEBI tumbled upon an email by Ajay. He was sharing crucial TBT data, clandestinely gathered from NSE that year, for commercial use to develop Algo software.²⁰

THE GREAT DATA THEFT

Inside the Cosy Club

Those handful people, occupying the upper echelons of NSE, were letting down the very institution where they were employed for a long time. This cabal with no regard for rules or fear of law were embodied in abuse of power with impunity.

On one side were Ajay and Susan, two leading stock market research scholars who were never formally employed by NSE, yet remained '*perfect insiders*' in the exchange for over two decades. They held key positions in crucial exchange committees and were awarded high profile board memberships in NSE's sister concern companies. Apart from further reach and network, it gave them a blueprint of the core working of the stock market ecosystem and complete insider knowledge.

On the other side were Ravi Narain and Chitra Ramkrishna, two of the longest-serving bosses of NSE with an image of erudite professionals who always played by the rulebook. They were what the Japanese call '*salary-man or woman*', popularly referred to white-collar workers with overriding loyalty and commitment to the corporation. These two, albeit were engaged with a lucrative remuneration package, bundled with perks and bonus that was the envy of any corporate honcho.

If Narain was cast as the Cambridge and Wharton-educated administrator, Ramkrishna, a chartered accountant, with extensive experience in resource management, treasury and project finance division of IDBI, was personally groomed by its executive director, Dr. RH Patil. She even had a brief stint with SEBI and was among the five of '*The cabal*' [sic] hand-picked by SS Nadkarni, then IDBI chairman, to set up NSE from scratch.

Interestingly, by her own admission, when Ramkrishna started her new assignment, she knew little about the business of bourses, yet slowly and steadily, out of the top five, Ramkrishna was to emerge as the more popular face of the organisation. The hush word around in NSE during her time was that the PR department had a mandate to run an ostentatious campaign for her, no wonder they were constantly hunting ways and means to put her into the spotlight and hype her corporate image. And they did a good job for her, as she was seen, heard and talked about in popular media and industry forums.

After being a deputy to Narain for many years, nearly two decades later, on April 1, 2013, Ramkrishna took over the corner room of the exchange as its MD and CEO. Just six months down the line she was ranked as 17th in the list of top global women business leaders by Fortune magazine, USA and the second most powerful businesswoman in India.

Her popularity can be gauged from the fact that in one of India's most-watched prime-time quiz shows hosted by Bollywood superstar, Amitabh Bachchan, a contestant was asked to name NSE's first woman boss. It was a stunt well managed by NSE's PR agency to promote Ramkrishna as a cult figure among CEOs in corporate India and also as a household name.

NSE's board was filled with Babus (a term used pejoratively to refer to bureaucrats and government officials) retired from the Finance Ministry or entities regulated by it. Political patronage served to ensure that SEBI remained friendly to them. To the outer world, the BSE was a brokers club but NSE's image was cast in its lofty standards of legal compliances.

Although they pretend not to, but globally, politicians, bureaucrats, corporate bosses and promoters of listed companies are all glued to the glitzy game of money and manipulation in the stock markets. Soaring stock prices and the market capitalisation greatly influences board policies and deal-making in companies.

A buoyant mood in the market often percolates into the social fabric of the society in terms of job creation and extra spending. Similarly, longer-term bear markets can create social unrest as it signals a failing economy, unemployment and a lack of growth prospects that can spoil the narrative for incumbent governments.

It is the kind of formidable power that the stock markets hold over politicians. In addition, it is also a cash-cow, which can be milked simply by price-moving policy announcements. Therefore, it is no wonder that those in power seek to control the stock markets. Brokers and fund managers are always ready to oblige as fronts to the political class and wily bureaucrats who control from behind the screen.

A lawyer by profession, Palaniappan Chidambaram was a seasoned politician who held the reigns of the Finance Ministry for a long time and was known to be as shrewd, although the popular perception was that he was more crafty than sharp-witted. No secrets in the balance sheet of any of India's large corporate houses could be hidden from him. When he was not a cabinet rank minister and sitting in the opposition in India's Parliament, he got appointed as a legal advisor to corporate houses irrespective of their credibility, argued their cases in the SC and was appointed on the board of several listed companies.

He was the legal counsel and advisor for Enron Corporation, the global energy giant that had an abysmal reputation for corruption, for a long time. Enron was a notorious company, it broke laws and engineered scandals in several other countries including the US. Even in India, it managed to hoodwink the government machinery to secure a sovereign guarantee for its

investment in a gas-based power project at Dabhol in Maharashtra and when the project went into a limbo they sued the Indian Government.

It was only when Enron dragged India to the International Court of Justice (ICJ), Chidambaram relinquished his position as its legal adviser at the time when he was made the FM by Sonia Gandhi.

Ironically, the Congress-led government in which Chidambaram virtually held the number 2 ranking, engaged a Pakistani-origin lawyer named Khawar Qureshi to present the Indian side at ICJ.²¹ Pakistan has the status as India's enemy and the dimwit lawyer lost the ICJ case in months. Harish Salve was fighting the case for India at ICJ before UPA came to power. Nobody knows why the erudite legal expert was replaced by an enemy country's inefficient lawyer.

In July 2005, the Indian Government made an out of court settlement and ended up paying millions of dollars in compensation to Enron promoters – USD 160 million to Bechtel and USD 145 million to General Electric respectively. The latter had provided the generating turbines to the Dabhol project, while Bechtel had constructed the physical plant.

Somewhere in the 1990s, when he was part of the government and commerce ministry, Chidambaram had to resign for his stock market dealings. He along with his wife had invested in shares of the scam-tainted Fairgrowth Financial Services Ltd (Fairgrowth). The then PM, Narsimha Rao, instantaneously accepted Chidambaram's resignation without the blink of an eye-lid, which the wily lawyer had submitted just as a publicity stunt. It was a lesson learnt and Chidambaram never repeated such a stunt even though controversies seldom left him.

Fairgrowth was promoted by maverick ex-banker Bilgi Ratnakar, who was linked to the share market scandal involving Harshad Mehta, the infamous big-bull of Indian markets in 1992. It was alleged by Janata Party President Subramanian Swamy that Chidambaram got 10,000 Fairgrowth

shares in 1991 out of the promoter's quota at a face value at Rs. 10 when its market price was much higher. The little trickery had multiplied Chidambaram's wealth by over Rs. 5,40,000 without much ado, Swamy had said.

Chidambaram and Swamy were arch-rivals – one became skilled at political manoeuvrings and another, a fiery corruption crusader. Sometime in January 1998, the Delhi High Court dismissed Swamy's petition seeking a Central Bureau of Investigation (CBI) inquiry against Chidambaram for his role in the Fairgrowth affair.²²

Interestingly, the lawyer who argued for Chidambaram in the case was another prominent lawyer-cum-politician and BJP leader, Arun Jaitley. Both Chidambaram and Jaitley took turns at becoming India's successive FMs, albeit from different political parties and governments.

But the grapevine goes that they were brothers in arms and often came to the rescue of each other. But that is a different story.

Due to his background as an eminent lawyer, Chidambaram knew the coterie of judges and their psyche well, he also had a keen insight into the functioning of fund managers and brokers. The world of finance was not new to him. Chidambaram had studied business administration from Harvard. He could move smoothly across the realm of legal eagles and finance professionals like he changed from his spotless white *mundu*-kurta to a dark suit before going on foreign trips.

Nalini was, of course, one, but controversy seemed Chidambaram's another wife. Be it India's 2G telecom scam or the high profile bribery case involving a media company heiress, fingers often rose in Chidambaram's direction. Usually brimming with confidence and standing tall against all the allegations, in recent times, Chidambaram had to scoot into hiding for nearly 24 hours when the CBI came to arrest him. It was for alleged facilitation of dubious foreign investments in a media channel, for which bribes were paid

to companies belonging to his son, Karti.

One of India's most powerful politicians, Chidambaram remained lodged in Tihar jail for 106 days in 2019. Jaitley had passed away then and none of the judges could find any argument to his rescue for rejecting the CBI custody over corruption charges levelled against him.

Chidambaram has a sizeable support base in Tamil Nadu especially in the parliamentary constituency Sivagangai, which elected him over seven times, and currently held by his son. His supporters saw him as a workaholic, razor-sharp lawyer. However, to the stock markets, he was their '*Prima Donna*'.

When Chidambaram was the FM, he once dialled Narain on his phone asking him to restart market trading at NSE just 15 minutes after it was halted sometime in October 2012. A little after 9 AM, NSE's Nifty index had crashed by 900 points out of the blue. A '*fat finger*' error had been discovered to have caused the fall. A fat finger error is a human error caused by pressing the wrong key when using a computer to input data.

Fat finger errors are often harmless but can sometimes have a significant market impact and this was one such incident which triggered a major crash. Since the 10 per cent circuit breaker was triggered, rules said that a mandatory two-hour shutdown of trading was necessary. After Chidambaram's call that day, Narain re-commenced trading at 10:22 AM. The streets were surprised.

"The goal was always to resume trading after a short shutdown. Everyone was on the same page," Chidambaram later told a journalist via text message.

When the journalist pointed out that it violated SEBI guidelines, Chidambaram said, "SEBI chairman was on board."²³

The same month that year after the '*fat finger crash*', Narain told the NSE board that he wished to step down from his position at the end of March 2013. Instead, he got ensconced as the board's vice-chairman and a shareholder director. There was nothing new in this. Some executives and

bureaucrats often found a way to remain in circulation and enjoy the perks of their position. There were several such who came to NSE after retiring from the Central Government and the RBI.

Meanwhile, on the other hand, rather than stepping in, market watchdog SEBI had remained quiet on re-starting of the markets by Narain on Chidambaram's order in blatant violation of the rule. But a few months after Chidambaram's ouster as the FM, post the general elections of 2014, the regulator said there was no communication between itself and NSE that day after the *'fat finger'* error halt.

What was the desperation for the country's FM to micro-manage NSE? SEBI went on to say NSE's behaviour was a systemic risk. Later, a former NSE board member told a journalist that the problem was that non-written communication travelled much faster at the exchange and left no audit trail.

It was apparent NSE had very little regard for rules and it breached them at will. For instance, rules required NSE to advertise when it made high profile appointments and only choose candidates after they cleared board interviews and obtained SEBI approval. Even when BSE wanted to promote its internal candidate and deputy MD, Ashish Chauhan, to the top job, it followed the procedure. However, all rules were flouted when Ramkrishna was promoted at the top in NSE.

To many, she came across as a luminous, charming and articulate personality although she spoke with a little bit of a south-Indian accent. Was it that, or something else, but she definitely had Chidambaram in her good books and vice-versa. Narain advocated the board that there was no better person than Ramkrishna for the top post at NSE. Be that as it may, the exchange saw its fortunes take a severe beating as she took the reins fully in her hands.

On the day she took charge in April 2013, Ramkrishna made one of the most controversial appointments at NSE. Bypassing the Human Resources department, she got in one Anand Subramanian as a strategic advisor to her

at an exorbitant pay and gave him a cabin next to her own.

Prior to joining NSE, Subramanian was employed at a Chennai based logistics firm Transafe Services Private Limited, a joint venture between the state-owned Balmer Lawrie and the ICICI group. His annual pay then was a little over Rs. 14 lakhs.

Ramkrishna offered him a mind-bogglingly Rs. 1.7 crore remuneration package.

NSE officials say Subramanian was brought in as practically the number 2 at NSE and had the privilege to work from Mumbai as well as from Chennai, his hometown. In just two years, he was promoted to the post of NSE's Chief Operating Officer (COO) with a rise in pay to again a mind-numbing Rs. 4.25 crore. The package was just a tad lower than Ramkrishna's Rs. 4.78 crore. (xxii)

This kind of unbelievable step-up in salary has perhaps never been ever heard, at least not in India, however, it seemed at NSE it was pretty normal.

There was no doubt this raised more than just eyebrows, and to quell the rising dissent, Ramkrishna had a rather lame justification. "I propose to use the facility of our chief strategic advisor to reduce my burden," Ramkrishna told NSE staffers.

Subramanian was mandated to handle a horde of responsibilities including people management, new business, corporate communication, marketing, business excellence, research and development, pricing, strategic planning and NSE's subsidiaries—IISL, DOTEX, NSE Tech and NSE IT. Yet, he was never designated as a key managerial person at the exchange. It was a blatant flouting of yet another rule of the exchange, which specifies the key managerial persons who have enhanced responsibility, their terms of appointment and salary are regulated by SEBI.

When confronted by SEBI on Subramanian's appointment, Ramakrishna said, "He was identified by HR consultant. Interviews were done by HR and

myself. Subsequently, the candidate also met Narain and Mathur (Chairman). I knew him through his wife Sunitha Anand.”

To the same question, Narain replied, “No, I did not have any role in appointment of Subramanian.”

Nearly a year after the Congress party lost elections to BJP, in January 2015, a whistleblower wrote an extensive letter to SEBI highlighting an Algo and ‘*Preferential Access*’ scandal at the NSE. Buried skeletons started to tumble out of NSE’s cupboard, one after another and the ugly secrets hidden away for long started to uncover.

The axe first fell on Subramanian as the probe by the NSE board found irregularities in his appointment and he was shown the door in 2016. Subsequently, in months, even Ramkrishna had to quit but insiders knew that she was given the ‘*golden handshake*’. She had netted nearly Rs. 50 crore in salary and variables in just about three years as MD and CEO.²⁴

For all the claims of high standards of transparency, NSE from within was a cosy club of people engaged in immoral and unethical practices. The whistleblower’s letter opened a ‘*Pandora’s Box*’.

On December 9, 2016, Minister of State for Finance in the BJP government, Arjun Meghwal, told the Parliament that NSE was found to have given some stockbrokers preferential access to its trading engines. (xxiii) The investigations had also revealed a “great data theft.”

Target NSE

Chanakya, the Algo software developed by Sunita’s company IFPL, derived its name from the third century BCE Indian teacher cum philosopher, jurist and economic scientist. Although born as Vishnugupta in India’s ancient knowledge city Taxila, he was called Chanakya, which in Sanskrit meant ‘*the most clever*’. Back then, the legend goes that Chanakya with his intelligence dethroned the powerful Magadha Kingdom rulers of the Nanda dynasty and

replaced them with a more *'just'* Chandragupta Maurya. It was the dawn of the Mauryan empire. In our age, Ajay used the intelligence of Chanakya to gain *'Preferential Access'* to NSE.

With unimpeded access, Ajay and Susan were being discreetly fed vital trade data from NSE all through 2009, just weeks ahead of the mysterious giant leap of Sensex and Nifty on May 18 that year, which shut down trading at the BSE and NSE. There was so much information in the files NSE shared with Ajay that he passed it to IFPL for improving and back-testing Chanakya's Algo codes. The data NSE was sharing with Ajay was nothing but TBT data. All this came to light after the 2015 whistleblower exposé.

To bargain a good price is the inherent nature of stock market traders. Their game of bargain hunting involves an element of bluff as nobody wants the other trader to know his cravings for demand or supply of stocks, which can potentially determine its price. Snappy computer screens at the broker office display the top five bids and ask price quotes. But their view is only the tip of an iceberg. Below the top five quotes on display are hordes of traders eagerly waiting in the queue looking for a match to their bid or ask price.

Millions of price quotes keep flowing throughout the trading session into the sea, i.e., an exchanges' master order matching engine, as dealers in the brokerage house keep punching trades. The display screens constantly flicker as price quotes keep changing and the stock prices move up and down. Both the top five quotes on display and those hidden below them, together constitute an exchanges' electronic order book, the iceberg.

Each change in the electronic order book involving the price or quantity is known as a *'tick'*. The real-time information of change in each tick is called the *'tick-by-tick'* or TBT data. Those in possession of the TBT data hold insight into the true size and shape of the iceberg, i.e., is the electronic order book.

The horse racing gambler duo, Benter and Coladonato, who had cracked the code in the late '90s, gathered information on unforeseen probabilities at

the racetrack and even employed experts who suggested more of it. They then constantly fed these probabilities to their Algos, which sharpened its prediction. Similarly, TBT data could be used to better the Algo codes and back-test them against those million probabilities, which is nothing but the order flows. Such Algos, when backed by speed bots like HFT tools and COLO, had an edge.

Since stock markets are severely marred by fundamental uncertainty at each passing second, complete order book data reigns supreme over any other tool of price discovery. TBT data gave Ajay an insight into the probabilities of an exchange order book. Benter and Coladonato used Algos to beat '*gamblers ruin*'. Ajay used it to build his *Chanakya*'s intelligence and beat '*other gamblers to ruin*'.

“The ‘data’ from which the economic calculus starts are never for the whole society ‘given’ to a single mind, which could work out the implications, and can never be so given.” – Friedrich A Hayek, Nobel Laureate, 1974.

Hayek was an old-world Austrian-British economist, who pioneered the study on '*Use of Knowledge in Society*'. In his theory, “The knowledge which is almost exclusively of temporary opportunities due to the circumstances of the fleeting moment, something that traders could benefit from, is never even shared with '*economists*' who regard themselves as definitely immune to the crude '*materialist fallacies*' and could work out its implications.”

In 2009, when NSE shared TBT data with Ajay, wider market players in India had no clue of its use.

“... on Day 1 Anupam is a finance guy. He is not a programmer. So drive him appropriately. He can go into all your existing projects—but in a domain knowledge role e.g. he can start working on trading strategies, which can go into all algorithm trading work,” Ajay wrote in an email to Sunita, which was revealed during a SEBI investigation. This is nothing but *Hayek* galore.

Below, the two tables show the data on display to a normal trader and those holding TBT price information up to the last tick. The difference in information on price quotes and quantity that each class holds is self-explanatory.

The shorter table is the usual price quotes available to the entire market. The longer table is the price and quantity information available to the one holding TBT data.

Order Book View for Retail User				Order Book View for TBT User			
Buy Qty	Buy Price	Sell Price	Sell Qty	Buy Qty	Buy Price	Sell Price	Sell Qty
25	50.00	50.05	115	25	50.00	50.05	115
135	49.95	50.10	274	135	49.95	50.10	274
200	49.90	50.15	34	200	49.90	50.15	34
35	49.85	50.20	84	35	49.85	50.20	84
15	49.80	50.25	275	15	49.80	50.25	275
26,875	TBQ	TSQ	32,754	238	49.75	50.30	398
				517	49.70	50.35	497
				891	49.65	50.40	364
				1580	49.60	50.45	294
				2536	49.55	50.50	3687
				35	49.50	50.55	691
				97	49.45	50.60	36
				129	49.40	50.65	291
				3347	49.35	50.70	443
				29	49.30	50.75	109
				1	49.25	50.80	86
				34	49.20	50.85	234
				12	49.15	50.90	684
				75	49.10	50.95	542
				15	49.05	51.00	7684
				5367	49.00	51.05	391
				538	48.95	51.10	761
				349	48.90	51.15	389
				274	48.85	51.20	26
				689	48.80	51.25	4268
				8345	48.75	51.30	792
				372	48.70	51.35	829
				995	48.65	51.40	95
						51.45	684
						51.50	7697
				26875	TBQ	TSQ	32754

TBT data can reveal the precise point at which large volumes are pending to be executed. In May this year, the share price of Vodafone Idea Ltd moved

up by 30 per cent on the news of Google's likely investment in the company.

In such a scenario, since TBT data can reveal the total quantity of buy orders up to the last tick, Algo bots may keep punching higher price of the stock and simultaneously also rig the F&O premiums until the last buyer is exhausted. The options premium for Vodafone Idea doubled and tripled that day. Damnable non-bot traders had to buy the stock higher. Later the same day, Vodafone Idea informed the stock exchanges that there was no immediate proposal from Google under the consideration of its board. Similarly, all the block and bulk data in any stock first reach the Algo and HFT traders. Retail investors come to know of such data only after the exchange uploads its daily files. This is price moving information if received on time.

SEBI deploys a skewed logic that price manipulation can only happen in non-derivative counters. Hence it has imposed no limit on the rise and fall in derivative stock counters. They can sky-rocket or fall to zero in a day. But the regulator has choked cash market trading with its heavy-handed approach by limiting the movement in stock price with various circuit filters and surveillance measures, which make the price discovery of a non-derivative counter impossible for months and years despite there being legitimate news flow or reasons.

Ajay and Susan even argued for such a mechanism through their research papers on the subject called '*call-auction*', which was nothing but restricting trading in cash stocks. SEBI's policies led to more volumes flowing into derivative trading and BSE became the chief casualty of such a brutal regime as it lost business to NSE. As a matter of fact, BSE's market share in equity trading fell to just around 2-3 percent from around 20 percent in 2008. Currently, it is around 8-10 percent.

In the derivatives, even when there is no news in a particular counter but since HFT Algos inflate or suppress its price based on advanced knowledge, lay traders are never able to catch a stock at their desired price. If the

intention is to buy a stock at Rs. 10 they end up buying at Rs. 10.5 or even higher and the scenario is reversed in selling.

Many old-time brokers will tell you that this was not the case before HFT tools came into play. The difference in price-earnings for HFT bots are measly in isolation but when played to one's advantage several 100 times a day in huge quantities, the economies of scale become mind-boggling. Also, a longer duration study of TBT data reveals a trading pattern during crucial events like election results or budget presentation, upping the ante.

In Hayek's words: "It is a curious fact that this sort of knowledge should be generally regarded with a kind of contempt and that anyone who is better equipped with theoretical or technical knowledge gains an advantage over somebody is thought to have acted almost disreputable and regarded as dishonest."

Certain other email exchanges between Ajay and Sunita, which were found during the SEBI probe into NSE's Algo scam, revealed their clandestine operations.

"You have to swear everyone to silence about the fact that the data we are getting out from the NSE is going into algorithmic trading work. It would be a severe problem if this fact comes to light, since NSE has not given anyone else this data,"

Ajay wrote to Sunita in 2009. This and other email exchanges are part of SEBI's investigation report.

"Data was shared with Ajay on the request of MD and DMD," Ravi Apte, NSE's chief of technology between 2007 and 2013, said during his interrogation by SEBI.

His fingers pointed towards his two bosses, Narain and Ramkrishna. Oral orders came from the top, other NSE officials too divulged during audit investigations.

“I was under the impression that an appropriate confidentiality agreement was signed with him (Ajay) by NSE,” Apte told SEBI. (xix)

The Agreement Fraud

Data theft under the guise of research was an act of cyber-crime but, falsifying legal documents for any purpose is a *‘white-collar offence’*. SEBI’s inefficiency was at its peak when the regulatory officials came across a *‘fake’* agreement between NSE, Ajay and Susan but it did not refer the case for a thorough criminal probe.

In all the probabilities, no data sharing agreement ever existed between NSE and the two researchers. Ajay was interrogated by SEBI after Meghwal revealed about the scam to the Parliament. But SEBI treated the swashbuckling research professor with kid-gloves. The regulator took the documents shown by him at face value, as the officers involved in the probe clearly lacked the heart and honesty to question them.

In the two agreements for sharing of data between NSE and Ajay that were shown to SEBI, no dates were mentioned. The foremost chapter, which any student of law is taught on the first day of attending college, is *‘date’* on agreements. The documents even though signed between the two parties, if it lacks the essential legal element in it, like the mention of the date of signing, it is never complete and non-existent in the eyes of law.

In its final order against Ajay, the SEBI WTM, just a rank below the chairman in the organisation’s hierarchy, failed to hold the agreements as *‘fake’* and went on to issue mundane commentary over missing clauses in the data sharing documents.

WTM Santosh Kumar Mohanty said there would have been no harm had the agreement between Ajay and NSE mentioned that the researcher could pass information to a third-party for commercial use. Who even digs into the clauses of *‘fake’* documents? Well, the super-cop SEBI did.

“In my view,” Mohanty said in his order, “There is no difficulty in sharing data by Ajay with an outside entity for commercial purposes as long as the agreement spells out clearly its objectives and provides for the commercial terms and conditions in the agreement in compliance with the stated data sharing policy of the exchange. However, when a data sharing agreement purely meant for research assumes the colour of a commercial agreement, thereby providing privileged access to confidential and exclusive trade data of the exchange to a select few, who, clandestinely exploit it for their commercial gains, it leads to serious issues and compromise of market integrity.”

A further look at the agreements revealed a number of discrepancies and irregularities that made them just reams of paper devoid of any legal value or *locus standi*. Lack of date in these ‘so-called’ agreements was followed by an absence of any ‘notary’ attestations. A notary seal on a document is an assurance of its authenticity and legal verification of the basic principles that turn a paper document into a legally binding agreement. Without the mandatory notary, the documents that Ajay passed as his agreement with NSE, was another clear evidence of the scheme of ‘*fraud*’, for stealing data from the exchange. Undoubtedly, it was done with the connivance and sanction of NSE’s top brass.

Despite the fact that the validity of the so-called agreements remains questionable, let’s examine what the agreements said? “Researchers Mr. Ajay Shah and Ms. Susan Thomas irrevocably and unconditionally agree to abide by and be bound by the terms and conditions of the undertaking, that they shall keep the data obtained from NSE confidential and shall not divulge to any person.” That’s true. IFPL and Sunita were just a front.

Earlier filings of IFPL in 2008 and 2010 to the Ministry of Corporate Affairs (MCA) show Sunita’s residential address at IGIDR campus in Mumbai. But it was Susan, who then resided at IGIDR as she was a PhD research professor there. In the later filings, IFPL changed Sunita’s address

to a village in Kerala.

Among other shareholders of the company also include one Mathew Verghese, a resident of Michigan in the US. The Thomas sisters had Verghese as their middle name. Yet another IFPL shareholder was named Suja Verghese Thomas with her address at IGIDR campus.

Then there was a Michigan company CompuSystems, Inc., which held 33 per cent stake in IFPL. Gemware Technology, a company registered at a Dubai post-box address and owned by one Deepak Raj Kholi too held IFPL shares.

IFPL was registered at a residential complex – Sungrace, Raheja Vihar, Powai in Mumbai. Who did this address belong too? SEBI has failed to ascertain this, but reportedly this address belonged to a senior NSE official.

The company statutory filings show that Sunita, who is shown as the key director of IFPL and her only other partner, Dagli, who joined the company in 2006, both survived on a measly take-home pay. For the company that made several lakhs in revenue every year, key directors surviving on a few thousand salaries? This is not quite palatable if not pretty ridiculous.

The scheme under which NSE said it gave data to Ajay and Susan, involved a study on liquidity index to be undertaken by research institute IGIDR. Yet, there were no cross-references provided to the institute in the agreement. Then, Susan was employed by IGIDR but Ajay was ensconced at NIPFP.

Ajay told SEBI that he had moved to Delhi in 2001 and his engagement with NSE has reduced since. Yet, the data flowed from the exchange to him unhindered.

Other discrepancies in the agreements were the missing date of undertaking, date of NSE's internal approvals, etc. Ajay was only flirting with SEBI's wisdom of something as casual as the date of events when he told them that the agreement with NSE was entered into in 2013. Wasn't the

regulator just questioning him about data he got from the exchange in 2009?

Witty enough, SEBI held Ajay and NSE guilty of a mere lapse in following the data sharing agreement. It ignored the fact that there existed no agreement in 2009, devoid of which, data extraction was just theft by design.

Nowhere does Mohanty mention that he considered the agreements ‘fake’ in the very first place, hence in his wisdom, there was no forgery involved. Faking of an agreement involving third-party dealings does not fall under the ambit of SEBI law. It is a violation to be decided under India’s criminal procedure code. Also, nobody in SEBI’s legal department may have told Mohanty that the theft of property, be it data or any other intellectual asset of NSE, cannot be dealt with SEBI’s ‘*paper-tiger*’ approach. SEBI had no powers to dispose of property theft cases. Like Bhavé, Mohanty too seemed to be unaware of his prime duty, the *Dharma*.

SEBI was deciding cases not within the realm of its authority. Mohan Gopal’s assessment of the regulator galore.

“There was no serious attempt by NSE and its officials to bind researchers to various terms and conditions of confidentiality, third-party sharing, etc., that have been stipulated in these agreements,” Mohanty said.

The pertinent questions to ask were, how could NSE bind Ajay and Susan to anything in 2009 for lapse of agreements executed in 2013? What about the data theft that occurred in 2009, which involved no agreement? If NSE kept sharing data with Ajay post-2013, still it was a theft of intellectual property as the agreements shown to you had no legal standing. Nobody put these posers to SEBI’s WTM.

Like G Anantraman, the former SEBI WTM who laid bare the full ramifications of the scam at NSDL, Mohanti too was a former tax officer, who usually had a very keen sense of inquisitiveness. But he could not see through the larger picture behind the fake data agreements and hence wrote

his orders by scratching the surface. Quite reminiscent of Bhave in 2008, laxity of Mohanty and his boss, Ajay Tyagi, in investigating the NSE also encouraged a slew of scandals at another large commodity exchange.

No criminal complaint was initiated against Ajay and NSE for presenting ‘fake’ documents or hatching a conspiracy to steal crucial trade data. In the past, the same regulator has filed several police complaints with regard to document forgery.

Ajay was banned from stock markets and holding a position on bodies connected to it for two years. Susan’s name does not appear in the SEBI diary nor was any order passed against her. Narain and Ramkrishna too got away lightly with a ban on associating from any listed company or stock exchange for up to five years.

As per the Hindu tradition, if sinners bathe at the confluence of river Ganges in the holy town of Benares, it would wash-off the evils of their wrongful conduct in the mortal life. In the stock markets, for long, SEBI had played the sacred Ganges where wrongdoers got away with a light rap on the knuckle. Its action against Ajay and the top brass of NSE set another precedent of how serious white-collar offences like conspiracy to steal data and submitting fake documents could be washed away by SEBI’s inefficacious smack. No account of their dubious commercial gains was done.

Experiment with Truth

May be due to his linkage to the Congress party ecosystem for long, Ajay held a ‘*Gandhian*’ view of himself and his cronies.

“My position has always been that we should be mindful of propriety even though there was no data use contract (with NSE) at that time. I have always, repeatedly told IFPL to go obtain TBT or CTC (computer to computer) data on their own, for use in Algo trading research and not utilise files that flow from the ‘relationship’ with NSE. On occasion, when IFPL

has done implementation work for us, some of the data that has come to us has gone to them for the purpose of software developments,” Ajay told SEBI during the course of his interrogation.

In the time of utter moral decadence, Mahatma Gandhi has vividly described in his autobiography *‘My Experiments with Truth’* as to how he often failed to protect his lofty standards of morally superior conduct. In Ajay’s testimony to SEBI, he hints that the data he got from NSE, even though could be exploited commercially, was never used to their advantage.

“Prior to the data use contract there were no limitations on what we could have done, other than our intuitive sense of fair-play,” Ajay said in his defence.

The blatant data breach at NSE, involving the exchange top brass, was unparalleled in the history of stock market scams in terms of sheer planning and scale. Ajay and Susan argued for a framework on Algo trading and even got the data on a platter which was obviously misused by them for their commercial gains by developing Algo trading products for sale in the securities market.

“I have been enthusiastic to the extent to which Algo trading can help increase liquidity and market efficiency of Indian financial markets. Hence, I have argued in creating the framework, through which this can come about. I have argued that the business will move to Algo trading to a greater extent than was envisaged,” Ajay said.

Sometime in February 2009, the research duo shifted to network-based data extraction from NSE and files flowed to them at the click of a button. It was like a virtual pipeline through which valuable data flowed free to the colluding couple.

“Susan and I were gatekeepers to data including physical access. We only make data available to *‘our collaborators’* for the purpose of authorising research papers. No *‘for-profit firm’* has even been our collaborator,” Ajay

said.

The Nobel laureate Hayek spent the majority of the 93 years of his life dwelling into the problem and repercussions of leaving unique economic and trade data that revealed knowledge of special circumstances in the hands of economists '*alone*'. His idea told him they could game the system. But despite unearthing of so much evidence against him, Ajay harped on his naivety and SEBI let him off lightly.

Virtuous gatekeepers were never required for guarding NSE's data if it was of any general nature. Also, there is no evidence, which said that IFPL was a '*not for-profit*' company working towards large public good. Yet, SEBI, the guardian of investor faith in India's capital market, never felt the urge to order a forensic audit and track down finances of entities linked to Ajay and Susan and other top rung NSE officials including Narain and Ramkrishna, who benevolently favoured them.

Secret data pipeline to Ajay, Susan and IFPL was NSE's worst kept secret since the exchange was never required to share any data for research purposes with them. Already, an extensive cache of NSE data that could fulfil any research goals was made available by the exchange in a legitimate manner.

For long, NSE produced two outstanding books every year titled *Indian Securities Market Review* and *NSE Fact Book*. These served excellent material on the equity market and all the other activities of NSE as they covered market design, regulations, traded products, NSE member firms and liquidity. In addition, the NSE also produced a monthly newsletter about the derivatives market, all of which was available for free. Besides the book, there was a CD NSE handed out to researchers each month at a nominal cost. This contained information on intraday trades, market index values and offered four snapshots of the entire limit order book every day at 11 AM, 12 PM, 1 PM and 2 PM.

When the exchange doled out such comprehensive data, nothing else except live '*order book*' remained hidden.

Despite this, Apte said his bosses Ravi and Ramkrishna wanted him to share more data with the researchers, which he obliged. SEBI's *happy-go-lucky official*, who has conveniently downplayed the ramifications of the data breach in his order, compellingly said, "Ajay was excreting implicit personal influence on NSE. Data they got was confidential and exclusive in nature."

IFPL Above Tata Consultancy Services (TCS)

TCS is India's leading information technology giant that wrote NSE's screen-based trading programme when the exchange was being launched. Screen-based trading was the first offering of NSE and a novel experience for Indian markets then. But TCS, a company that came from the most reputed house of the Tata Group, was already at its pinnacle in the country's tech scenario and hence the software it created for the exchange in the early 1990s was touted to attract faith in its offering.

Sometime around 1997 when the NSE was writing its much-valued risk management software and even moving ahead with its work on demat share project under NSDL, the exchange chose a start-up firm IFPL for the coding job.

"I began my conversations with the NSE in the early period and got involved in their thinking. This was done in an informal way, mostly without formal contracting. This led to some useful initiatives such as: construction of Nifty, building of *Parallel Risk Management Software System* (PRISM), which can calculate real-time value-at-risk of brokers and even at the level of their clients at a speed of over 100 trades per second. It also led to setting up of NSE's clearing corporation, launch of derivative trading and NSDL etc. This relationship was at its peak in the early years i.e. 1994-2001. I moved to Delhi between 2001-2005 to work for the Ministry of Finance and since then the engagement with NSE declined," Ajay told SEBI.

PRISM is NSE's risk management software for F&O trades that mainly

calculates the margin and its shortfall, etc. IFPL and another firm called Starcom Software were roped in to write the code of PRISM.

“Starcom Software, led by Shuvam Mishra, has done a name change and I do not know their present name,” Ajay said.

Nothing could be further from the truth. In reality, Ajay and Susan’s friendship with Mishra goes back to their campus days at IIT Bombay, when they were colleagues and they studied coding together.

Starcom Software Pvt Ltd was rechristened as Merce Technologies Pvt Ltd and lists NSE, NSDL and stock brokerage cum financial services firm, Edelweiss group, among its clients. Earlier filings of the company mentioned Edelweiss Capital as one of its shareholders. The equity broker has been one of Merce’s oldest shareholders.

Like IFPL had a shareholder from Michigan, one of Merce’s largest shareholders’ Florida Properties, which held 33 per cent stake, was registered at an address in Cyprus, a haven for hiding black money. Edelweiss Capital has retained its holding but Florida Properties was moved out from the list of shareholders in 2015 after the lid at NSE’s COLO and Algo scam was blown. The 33 per cent stake of Florida Properties was transferred to Mishra. Merce now has an office at Pearl Street, Denver, US.

Both, IFPL and Starcom (now Merce), which had access to crucial data from NSE also had brokers as their clients. In fact, IFPL has been selling a suite of programmed trading products. Some of its marquee broker clients from the very initial days of the company included Kerala’s largest and first internet trading company Geojit, which ironically had Susan and other family members of Ajay on its board.

Other client list included NSE, NSDL, DSP Merrill Lynch, Kotak Securities, Coimbatore Capital, Birla Sun Life Asset Management Co, Templeton, LIC Mutual Fund, IDBI Principal, Middle East Financial Network, Gemware Technologies LLC, Dubai. Geojit Securities had

launched trading operations in Dubai too.

A long list of brokers who used IFPL's Algo software is nowhere disclosed but it surely included some of India's large proprietary traders.

"In 1997, my friend Deepak Kohli (who was then in Abu Dhabi, with Standard Chartered) asked me for suggestions on cheap Indian software development. I connected him to Sunita and he ended up becoming a customer and an investor even, taking an equity stake. In February 2009, my friend, Matthieu Stigler, asked me for a suggestion for cheap Indian software development, and I suggested he speak with IFPL. I am a person with strong connections into securities firms and could potentially have done differently in terms of helping them (IFPL) in business development, but did not," Ajay said.

His actions were contrary to his words. If not for Ajay and Susan and CMIE, who knew IFPL and Mishra at all? Ajay told SEBI that NSE had contracted IGIDR for a project on building risk management systems, a *La-RAND* Corporation style arrangement.

IGIDR roped in Ajay's sister-in-law, Sunita, and friend, Mishra, for the front-end and back-end job, for which both their firms got an initial payment of Rs. 6 lakh each. The money seems small if looked in isolation but it should be noteworthy that both firms managed to retain NSE as its client despite the exchange having re-written the source code for PRISM.

The source code is the original data, which is the starting point of any software programme that helps in controlling the entire system. Those who can make changes to the source code can dictate the terms of the software. The point here is, for all the denial about Starcom, it was not without Ajay's influence that NSE would have awarded such a project to any novice.

SEBI treated stock exchanges and related companies as key market infrastructure institutions and kept a tight restraint on their working and shareholding. When Chidambaram was the FM, he even got committees

constituted to diversify shareholding of stock exchanges, depositories and clearing corporations so that no private vested interests could exert control over them. But a web of companies that did sensitive work like writing risk management and derivative software for NSE, remained exempt from scrutiny even though they served brokers or had their owners coming via foreign tax jurisdictions.

Ajay said IFPL needed the data for a study on liquidity index. But the company never finished its task. Liquidity index was later developed by NSE's subsidiary company, India Index Services Pvt Ltd. NSE even had a tech arm called NSE IT and quite a lot of work that was outsourced by the exchanges' top brass was just in the domain of its sister company. SEBI had abandoned its *Dharma* as it raised no questions or dived deep into what was going on at NSE. The data goldmine was left unguarded by the supposedly chesty exchange officials and an ineffectual regulator.

Shareholding of IFPL

Ledger Folio Of Share Holder	Share Holder/ Debenture holder Name	Address	Father's/ Husband's Name	Type of share/ Debenture held	No. of Shares / Debentures Held	Amount Per Share
2	AVDHOT MADHAV DESHPANDE	3A, 2 ND FLOOR, GAUTAMI CO-OP HOUSING SOCIETY, KOTHRUD, PUNE- 411029	MR. MADHAV DESHPANDE	EQUITY	500	10
3	SUNITA VERGHESE THOMAS	E7, INDIRA GANDHI INST. OF DEVP. RESEARCH, SANTOSH NAGAR, FILM CITY ROAD, COREGAON (E) MUMBAI- 400063	THOMAS VERGHESE	EQUITY	66499	10
4	DEEPAK RAJ KOHLI	C/O GEMWARE TECHNOLOGY LTD. P.O.BOX 35344, DUBAI, U.A.E.	RAJ KOHLI	EQUITY	30000	10
5	SUNITA VERGHESE THOMAS	E7, INDIRA GANDHI INST. OF DEVP. RESEARCH, SANTOSH NAGAR, FILM CITY ROAD, COREGAON (E) MUMBAI-400063	THOMAS VERGHESE	EQUITY	1	10
6	RAJ KOHLI	C1/1426, SECTOR C VASANT KUNJ, NEW DELHI	THOMAS VERGHESE	EQUITY	3000	10
7	COMPUSYSTEMS INC.	1955, WEST HAMLIN, SUITE 200, ROCHESTER HILLS, MICHIGAN - 48309	N.A.	EQUITY	50000	10
				Total	150000	10

BATTLE OF THE BOURSES

The Rivalry – NSE vs. FTIL

Dholidas warmed up as the men stepped out of the car at the Exchange Square in Andheri, the Mumbai headquarters of Multi Commodity Exchange (MCX). *Dholak* beats filled the sultry air in festivity. There was no wedding *baraat* but it was Jignesh Shah's moment.

In the customary Gujarati style of celebration, Shah greeted the legal team that brought the good news. The courts had finally given the green light for the stock exchange that Financial Technologies (India) Limited (FTIL), the group Shah founded, was sponsoring.

As a jubilant staff gathered around him, the man, with a penchant for Bollywood dialogue rattled off the Salman Khan one-liner, “*Ek baar jo maine commitment kar di, uske bad main apne aap ki bhi nahi sunta...*” (Once I make some commitment, after that I do not listen to even my own self). That was Shah in April 2012.²⁵

In 1995, when NSE was gearing up for equity trading, that same year, Shah also launched FTIL. Unlike Ravi Narain, the home-grown local boy, Shah was a Bombay University engineer who did grunge work for many years in the bowels of the BSE before promoting FTIL.

Just before NSE was to start derivatives trading in 2000, Shah's company came up with its flagship product *ODIN* software that offered trading on both NSE and BSE via a single screen. It was an instant hit as it cut a brokers' hassle of maintaining two separate screens to trade on both the exchanges. FTIL gained more than 80 per cent market share in the trading software market and Shah developed a deep bonding with investors and brokers.

A middle-class Gujarati boy from Kandivali, a distant suburb of Mumbai, whose father dealt in iron and steel trade, Shah shot to limelight in 2003 when FTIL introduced its first commodity derivatives trading platform, the MCX. The fireworks that began at FTIL's Andheri office that year would continue to light-up the skylines of Mumbai's financial markets for a long time to come.

Business baron Mukesh Ambani entered the first buy order for gold futures contract to formally launch the MCX in 2003. Since then, a high volume churn in bullion, base metals and oil and gas contracts on the exchange became a major draw for those who wanted to wager in the commodity segment in India.

MCX derived its price quotes from the Chicago Mercantile Exchange (CME) in the United States and London Metal Exchange (LME) in the United Kingdom, two of the most liquid global commodity pools. The price spreads on the MCX nearly replicated the United States grade crude oil that is benchmarked on CME's NYMEX trading platform.

The gold and silver quotes too were closely linked to the global benchmark prices. Since MCX took away the hassle of Rupee-Dollar conversion for the domestic traders, the exchange was a hit. It was widely believed that Shah enjoyed the backing of the Reliance group, since Mukesh, the eldest and the more reticent son of Dhirubhai Ambani came for the MCX inauguration event. But Shah fought his own battles; often so intense that it trumped his personal antipathies.

If Ravi Narain and Chitra Ramkrishna had a reputation of salaried

executives playing by the rule book, Shah, in contrast, came across as a *'business-man'* with the permanent air of outsider intent on crashing into the party. After his two successful ventures, the FTIL and MCX, Shah declared that he would be a billionaire before he turned 40, a goal he missed narrowly.

Friends and foes compared him with Dhirubhai, for different reasons of course; Shah merely shrugged that the system was not friendly to first-generation businessmen. Dhirubhai had built his empire by waging an intense corporate battle with established business houses like the Wadias and Tatas. When Shah entered the exchange fray, NSE was an established corporate rival.

The similarities between Narain and Shah were that the two had a fierce fighting spirit. Since they were in the same business space, it was inevitable that their paths would cross. And they did. Like NSE was in the equity markets, MCX emerged as a virtual monopoly in commodities. The exchange grabbed 85 per cent market share in the business comfortably beating the NSE-sponsored NCDEX.

In the equity segment, since most brokers used FTIL's proprietary technology for trading on NSE, Shah had the dream of beating the exchange in its own game. This ambition saw him starting MCX Stock Exchange (MCX SX). He believed that MCX SX could put NSE in its place, in the same way that MCX bested NCDEX in the commodities space, if it was allowed to start trading in stocks. The NSE-FTIL fight was thought a potential successor to the Ambani-Wadia saga.²⁶

MCX SX applied for an equity trading licence with SEBI sometime in 2008 but Bhav, the chairman of the regulatory body, gave it permission to trade only in the currency segment. SEBI kept MCX SX waiting for long before it could start trading in equities and said that its application was under examination. MCX SX launched currency trading on its platform in October that year and a few months later, the exchange took SEBI to the Bombay High Court (HC) for the delay in equity licence.

MCX SX told the court that the regulator was dragging its feet in allowing it to launch equity trading, even though it had fulfilled all the pre-conditions, just to protect NSE's monopoly. Shah even tried to cajole the FM but was unable to read Chidambaram's mind.

The NSE, on the other hand, started applying a squeeze on FTIL. The exchange not only enjoyed a near monopoly in stock derivatives but its benchmark index, the Nifty-50, was becoming a global asset. It attracted marquee foreign financial houses as its shareholders. And most important – the exchange was planning to launch high-tech trading tools involving HFT-driven Algos and COLO grid under its Direct Market Access (DMA) scheme. Although the move had many facets, it remained a closely guarded secret in the exchange.

In August 2008 when NSE launched currency trading, it put FTIL's software ODIN on its watchlist and rejected the company's application for new installations or licences. NSE said ODIN had bugs. For the distant observers, it was a skirmish between the two rivals but only insiders had the knowledge of the precise gains and loss. Highest stakes in the battle were over the trading software.

The same year, the NSE also picked a 26 per cent stake in a software company, Omnesys, which would promote HFT and Algo trading tools for the exchange and counter FTIL's ODIN.

MCX quickly went to the court, alleging lack of fairness by NSE and said the exchange had acted with mala-fide intentions of stifling competition. It had no knowledge of the tech grid that was coming up. But even as the court case continued, NSE let Omnesys march ahead with its DMA offering to brokers. Since FTIL remained blocked by NSE for nearly two and a half years, before the two parties could settle the case outside the court, Omnesys and Ajay-linked IFPL had penetrated deep among brokers with their HFT and Algo offering.

The much-coveted TBT data came handy and the back-tested Algos gave

them an edge. Meanwhile, FTIL remained entangled in the court battles.

At its peak, FTIL's total market capitalisation was Rs 13,500 crore in June 2007 (nearly three billion USD based on the currency exchange rate) and the company's revenues that year were around Rs 1,350 crore. The figures are a testimony that reveals how high the stakes were and what awaited those who could gain an edge with COLO operations in the years to come.

With an attack on FTIL, the NSE had not only under-cut Shah's revenue base and the foundation of his business, but also stole ODIN's first-mover advantage in COLO trading that went to Omnesys. Since NSE was the monopoly exchange, FTIL had nowhere else to go and once volumes increased via DMA and COLO grid, no other exchange would be able to steal a march over NSE ever. The dominance of those who controlled NSE would also remain intact. It was a similar script.

When NSDL was established, as the head of the depository, Bhavne made demat of shares compulsory for listed companies and refused to handle any delivery of physical or paper shares by institutional investors. It led traders to shun BSE as they were forced to punch orders only on the screens at NSE. But then, BSE was a notoriously tainted club of brokers and nobody cared including the government, which played ball with NSDL.

Demat of physical shares did good for the Indian markets but had it been planned fairly, it would have put BSE on an equal footing to NSE and not created a monstrous monopoly that the latter exchange became – bread crumbs are all that is left for other exchanges to collect.

Dethroning Jignesh Shah

“When regulation is weak, it encourages players who have strengths in fixing the regulatory system to their own advantage. Even a few years of faulty regulation can do long-term damage to an industry by killing off firms with high ethical standards and endowed with skills in running a business as

opposed to skills in fixing the system.” This is not about NSE, Bhavé or any secret TBT data expedition. It is what Ajay wrote in his column ‘*The X factor in Regulations*’ about the meteoric rise of Shah and MCX in a leading financial daily.²⁷

Apart from being close to NSE then, Ajay was also a board member of NCDEX, the rival of MCX. Shah sued Ajay for calling MCX business dubious and terming him a fixer.

Ajay told the Bombay HC that he had written the article in ‘*good faith*’ and in ‘*public interest*’. The Judge noted that Ajay had failed to disclose in his column that he was associated with NCDEX. Ajay had spilt his anger and fury in his column against MCX, as the Forwards Market Commission (FMC), a regulator to commodity markets like SEBI was to equities, had disallowed NCDEX from cutting transaction fees on its platform.

Ajay said FMC was protecting MCX valuations and had acted in a mischievous manner. Shah knew a thing or two about influencing the FMC since MCX was the largest exchange in that space but the commodity regulators shenanigans could be termed ‘*lilliputian*’ when it came to SEBI’s prowess that aided NSE. ‘*In the good faith*’, Ajay missed this point while lashing out at the FMC and MCX in his article.

“There is no presumption that eminent people would write articles only for educating the public at large and in good faith,” Justice D G Karnik of Bombay HC told Ajay. “Experience has shown otherwise,” he said.²⁸

What happened to the defamation case is a less interesting story but an idea floated by Ajay in that article in February 2009 was adopted by Bhavé the same year. It would change the face of India’s stock exchange industry and even the thought would be nipped in the bud, if it occurred to any private player, to start an equity exchange in India. NSE was fortified.

“In recent years, a new animal has come about: The for-profit exchange,” Ajay wrote in his column ‘X factor’. “This raises difficult problems for

regulators, for the incentives of a for-profit exchange often run contrary to the interests of the economy. These conflicts are particularly accentuated when a for-profit exchange becomes focused on getting a high valuation for its shares. In India, MCX is a for-profit exchange. It is owned by a software company named FTIL. Unlike most Indian software companies, FTIL does little by way of software exports; it derives its primary revenues from business related to exchanges. '*Ownership and management*' of the exchange business must be separated from other financial businesses."

He all but completely ignored NSE's supernormal profits that the exchange was making.

Surprisingly, in this case, Bhavé took the cue from Ajay's article and responded with alacrity. In December 2009, SEBI proposed a high-powered committee to review the '*ownership*' norms of stock exchanges. The proposal by MCX SX to launch equity trading was further delayed until the committee came up with its report.

Former RBI governor Bimal Jalan was chosen as the team's captain and others who would play the game as part of the team included KP Krishnan, who had then moved from the Finance Ministry to PM's Economic Advisory Council. KM Abraham, SEBI's WTM who worked under Bhavé and banker Uday Kotak, whose Kotak Mahindra Bank would eventually pick-up 15 per cent stake in MCX after Shah's fall and even become a banker to the NSE.

When Uday was made a panel member of the Jalan committee, his group held a majority stake in Ahmedabad Commodity Exchange (ACE). It had been operational since 1954. Kotak Group had applied to the FMC to buy a 51 per cent stake in ACE in 2008, which was approved in 2009. MCX and Shah were direct competitors to the Kotak Group in the commodity segment.

The Jalan team came up with its report in November 2010 and said that private companies and individuals should own only 5 per cent stake in a stock exchange. But it allowed financial institutions, including private banks, to

hold 15 per cent as anchor investors. Foreign entity holding too was restricted. Among other things, the committee even capped any incentives like stock options that an exchange could provide its top management. Silly, only paying high salaries with SEBI's approval was left open.

For the exchange industry, the Jalan rules mimicked the period called licence-*Raj* in India when every industry was regulated and there was an elaborate system of permits and licences, regulations accompanying red tape that were required to set up and run businesses in the country. Jalan gave a caveat after his report; he said all norms – the licence *Raj* – should be reviewed after three years.²⁹

Jalan panel's thought process behind such regulations was that the exchanges are systemically important public institutions and should be protected from private vested interests.

Yet, private individuals like Ajay, Susan and their *'for-profit'* company IFPL, Starcom and Merce Technologies, Omnesys all of which were intensely connected to brokers, flourished and thrived on NSE's data and exclusive access. Jalan called exchanges as *'public institutions'* where it suited them, but when it came to sharing of information under the right to information act, exchanges were quick to retort that they are private organisations and cannot be compelled to adhere to the norms *'created for general public interest'*.

BSE had some of the marquee names in its list of shareholders including Thomas Caldwell and George Soros, all of whom were searching for an exit after the Jalan panel report was tabled.

The new norms, which were a restriction on shareholding of private entities into an exchange, were termed as Manner of Increasing & Maintaining Public Shareholding in Recognised Exchanges (MIMPs) regulations. FTIL group was not able to bring down its stake in MCX SX to adhere to the rules and Shah found himself in a peculiar situation.

The prospective buyers of a stake in MCX SX demanded that the exchange first secure permission to launch trading in equity, while SEBI said, the exchange promoters first cut their stake as per new rules. For NSE, the way to the top was a cakewalk in the 1990s as the bureaucrats and government were all bending their backs to make it a grand success story. But the same system and bureaucrats vitiated the scenario for the entry of any other new player in the segment.

London Stock Exchange (LSE) was interested in entering India and was even eyeing a stake in the Delhi Stock Exchange (DSE). The Jalan rules gave it cold feet. **Hirander Misra**, chairman of United Kingdom-based Forum Trading Solutions Ltd and a consultant on automated trading pulled out of negotiations for a top job at the Delhi bourse.

This was over regulatory restrictions on economic interest. A pioneer of the electronic trading business, he was the co-founder of Chi-X Europe, a small trading platform that challenged the mighty LSE to become the second largest equities trading venue in Europe in 2010. Jalan committee suggestion thus also led to the closure of 16 regional exchanges in India as the sector was dead for private and foreign investments. The only exchange standing now was NSE – ‘*too big to fail*’.

“DSE was finalised but new regulations don’t make economic sense as key exchange personnel globally are allowed economic interest but not in India,” Misra told me in an interview at *Business Standard*.³⁰

“Once competition took hold in the west among exchanges, it had brought innovation even from incumbents, lower fees, tighter spreads and demand for more competition.” In Misra’s view, a concern in India was that the NSE was too dominant.

“NSE’s innovations like order driven trading and central counterparty clearing, which once drove down costs, are no longer subject to serious competition,” he said.

A day after SEBI announced it was implementing Jalan team's recommendations, Madhu Kannan, MD and CEO, BSE, put in his papers in a fit of rage – nearly six months before his term was to end.

In a bid to adhere to the regulations, Shah simply cancelled nearly 90 per cent of equity rights of MCX SX promoter entities FTIL and MCX and converted them into warrants without voting rights. It ensured that shareholding of two promoter entities dropped to 5 per cent each and 90 per cent stake went to other government-led public entities, private sector banks and financial institutions. The warrants, Shah thought, could be sold later.

NSE, on the other hand, had 60 per cent private and foreign investors holding its stake. Bhavé did not budge then too and opposed the restructuring plan of MCX SX. He said SEBI had discovered that Shah was acting in concert with some of the shareholders.

Subsequently, the matter moved to the SC where MCX SX told the court that SEBI had tricked it by first suggesting the warrant route and then rejecting it. It said that Bhavé and his officials had themselves suggested the idea, which was also approved by the Bombay HC.

“Despite the fact that the entire scheme was suggested by SEBI on October 5, 2009,” MCX SX told the court, “the regulator had raised issues almost a year after.”

“The silence of SEBI, even after having full information since October 5, 2009, appears to be designed to damage the applicant's business and reputation. The advantage of this delay and the damage to our business and reputation is only to NSE,” MCX SX said.

Finally, after Bhavé moved out, SEBI agreed to give Shah the licence for equity trading. The Dholidas were called to celebrate the end of a long-drawn battle after the SC approved the warrants of MCX SX to comply with the new shareholding norms.

That year turned out to be a watershed year for Shah; not only had he

streamlined the structure of one exchange, but he also went public with another. The MCX IPO saw stupendous success with 54 times over subscription raising Rs. 663 crores. The share price of the company rose 26 per cent on the listing day.

Before Bhavé's ouster from SEBI, an OOH advertisement of MCX SX placed a hoarding, mounting it strategically at the opening of BKC near the western express highway. It teased the blithe regulator twice every day; in the morning when he travelled to work and evening while retiring for home. Shah said it was his way of showing his commitment to his cause.

But while Shah was beating his own drum, NSE had already secured a monopoly over HFT-driven Algo trading. All the high-pitched legal battles that started at SEBI's BKC office and spilled over into the courtrooms in Mumbai and Delhi had delayed equity trading launch of MCX SX by four years. The exchange was miles behind NSE, already. Like the '*Three Musketeers*' of Dumas; Narain, Ajay and Bhavé ensured a sobriquet for NSE as the world's largest derivative market place. HFT-driven Algos and COLO grid were the currency.

In 2013, Chidambaram came for the launch of MCX SX and mispronounced 'SX' several times during his entire ceremonial speech, which I felt was very dull for the FM's own taste, who could otherwise coin words and phrases articulately.

His constant slip of the tongue on 'SX' made many, including the journalists in the crowd, wonder if Chidambaram was in a state of somniloquy, also known as sleep talking. However, to some, it was an indication of a bad omen that the stars foretold about Shah.

At the launch party that evening, a beaming Shah said in his customary, filmy style, "*Picture abhi baki hain*" (The film is still not over). Not knowing what it meant for him. There was no scant idea that Dhirubhai's clone had about New Delhi's plan. Here, he digressed from the personality traits of the proficient Reliance group founder.

The National Spot Exchange Ltd (NSEL) Fiasco

The battle for supremacy in India's financial market did not end abruptly since MCX SX fell far behind in the race. The story moved at a blinding speed once Chidambaram left the porch of the five-star hotel in Mumbai after a final handshake with Shah.

The NSEL was India's first electronic commodity exchange that facilitated the purchase and sale of commodities, including agricultural products, by providing '*spot delivery*' of goods. NSEL commenced trading in October 2008 in just a week after MCX SX went live with currency trading. Then, the exchange said it had secured permission from the Ministry of Consumer Affairs under the Central Government to start trading.

The arguments that NSEL gave to the government was that it would ease the pressure on the farmers to sell their proceeds at a lower price due to lack of warehousing. The exchange was allowed to operate in 16 states in India and provide delivery-based spot trading in 52 commodities. But the trading on NSEL became frantic as people sensed an opportunity to make risk-free returns.

On the one side were high net worth individuals, companies and retail traders who were lured for trading on NSEL by brokers. On the other side were companies, mainly processors of agriculture goods, who would enter to buy and sell contracts with the clients of top brokerage houses. The trading pattern was that the brokerage clients entered to buy contracts for agriculture goods and the processing companies would promise them the delivery of goods in the warehouse in 11 days. That was the NSEL contract cycle.

But often, the brokerage clients saw that they were getting higher prices for the goods they had purchased and sold them before the settlement could happen. Delivery of goods was seldom accepted. As the wager went on for

four years, brokerages and processing companies revved up a volume of close to Rs. 6,000 crores.

This did not hide for long and came to limelight catching the attention of the regulators and the government. When the trading was at its feverish pitch, the government decided to act and issued a dictate, which said that NSEL was barred from issuing any new contracts as it was violating the norms of the delivery cycle by extending the contracts beyond 11 days and allowing traders to enter into simultaneous pair contracts. It created panic among traders, who came to close their existent contract positions at once. The situation resembled a '*run on the bank*'. NSEL was fully owned by the FTIL group and obviously, it had to face the brunt.

Since the 1929's '*Great Depression*' in the Americas, history has shown that even some of the largest banks in the world have never survived a '*run on the bank*' phenomena. It occurs when panic-struck investors all rush to withdraw their money at once and no bank can make good payments. In the NSEL's case, the government's sudden dictate to close down contracts led to anxiety among the traders who were playing on the exchange for four years. There was a run on NSEL and the goods in the warehouse were woefully short as the retail clients often sold back their contracts in days and never demanded delivery. Only this time, they could not close their contracts and hence all demanded delivery of goods.

NSEL contracts were high leveraged trading bets as brokers even cut financing deals, which they were not supposed to, and no delivery of goods was anticipated. Brokers even told their clients that they had surveyed warehouses linked to NSEL and goods were all present. But the rush of investors to close their contracts at once and take delivery birthed a payment crisis to the tune of Rs. 5,600 crores.

The frantic trading so far had generated volumes far more than NSEL could keep pace with deliverable goods. The government said the exchange was extending its contract settlement cycle to 25 days or even 35 days and

that was its violation. The NSEL told the Bombay HC that NCDEX too had contracts that stretched to 35 days or more than the stipulated cycle. But the government took no actions against it. The contract cycle case is still pending in the court.

Together, the FMC, Mumbai Police, CBI and Enforcement Directorate (ED) all turned the heat on NSEL and Shah's empire during the months that followed. Assets of over Rs. 2000 crores belonging to the FTIL group and Shah's personal belongings including his residence were seized by the Mumbai police. Among other things, an attempt was made by the police to attach even ODIN, the trading software used by the brokers, which was FTIL's key money-spinner.

Who was educating cops about ODIN? It could '*not*' have been a saleable asset but somebody who knew the exchange technology well was guiding the police. The court later stayed its attachment and even disallowed further seizing of FT and Shah's properties. Meanwhile, the HFT-driven Algos were hard at work and drove the share price of FTIL to ground. It was a stock being traded in NSE's derivative segment and there was no bar on how much it could move. The stock that was trading at around Rs. 1,000 at the start of 2013 crashed to below Rs. 100 in a few weeks as the noose around Shah tightened hard.

NSEL's MD and CEO at that time was Anjani Sinha, the man who was given a long rope by Shah. Sinha was responsible for designing products and the day-to-day running of the spot bourse.³¹ He initially took all the blame on himself only to later turn around and say that Shah was the real kingpin.

Shah was called in the Crawford Market office of the Economic Offence Wing (EOW) of Mumbai police for customary questioning by senior police officials and they arrested him.

"While Jignesh was waiting in the EOW office, a senior officer of Mumbai police was having an unscheduled meeting at the SEBI office where

a Finance Ministry official had landed up. When the EOW officer returned a little after 5pm, Jignesh was told that he was being taken in. Sinha's damaging statement, follow-up investigations by EOW, the appointment of Rakesh Maria as Commissioner of Mumbai Police, some of the invisible forces and a few powerful enemies in New Delhi – who pushed for action before a new government took over in a few weeks – went against Jignesh," *The Economic Times* reported quoting a source.³²

A twist in the tale was that brokers who propped up the clients to generate huge volumes were never even touched by the FMC despite the fact that it was their fiduciary duty to protect their clients. If NSEL volumes were quietly revived up to dizzy heights, some of the brokers in Mumbai played a role by luring clients to trade on it.

Messages like "13 per cent (return) in raw wool February" were sent out by brokers to entice clients.

Investigations by EOW found that brokers traded without client permission, illegally changed unique client codes, engaged in market-capturing activity, traded under the names of employees, had nexus with defaulters, re-routed funds through multiple accounts, enrolled and financed low-income people as clients, misled submission to the EOW about clearing and forwarding services, gave false assurances to investors and were involved in short-selling that was not allowed.

That report, submitted by Rajvardhan, the then additional commissioner of police, EOW, about brokers' *benami* trading operations and money transactions was brutally suppressed by the then FMC chairman Ramesh Abhishek, the 1982 batch IAS officer. Abhishek had come to FMC on deputation by the Central Government first as a junior officer but was made the regulatory chairman. His actions were directed only against Shah and he gave the brokers a free pass.

The Delhi-based Serious Fraud Office (SFO) that mainly probes

corporate crimes, made scathing remarks in its investigation report against commodity arms of top five brokers including Anand Rathi Commodities, India InfoLline Commodities, Motilal Oswal Commodities, Geofin Comtrade (Formerly known as Geojit Comtrade Limited) and Phillip Commodities among others on several issues.

SFIO said investors like Achal Agarwal (of Geofin), Borosil Glass Works, Encore Natural Polymers (of Anand Rathi), Vishvanidhi Dalmia (of India Infoline), Moti Dadlani (of Motilal Oswal) had complained against their brokers for fabricating documents, forging their signatures and trading without authority.

Agarwal told SIFO that he discovered that Rs. 5 crores were invested in his name even when he had given only Rs. 97 lakhs to his broker. Dadlani discovered Rs. 1.82 crore invested on his name against an amount of Rs. 50 lakhs he gave to Motilal Oswal.

A figure of Rs. 5,600 crores and 13,000 investors became synonymous with NSEL scam but only 7,217 responded to SFIO. A Bombay HC committee received claims worth only Rs. 650 crores from 4,697 entities. Out of the 12,735 claimants registered with NSEL, 67 per cent of the outstanding amount was represented by 781 traders. Simply put, 6 per cent of them were claiming 67 per cent of Rs. 5,600 crores.³³

There were 1,200 companies that accounted for Rs. 2,800 crores. The HC committee found that 32 companies that had submitted their claims were already declared as '*shell companies*' by the government. The 781 traders that had an exposure of over Rs. 1 crore were claiming Rs. 3,606 crores.

Also, investigations showed that most of the large NSEL traders had transactions disproportionate to their known source of income. A lot of the NSEL investors claimed their money by raising demands via social media platforms but dragged their feet when asked to appear or submit their full claim of money to the court.

Abhishek, on the other hand, had not only suppressed the vital probe report against brokers that held them accountable for various violations but declared Shah '*not fit and proper*' with a vengeance. It meant Shah could not hold a stake in MCX and MCX SX.

The middle-class Kandivali boy fell by the wayside and sold his stake in the two exchanges. Kotak Group and few other high net worth individuals including billionaire stock-picker, Rakesh Jhunjhunwala, picked up Shah and FTIL's stake in MCX. Shah was left only with FTIL, which has since been rechristened 63 Moons Technologies Ltd.

Shah's troubles did not end here. After he was released from the prison, Shah had to wage another battle to keep his holding in FTIL. Post Chidambaram's ouster from government, his former lawyer friend Jaitley took over as the FM. The ED, which fell under the Ministry of Finance, called Shah for questioning and arrested him again.

This time the Ministry under Jaitley was forcing a takeover of FTIL's board and seeking to merge the company with defunct NSEL.

The SC came to Shah's rescue and thwarted this attempt by the Ministry of Finance and also came down heavily on FMC for protecting private interests and aggravating a '*resolvable*' NSEL crisis.

"FMC was acting for protection of private interest of a group of investors / traders, distinct from public interest," SC noted. The judgement, delivered by a division bench of **Rohinton Nariman** and **Vineet Saran**, was a tight slap for Abhishek as it ridiculed his ill-thought-out recommendation to the Ministry of Finance to order a merger between FTIL and NSEL.

When Shah was released from the ED's custody, he filed a Rs. 100 crores' lawsuit on the agency. That case is still pending trial.

The equity exchange MCX SX is now a dying entity kept alive on a ventilator. It is currently engaged in a battle of survival but also a victim of delayed justice. Way back when MCX SX was being launched, NSE had

embarked upon massive transaction charge discounts to brokers.

Shah dragged the exchange to Competition Commission of India (CCI), which held NSE guilty of monopolistic practices in 2011 and imposed a fine of Rs 55.5 crore for abusing its dominant position. The MCX SX calculated its business loss and planned to seek a compensation amount of more than Rs 800 crore from the NSE.³⁴ The case is now in SC, where it is languishing, pending trial.

Nearly nine years have passed since the CCI verdict but the case has just moved from one date to another in the SC leading to a slow and painful death for MCX SX. NSE's lawyer, who has managed to delay the case thus far is another Congress party veteran, Abhishek Manu Singhvi. Once he even sought date from the apex court saying that he was busy in general elections.

Meanwhile, Susan was slowly but steadily targeting MCX. She established a secret '*data pipeline*' at the exchange, the one similar to what Ajay had done at NSE. Since Ajay was tainted for his role at NSE, Susan was at the forefront in quietly retrieving crucial trade data from MCX this time.

Had SEBI acted against her in the NSE case, the story would have been different.

Data Heist II – Target MCX

As the chaos over MCX and Shah started to recede, MCX began scouting for a new MD and CEO. Mrugank Paranjape, the former head of Deutsche Bank in India was shortlisted for the job.

Call it a coincidence, Paranjape who joined MCX in mid-2016, was another campus mate of Ajay and Susan from the good old days of IIT Bombay.

Paranjape joined the exchange as the hullabaloo over MCX and Shah's hold had receded in 2016. By the time the FMC too had been merged with

SEBI and the officials who were involved with NSEL investigations crossed over to the larger regulatory entity.

Now MCX was under SEBI's preview and former FMC official SK Mohanty was given the job to keep an eye on the commodity exchange after he was promoted to the post of WTM from ED in 2017. Just a month after Paranjape came to helm MCX, the exchange started a data pipeline to Susan, the pretext was the usual research by IGIDR. The hidden aim, of course, was the extraction of trade data for HFT-driven Algo software.

When I first broke the story at *The Hindu Business Line* about the secret data pipeline at MCX, it felt like deja-vu to me.³⁵

The script and the characters all came dangerously close to the scandal at NSE, which was yet an unfinished job for SEBI. HFT-driven Algo trading volumes were on a rise at MCX and I followed up the story vigorously. My source, an ingrained MCX insider, gave me the forensic audit report by Delhi-based accounting firm TR Chadha & Co, which had just been presented to the board members.

In just a few months after *The Hindu Business Line* did the first story in 2018 about the data pipeline to Susan, the exchange board came under pressure and somehow a forensic investigation was ordered. T R Chadha & Co's report turned out to be the most damaging to have been written by any forensic auditor against data theft at any large exchange in the world.

A simple Google search could have revealed to the MCX officials the big picture details of the scandal at NSE and also the role of Ajay, Susan and IGIDR that was under SEBI scrutiny since 2015 after a whistleblower exposed the COLO scam.

As a rule, the exchanges themselves are the first level regulators, and then comes the role of the apex regulator SEBI. MCX failed in its duty to check on those with whom it shared exclusive data. The story of agreement between MCX and Susan was all but similar to the one the research duo had

with NSE. TR Chadha's report virtually says that it utterly disbelieved MCX officials when they pleaded ignorance.

"Why a background check of the researchers was not done before sharing the data. And how was the data being shared important for research," the auditors started with asking the officials at MCX. These were simple but key questions that SEBI had missed during its investigations at NSE.

"NSE scandal was out of my knowledge. The operational aspect (of data sharing) was completely monitored by the research department and they should be in a position to answer," Paranjape explained his point.

"We believed in the capability of the IT team to advise us, the IT team being the guardian of data to be shared with the outside world," said V **Shunmugam**, the Head of Research at MCX. Paranjape tossed the ball to him and Shanmugam to the IT guys.

"It seems '*researchers*' want to gain access to data not available to other traders and not required to be available," TR Chadha & Co said.

Like NSE, MCX too maintained '*no trail*' of the data it had shared with Susan. The files uploaded on the FTP server by MCX IT department were all deleted. FTP is an acronym for File Transfer Protocol, where one computer is the data storage bank and another receiver. It is about making real-time connections from a personal computer to the storage bank. The auditor observed discrepancies between the time of data files shared with Susan and log records. MCX expressed ignorance about this.

On review of logs for FTP server, it was observed that certain key fields such as bytes received, bytes sent, time taken were not enabled resulting in not knowing the size of the data file. Like in the case of NSE, nobody knew the exact amount of data that was shared by MCX.

The policy of MCX was to store data that it shared with others for three years, but auditors observed that the exchange stored data shared with Susan and her associates only for 7-10 days. This made it difficult to verify if the

data shared was masked, meaning had client information or not.

Rahi Racharla, IT official at MCX admitted that he was not aware of the data-masking policy. Another official also said he could not recollect any data-masking policy. Eerily reminiscent of NSE like agreement fraud at MCX.

Both Susan and Paranjape claimed there existed a data sharing agreement between the exchange and the researchers. It was NO different than the agreement that Ajay had with NSE – basically fake.

The audit report says various people outside IGIDR were involved in receiving data from the MCX and it was not just Susan. Curiously, Susan directed the exchange to share data directly with one Chirag Anand, whose name was nowhere mentioned in the agreement. Susan told the auditors there existed another '*undertaking*' that she had given to MCX and those points not mentioned in the agreement were part of the undertaking – really ludicrous.

Audit findings suggested data was shared based on the '*undertaking*' even though it had '*no role with IGIDR*' research. Both the agreement and the undertaking had no proper names, no defined role and contact details mentioned in it of people with whom data was to be shared. Needless to say, the signing of agreements and stamping on the document had different dates. It was just another lie to hide the previous one.

Based out of New Delhi, Anand worked to design Algo software and programmes. In the past, he had worked with IGIDR, NIPFP and was a contributing author to Ajay's blog.³⁶ He also played a role in some of the government committees where Ajay worked.

'*Data fields*', which were not part of the undertaking, were also shared in an unmasked manner with Anand. He sought 22 fields of trade data, 24 fields of order data, 37 fields of *bhav* copy (historical contract-wise summary) and 64 fields of master file data. According to the auditor, the Algo expert was

directly in touch with the technology department to extract the '*required data without involving people from operations and compliance departments*'.

TR Chadha & Co meticulously detailed data fields that constituted '*live data*' being shared by MCX. The audit report said that the exchange shared details of pending orders, still valid up to a certain date at the exchange level, actual order quantity and display quantity along with quantity filled '*today*', all of which reflect actual transactions done against orders on that date. Access was also given on '*disclosed quantity*' and '*actual quantity*' on a daily basis, actual quantity placed by the trader.

"All this was something the trader did not want others to know as it would result in knowledge of sensitive data," the auditor said.

In her reply on why Anand wanted so much data, Susan simply said, "Chirag's inputs are required on this."

Susan's ignorance was scarcely inspiring as she was the lead researcher. She even accepted that '*live data*' was never needed and **said** she was not specifically '*authorised*' by IGIDR for signing an undertaking with MCX.

"But the faculties are allowed to do so," she contended.

No records existed to suggest if MCX verified Susan's eligibility to sign an undertaking on behalf of IGIDR. "No such check was done," Shanmugam said.

"The faculty at IGIDR are permitted to receive the data that is required to fulfil research. I gave undertaking on behalf of IGIDR. It had provisions of data fields, mechanism of data sharing and confidentiality requirements," Susan said as she defended her stand.

But did she at least inform IGIDR about the '*undertaking*'? "It was not required," Susan added.

MCX legal officials told the auditor that neither the '*agreement*' nor the '*private undertaking*' was presented to them for a review. All this came close

to NSE technology chief Ravi Apte's assertions about NSE's agreement with Ajay and Susan, which he thought was vetted by the legal department, but was not.

"Agreement was done to utilise their research capabilities to take forward our policy advocacy work in the area of Commodity Transaction Tax (CTT), bullion markets and analyse 'HFT' data for regulatory policy feedback. Research and data undertaking are independent except that HFT data was used in accomplishing the research work as per the agreement," Shanmugam averred.

Even though naively, Shanmugam had revealed Susan and Anand's hidden agenda, which was essentially only to extract HFT data. Such a breach had its parallel only in NSE. Audit probe found no mention of sharing of HFT in the agreement or the undertaking. According to Paranjape, there existed no common context or link between the undertaking and agreement.

"One was for daily data sharing and another was paid research work as per specific deliverables," he said.

What were the deliverables? MCX said it had rolled out a '*pipeline*' of the live market and HFT data in 2016 to IGIDR for research on CTT that was imposed by Chidambaram in his 2012 budget.

Then, the exchange was owned and managed by Shah and he had fought tooth and nail to oppose the FM from imposing the tax. Nevertheless, the FM was determined. Chidambaram's argument was that a transaction tax existed in the equity markets and it was only fair that the same be imposed even in commodities. Shah thought it could have affected the valuations of the impending MCX IPO but it did not.

For a similar study on the impact of CTT on commodity market volume, researchers from IIT Kharagpur and Indian Statistical Institute, Kolkata were asked by MCX to use data that was already in the public domain.

MCX however showered its benevolence only on Susan and Anand. At

NSE, the pretext for sharing of data was study on liquidity index, at MCX it was CTT – as a matter of fact, both were hoaxes.

The deliverables promised by Susan and IGIDR included a research paper on CTT's impact on growth, conducting a round table to discuss relevant topics on the Indian commodity market and a policy paper on the impact of gold futures and its ecosystem in India. So what was delivered?

The IGIDR assignment paper was submitted on August 23, 2017 against the March deadline hence it was not accepted by MCX. A secret round table conference was held but the researchers were so occupied, they forgot to note down the minutes, who cared anyway. Policy paper on gold too never came.

TR Chadha & Co said in its report, “No specific deliverables with concrete timelines were documented in the agreement dated August 29, 2016. Undertaking was unknown to anybody in IGIDR or MCX. The research topic is open-ended and generic like studying issues of relevance to the commodities market.” The auditor also said that MCX data was accessible to the ‘*so-called researchers*’ beyond the dates set in the ‘*questionable*’ agreement.

What happened to the HFT data that Anand got? It remained a mystery like what IFPL did with the data it got from the NSE.

Four additional fields viz. client identity flag, trading member ID, volume filled today and spread combination type were also shared though this was not part of the undertaking. As per MCX, this was done in pursuance of discussions with IGIDR to ‘*reconstruct live market conditions*’. However, no written records mentioned it.

Finally, TR Chadha & Co concluded that there was a possibility that the data MCX shared with Susan and Anand was used for some other purpose.

“Prima-facie indications are that the objective behind seeking trading-related data seems more related to developing Algo strategy,” the auditor cast

aspersions after the probe.

The story, however, came alive as the auditor found an email dated June 6, 2016. It was sent by Susan to Paranjape.

Part of the text was, “My sister Sunita runs a software company, which does work for a few securities firms, in algorithmic trading for the NSE/BSE. It would be useful for you to meet her and her partner and look for areas for cooperation.”

Both Paranjape and Shanmugam refused the forensic auditors access to their personal email accounts. Nor did SEBI or MCX board members insist on it too.

Only keyword search of enterprise emails of Paranjape and Shanmugam was allowed to the investigators. Auditors used HFT, Algo, IGIDR, Susan and Chirag Anand to go through email accounts including ‘mrugank.paranjape@mcxindia.com’ and ‘shanmugamv@mcxindia.com’. No email probe of Susan or Anand was ever done. In a high-tech scandal, SEBI and MCX did not see the need to allow the forensic investigators access to personal emails, which are the most preferred mode.

There was screaming evidence about the lack of will to allow a deeper probe. Again, parallel to NSE.

Yet, the forensic auditor managed to find an email exchange between Mrugank Paranjape and Shanmugam wherein they discussed the sharing of ‘*data pipeline*’ with Susan and Anand. “All the correspondence with Susan and her team was on their personal emails and the team members were not on the rolls of IGIDR. Because personal emails were used, the audit trail may be thrown off and IP tracing of the same may become difficult,” the auditor noted in his report.

Two of India’s largest exchanges suffered an unprecedented data breach. Nobody, including super-cop SEBI, wanted to conduct a thorough probe. It speaks volumes of how corrupt the system has been reduced to. The Harshad

Mehta and Ketan Parekh stock scandal often evoke heightened emotions in India but were never as blatant a saga of the data breach that was perpetrated at NSE and MCX by deeply vested interest.

Both the rogue stock traders, Mehta and Parekh were accused of stock rigging and their source of funds that aided them in keeping their wager high was traced to loopholes they exploited in India's banking system. The stock market infrastructure largely remained unharmed. But the data breach by Ajay, Susan and their associates was India's first fraud that laid a direct siege on NSE and MCX, where billions of dollar worth transactions take place daily. The scale of their scheme was simply mind-boggling, in fact overwhelming.

Despite this, till date, SEBI has not issued even a single SCN to anybody for data heist at MCX. Both Paranjape and Shanmugam have moved out of the exchange since. Their taint was all but washed-off as SEBI took no action on the T R Chadha & Co report and it was just buried.

On May 29th this year, the State Bank of India, the country's largest government promoted bank said it had found Paranjape 'fit and proper' to be appointed as its board member (shareholder nominee). No clearance from Central Vigilance Commission (CVC), the overzealous organisation filled with bureaucrats that probes every matter of corruption allegation, was required?

Shanmugam is rumoured to have joined the National Institute of Securities Management, promoted by SEBI and currently helmed by Mohanty – if true, SEBI's benevolence knew no bounds. During the data breach at NSE, Bhavne reliniquished the path of Dharma. During MCX, it was the incumbent chairmanm Tyagi, who had failed in his duty.

Tyagi's appointment as SEBI chairman too involved some backroom drama. The 1984 batch IAS officer was first hand-picked as a replacement to Krishnan in the Finance Ministry by Jaitley. Then the FM also recommended

him for the post of SEBI chairman and convinced the PM's Office (PMO) for his appointment, like Chidambaram did for Bhavé.

In a rare instance, Tyagi was first accorded a five-year term when Jaitley proposed his name for the SEBI chairman's post in 2017. The term was reduced to three years by the PMO. But again in 2020, the government twice extended Tyagi's term as SEBI chairman till February 2022.

THE GOLDMINE'S CODE

A Letter from Singapore

The financial market community saw a glimmer of hope after the Congress-led government in the centre was dislodged in a humiliating election defeat by the BJP in 2014. Consequently, Chidambaram's ouster from the seat of power was reason enough for the beleaguered few to take on the unscrupulous elements at NSE's COLO trading farm. The elite at the exchange started to witness sudden, sharp vicissitudes of luck since losing their political patronage and it caused considerable anxiety in the ranks.

In the midst of this, somewhere in January 2015, SEBI received a letter from Singapore that was written by a whistleblower under the pseudonym – Ken Fong. The elaborately detailed letter was enough to set the cat among the pigeons.

Ken Fong started his letter with a straight point, "I wish to draw your attention to a sophisticated fraud at NSE co-location. The market manipulation I'm referring to has been occurring by enabling certain vested brokers to get price information ahead of the rest of the market and thus enabling them to front-run. The most shocking aspect is that when the matter came to the knowledge of NSE management, they chose to hush it up rather than come out in the open against the same."

In the eight-page letter, the whistleblower described the contours of a ‘*Preferential Access scam*’, through the use of the COLO grid and TBT data that was brewing at the NSE for the past few years. Suddenly, the exchange became a grimy cauldron and what began to unfold was the tale of SEBI’s prolonged apathy to the happenings inside the NSE – a.k.a. the CB Bhave galore.

The informer told SEBI that certain brokers, who had booked the server space in NSE’s COLO farm got favourable treatment in ‘*first connection*’ to the exchange data dissemination engine. They would get market data a few seconds in advance than others even in the COLO farm and were able to hit and run trades ahead of anybody else. A crucial part of the letter was where it gave a low-down on how the COLO grid at NSE was inherently built to be exploited and that the exchange staff (themselves) tipped brokers about the loopholes in the system.

The idea of COLO itself had created an Orwellian state where brokers who were within the farm were higher among the equals, as the whistleblower’s revelations now revealed that the entire infrastructure was in fact defective.

A copy of the letter, among others, was also marked to veteran financial market journalist, Sucheta Dalal, and naturally, she lapped it up and published it in her personal finance portal *Moneylife* even when other media houses refrained doing the same. There could have been multiple reasons. Maybe because the revelations were far too explosive in nature, or maybe because of peer pressure from management, after all, NSE was not an ordinary run-of-the-mill institution, not in the eye of the public at least.

As an investigative journalist who blew off the humongous Harshad Mehta scam of 1992, Dalal had a slew of questions for the exchange and the regulator. Perhaps, as anticipated, both the NSE and SEBI decided to bury their heads in the sand and did not reply to Dalal’s diligent queries on the letter, for weeks. However after she published the story (without any

reactions or comments from both, of course), NSE hit back at her with a Rs. 100 crore lawsuit in the Bombay HC.

The move of NSE, however, backfired miserably. Like jurist Mohan Gopal who could not hide his fury against the guile of SEBI's high profile board members in the NSDL matter, Justice Gautam Patel of the Bombay HC also came down heavily on the NSE for attempting to muscle the press freedom. He imposed a Rs. 50 lakh penalty on the exchange. Perhaps, something which has not been heard of before. Chitra Ramakrishna was at the helm of NSE then.

"I do not believe a defamation action should be allowed to be used to negate or stifle genuine criticism that is harshly worded; nor should it be allowed to choke a fair warning to the public if its interest stands threatened in some way," before adding, "It is to me a matter of great dismay that the NSE should have attempted this action at all," Justice Patel said in his order.³⁷

The reprimanding order of such a nature rang the alarm bells in the Finance Ministry, after all, it was concerning two premier institutions of the country and New Delhi dispatched a team to inquire about the scandal in the Mumbai markets. But much before the '*bell hath no fury*' verdict of justice Patel against the NSE in September 2015, there was one official in SEBI who had initiated an inquiry into the whistleblower allegations.

Rajeev Kumar Agarwal, SEBI's then WTM, was trying to understand the mystery of the COLO grid at NSE since February 2015 and advised his department to seek NSE's response on the letter. Subsequently, NSE responded to SEBI in April. But SEBI's Market Regulations Department (MRD) that fell under Agarwal, was not happy with the lame-duck answers from the exchange, and it asked NSE to re-examine the matter and this time also to submit a report to its own '*technology committee*'. The exchange submitted its report on July 31st and said that the technology committee of NSE was convinced that the informers' allegations were all outright bogus.

Although put off with this answer and attitude, Agarwal though had to lie low; he continued pursuing the allegations.

NSE officials would have heaved a sigh of relief, but it was not to be for long. On August 10th, the informer Fong shot another letter bomb to SEBI and described the rigging of the COLO grid in much finer details. This time, Agarwal constituted a team of officials from across the departments to probe. The SEBI Cross Functional Team (CFT) in 2015 was the first batch of regulatory officials that went into the details of NSE's COLO farm since it was established over five years ago.

Agarwal went a step further. He took the matter to SEBI's '*Technical Advisory Committee*' (TAC), which had IIT Bombay professors as its members. The SEBI for the first time engaged IIT Bombay to conduct a forensic audit of NSE's system under the overall supervision of TAC. "India's premium tech institute was involved to understand the matter in a fair manner," Agarwal later told the press. He was getting SEBI back on the path of 'Dharma' that Bhave had abandoned.

The primary findings of the IIT Bombay and CFT team were sensational. It said, "The speed of some machines in NSE's COLO farm that disseminated TBT data was higher than others. A broker who was first in the cue in connecting his trading bot to the fastest machine, received the TBT data, seconds in advance, for the entire day against those behind in the queue."

The information by the whistleblower that NSE's COLO farm was built to be rigged was coming true, as it became evident by SEBI's initial findings on November 30, 2015 and later the TAC's report of 2016.

The informer in his letter had revealed about a Delhi-based broker, OPG Securities, being highly favoured by NSE. CFT and IIT Bombay found that the said broker could connect his bots first to fastest servers in the COLO on a maximum number of days. Also, OPG's trading volumes saw a massive

spike when he connected first. This was a startling revelation, but not unexpected by Agarwal.

More than before, he was now determined to take on the mighty NSE as the high-tech trading grid of the exchange had uncomprehending defects. To lift the veil of NSE's COLO, Agarwal initiated a series of probes. He asked the NSE to bring an external forensic auditor to look at it. Deloitte Touche Tohmatsu and Ernst & Young (E&Y) two of the big four forensic auditors were appointed.

In 2017, on his last day in office, Agarwal's boss and the then SEBI chairman, Upendra Kumar Sinha, asked NSE to expand its probe to the cash segment as well. Deloitte's report was the most crucial one since it looked at nearly the entire of NSE's COLO. However, weeks before Deloitte was to submit its first report, alas, Agarwal was on his way out.

Shooing Away a Change Maker

The Central Government recruits Income Tax officials from the Indian Revenue Services (IRS) stream. The 1983 batch IRS officer Agarwal already had 28 years of experience in investigations, assessment and enforcement in the Income Tax department when he was appointed to SEBI sometime in 2011.

Agarwal was an IIT Roorkee alumnus, he had studied electronics engineering at the prestigious technical institution. After a long stint as a commissioner in the Income Tax department, Agarwal was first appointed as a member of the FMC – the regulator for the commodity futures business for a period of five-and-a-half years, where he was responsible for laying down the regulatory mechanisms for commodity futures markets and the development of the markets, which had just started their operations. From FMC he was brought in at SEBI.

As the in-charge of market regulations, Agarwal started with taming Algo

trading that had remained outside the purview of SEBI for long. In over five years since they had been in existence, SEBI came up with the first set of regulations on HFT and Algos in 2012 due to Agarwal's initiative. Regulating COLO was still way away as there was no understanding of the game in SEBI due to Bhavé's long apathy towards it during his stint between 2008 to 2011. Agarwal's former colleagues at SEBI say that his approach towards COLO was also slow as Chidambaram was occupying the FM's chair and he would have risked upsetting the equations.

The scenario, however, changed in 2015 as the Congress party lost elections and Chidambaram was no more the FM. After the whistleblower's first letter in January, Agarwal was seen pushing SEBI to investigate all the allegations. Once IIT Bombay and SEBI's CFT confirmed the glaring laxity in NSE's systems, Agarwal set the ball rolling for a further probe.

He directed the NSE to fix management responsibility with regard to the lapse in COLO systems and asked the exchange to deposit all earnings in fees and transaction charges from the segment in a separate escrow account until investigations got over. NSE had deposited over Rs. 2,344 crores until May 2019. But while the probe was on, Agarwal was nearing the end of his five-year term at SEBI and it could have been extended, however, at this time, his detractors who were waiting in the wings, stuck at him.

Despite earning an initial breakthrough for SEBI with findings of CFT and IIT Bombay into one of the most intriguing market scandals, Agarwal was denied an extension to his term. Not only that, but he was also made to quit SEBI in an undignified manner.

His exit came just days before Deloitte submitted its first report on the NSE COLO system that largely confirmed the whistleblower allegations and all the CFT findings. A report by SEBI's TAC committee too confirmed that NSE's COLO grid was prone to manipulation and gave a split-second advantage to those who could exploit it.

Agarwal's ouster from SEBI was a peculiar case that shows the power play

of the 'bureaucratic brotherhood' in India and the cabal that was accustomed to arm-twisting the administrative system.

Sometime in 2016, the SAT, the statutory body that adjudicates SEBI related cases, passed strictures concerning a listed company's matter that was being handled by Agarwal. He was then travelling on an outstation trip and could not present SEBI's side. The SAT order was passed by the bench comprising Dr. CKG Nair, the former Finance Ministry official and the then junior of Dr. KP Krishnan.

It was an open secret that both Nair and Krishnan were close confidants of P Chidambaram during their stint at the Finance Ministry. SAT imposed a cost of Rs. 1 lakh on SEBI for technical deficiencies in the order passed by Agarwal in July 2016 against the Kolkata-based company. The tribunal also directed the regulator to get the matter heard by a different member.³⁸

But after he returned, Agarwal filed a review application in SAT and the tribunal agreed to drop the cost of Rs. 1 lakh and asked SEBI to pass an appropriate order within a week. The SAT also allowed Agarwal to hear the same matter. But the earlier SAT order had set the plan of Agarwal's detractors already in motion.

The CVC, a body that mainly probes wrongdoings by civil servants, started a probe against Agarwal based on the first SAT order. Even when SAT reviewed and reversed its order against SEBI and Agarwal, the CVC, consisting of the bureaucratic cabal, held back the clearance for Agarwal's reappointment at SEBI when his stint came up for renewal.

Surprisingly, the CVC overlooked the fact that the SAT's first order itself involved a grave conflict of interest as it was passed by Nair who was occupying the division bench. At that time, Nair's daughter worked with the NSE and yet he went ahead and adjudicated a case against Agarwal, who was supervising investigations into the COLO grid of the exchange.

Years later, Nair had to recuse himself from hearing any matter related to

the NSE COLO scam on the same ground of conflict of interest. Nair's daughter was appointed at NSE in 2013 when he was working in the Ministry of Finance, Capital Market division and the exchange fell directly under his preview of decision making.³⁹ As a judge at the SAT, he was still conflicted in presiding over cases related to the exchange.

Last-minute CVC complaints and probes are an old trick to stall or delay crucial administrative appointments and the *Babus* had prevailed in their chicanery, yet again.

It was quite interesting that the CVC clearance for the extension of Agarwal's stint came just a day after SEBI had selected another candidate to fill his post. Until then, the CVC kept raising mundane queries and delayed its assessment, which one would argue were very frivolous reasons. CVC's handling of the matter raises suspicion mainly as it picked one '*quasi-judicial*' order passed by Agarwal to hold back his clearance. Each day, SEBI passes several orders and there are many gaps in them but never the CVC had come to the middle in those matters previously. The point was, who was behind this? *The Market Mafia*?

"In one case, while hearing an appeal, SAT issued unfavourable directions against a whole-time member for delay in passing an order. Later on, these directions were withdrawn by SAT. However, based on this information published in newspapers, vigilance authorities initiated a proceeding on why this delay had taken place." This was written by the then SEBI chief Sinha in his book *Going Public: My Time at SEBI* to highlight how he was surprised at CVC's arbitrary action.

As anticipated, after Agarwal's exit from SEBI, the COLO probe did not move any further. But since the previously conducted investigations had yielded so much evidence about the defective COLO grid and information about the massive data theft at NSE, it became impossible to completely overturn the case even for the vested lobbies. In 2018, SEBI issued an SCN to more than a dozen NSE officials, Ajay Shah, and a few brokers based on

the past probe. But its half-baked orders showed that the regulator, however, had failed to take the COLO investigations any deeper and unravel the full ramifications of the scam.

SEBI's final verdict in the COLO and Algo scandal has several gaps and leaves enough loopholes to give escape routes for the accused. The investigation itself is incomplete in several aspects; like Algo software of IFPL was deployed by a broker at NSE's COLO in a 40 per cent profit-sharing deal. There was no clause in the agreement for sharing of loss – it was all about guaranteed incomes.

This has not been looked at a deeper level by SEBI. In fact, SEBI let go of Susan, even though IFPL, one of the company's directors under probe was registered at IGIDR on her address. Nobody has questioned her qualifications or tried to know if Susan ever held a PhD degree. She was a researcher and an insider like Ajay in NSE for well over a decade. SEBI even turned a blind eye to other markets' connections of Susan and Ajay that have severe ramifications to their misuse of data and position at NSE.

A money trail involving the researchers is also grossly left untouched. SEBI has not answered the most crucial question in its investigation: How did Ajay and Susan benefit from the data they gathered from the NSE in a clandestine manner?

Like Bhav, the apathy of the incumbent SEBI chairman Ajay Tyagi and his colleague SK Mohanty, who was handling Agarwal's portfolio, had encouraged an NSE like scandal and data theft at MCX.

Solving the '*Flash Trade*' Enigma

The shiny allure and opulence of gold is enough to charm anybody towards it. In the financial markets, although they say that data is the new gold, not everybody falls for it like they would for the bling of the yellow metal. This, since all are not capable of turning the data into a business opportunity as it

requires special skills. Just like ore has to be processed to extract gold, raw data must be processed to extract its true value. Who else could have known this better than Ajay and Susan?

Since 1987, CMIE, Ajay's family enterprise made all its profits from the mining of data that was later sold to brokers, bankers, policymakers and other finance professionals. Yet the NSE ignored a grave '*conflict of interest*' that existed in sharing data with Ajay and Susan, needless to say, it was without appropriate non-disclosure agreements. All this came out in SEBI's investigations that followed the whistleblower letter.

Data that Ajay and Susan extracted through doubtful means from NSE and MCX could be worthless if seen in isolation but a virtual gold mine when shared for developing and back-testing of Algo software. And Ajay's email to Sunita, in which he seeks her to swear by a code of silence on the fact that the data they extracted from NSE was going into Algo work, is evidence enough to show that the researchers used academia as a smokescreen to reap commercial gains.

It was in February 2009 that Ajay was extracting crucial data from NSE, which went into preparing his Algos. His mission reveals itself when seen in conjunction with a key event that hung upon India's stock market some 12 weeks into the future – results of that year's general elections that would bring Manmohan Singh back as the country's PM for the second consecutive term. In either of the scenarios that the results were to present, the stock markets would react sharply, on the up or down.

So what caused the markets to come to a grinding halt on May 18, 2009? The play of the Algos, HFT boats and advanced knowledge from TBT data, which is together summed up as '*Flash Trade*'. It could hit and run markets in a jiffy before the human mind could decide if it wants to partake a trade.

For us, Ajay and Susan have detailed how India's market cannot shut down without the force called '*flash orders*'. In their book, *Indian Financial Market: 'An Insider's Guide to How the Markets Work,*' which was released in

September 2008, the researchers talk about the resilience of Indian markets prior to the advent of fast-paced HFT-driven Algos.⁴⁰

“In order to obtain some insight into the resilience of the Indian equity market, we closely examine(d) the experience of two large price shocks on the equity market—the 36 percent drop in the price of Infosys over two days starting 10th April 2003 and a 12 percent drop in Nifty index 17th May 2004,” they wrote in the book.

Their study confirms that a 2009-like market shutdown is highly impossible without the distortion caused by ‘*flash boys*’ orders’. But how?

The book narrates a similar scenario like the 2009 general elections when the Nifty index had crashed sharply and yet trading was not halted as there were no HFT and Algo tools that could trigger circuit breakers with their latency and speed. This is described in following two scenarios:

Scenario 1

“General elections took place in 2004 with an unexpected outcome. In response to the negative news, Nifty dropped by an unprecedented 21 percent in the first two hours of trading on May 17, 2004. At the end of the trading day, the market closed with a largest-ever one-day drop of 12 percent. Despite such a large drop, the equity market institutions did not collapse. Instead, markets stayed open for business. What is more, the liquidity in the market remained sound. Price discovery at this time was concentrated on the index derivatives market (with the largest activity being focused on Nifty futures trading). Here too the liquidity remained strong.”

Scenario 2

“On April 10, 2003, Infosys Technologies Ltd, one of the biggest tech firms in the country, announced gloomy earnings outlook. It basically signalled that it would be unable to maintain the high earnings growth associated with

the software industry. The valuation of the market dropped sharply and dramatically. Over the two-day period, the share price of the company fell by 36 percent, one of the largest-ever price changes over two days for a major stock in India. However, the supply of liquidity on the market remained steady.”

So, if this was the case in 2004, what had changed for the markets in 2009 to have triggered the circuit breaker in minutes of opening, which led to a shutdown of trade? Unlike 2004, HFT and Algos were playing in the market in 2009 and with the TBT data that gave them advanced knowledge of trades.

That year, the *‘flash boys’* could pre-empt every order that came into the system. They had the full order book up to the last tick and kept hitting a higher price for the Nifty index on May 18, 2009 with the help of the speed bots, which led to the market shut down.

Who were these *‘flash boys’*? Ajay and Susan? Since the data they got on a platter from the NSE was something that nobody else in the markets had and it went into IFPL’s Algo bot. The wider markets had no idea of the tech tools as Bhave was aloof during his time – it was front-running to its core.

“A program trade for the 100-stock CNX-100 basket gets executed in under a second on NSE. Hence, there are no practical impediments for doing direct spot-futures arbitrage for any of the three (NSE) indexes,” their book revealed on page 64 under the chapter titled ‘Liquidity’. (xxxviii)

The book is a pure insight into what Ajay was going to do with the exclusive data they were so clandestinely gathering from NSE. All of this was ‘key’ in writing software programmes for Algo trading. The idea of how HFT and Algos can trigger circuit breakers is also shared by the researchers in their book.

“Turnover in *‘stock futures’* and *‘stock options’* is dispersed over a large number of underlying stocks. In contrast, almost all the index derivatives turnover is concentrated in one index, the Nifty. As a consequence, a full 51

percent of the notional turnover (of the entire market) pertains to one underlying element — the Nifty. The near month Nifty futures is India's most liquid financial market. It is difficult to express negative views on the Indian equity spot market (in absence of effective mechanism)."

They further write, "This is a critical role that is played by the index and stock derivatives markets: the Nifty futures and the stock futures, feature a free play between positive and negative views. Hence, in practice, the lack of a short-selling and stock-borrowing mechanism in India only impedes reverse cash and carry arbitrage. NSE operates a limit order book. (They already had a liquidity supply schedule to know the impact cost obtained for all transaction sizes). As an example, a purchase of Rs.1 billion—i.e., a single market order of value Rs. 1-billion—gets executed at a one-way impact cost of 20 basis points. A sell order for Rs.1-billion of the Nifty futures gets a somewhat inferior execution, with a bigger impact cost. (The catch their study from real-time data gave them). In the real world, a customer might seldom place a single market order of a billion rupees or roughly \$25.5 million. A customer faced with such a requirement might place a limit order at the touch (i.e., halfway between bid and offer) and wait for the market to chip away at this order. Or, a customer faced with such a requirement might dribble out five market orders of \$5 million each."

Simply, what the mechanism uncovers is that: if the transaction cost, which is the money and margin required to be put up to play the trade, is ignored and large orders placed in quick successions, it would potentially drive the Nifty crazy. This is because the markets were not being given enough time to chip away at the small trades.

Like in 2009, the buy orders all came at high latency due to HFT and Algos and Nifty hitting a circuit before the counterparties could chip away at those orders. Comparatively, in 2004 when the Nifty fell 21 per cent on election results, no circuit breaker was triggered as sell orders lacked the speed of HFT and Algos and allowed the markets to chip away at the trades.

Further, in a research paper titled '*Measuring and Explaining the Asymmetry of Liquidity*', Susan details that in 2009, there existed more probability of being able to execute a single large order on the buy-side as the sell-side liquidity was worse than buy-side liquidity.

With this, the positive news flow of Congress-led UPA election victory was already supportive and it made it difficult for traders to express any negativity. At least on May 18, 2009, the next day of election results, it ensued a scramble among traders to buy, not knowing that HFT and Algo bots were already lurking to pre-empt their trade.

Boom! There were two consecutive upper circuits that day and a market shutdown. For the Congress, it made their victory historic even though the party was short of majority on its own. Chidambaram, for whom the markets were his policy barometer, could do his chest-beating.

But a dissection of that day's two consecutive market circuits suggested a lack of wider participation in the unexampled 20 per cent rally of the benchmark indices Sensex and Nifty. This was since the index futures turnover was only Rs. 1,000 crores (roughly USD 222.22 million) and almost all of that was concentrated in the Nifty index. Two exchanges, which had thousands of billion-dollar market-caps, were being shut down in minutes via orders worth USD 300 million in cash and derivatives combined.

The May 6, 2010 flash crash in the US was a market fall that started at 2:32 PM (EDT) and lasted for approximately 36 minutes. It shut down the trillion-dollar American stock market. On April 21, 2015, nearly five years after the incident, the US Department of Justice brought in 22 criminal counts, including fraud and market manipulation against Navinder Singh Sarao, a British financial trader.

Later, Algo spoofing, layering and front-running, all of it came out during an investigation by the US market regulator, SEC and it also found that HFT traders exacerbated the price declines.

The point is that often there is a fair investigation into the events in other countries that have large stock markets. People responsible are booked, criminal charges are framed and they are tried before a jury, something which is dexterously missing in NSE's COLO and Algo scam. This, despite the fact that there exist very severe grounds of serious criminal intent, which have to be investigated and tried by a law enforcing agency and are beyond the scope of SEBI.

How Can Shallow Market Orders Really Trigger Circuit Breakers?

For the sceptics, who will question as to how orders worth a couple of hundred millions could drive the Nifty index crazy in 2009, here is a live example:

On October 5, 2012, the Nifty index witnessed 920 points or 16 per cent *'flash crash'* between 9:49 AM and 9:51 AM. The market had to be shut. While Mumbai-based brokerage house Emkay Global Financial Services Ltd took the blame for the market crash and shutdown due to a *'fat finger'* error by one of its traders, market participants had whispered about a systems failure at NSE. On July 11, 2013, it was reported by me in *The Economic Times*: "Sebi blames NSE for flash crash and over failure to put in place adequate risk management system."⁴¹

NSE had declared that the crash was not an outcome of any technical glitch in the exchange system, but a *'freak trade'* a result of 59 erroneous orders placed by Emkay Global Financial Services Ltd. But a subsequent SEBI probe found that sell orders worth mere Rs. 917 crores (approximately USD 176.34 million, considering 52 as the October 2012 Dollar rate against the Rupee) led to a fall in the Nifty index. The worth of orders in this case that caused Nifty to go crazy was even less than the 2009 incident.

The trader at the brokerage had punched an order to sell 17 lakh (1.7 million) baskets of shares belonging to the Nifty index, instead of putting through a sell order worth Rs. 17 lakhs (around USD 33,000 as per the

prevailing dollar-rupee exchange rate). As a result, shares worth Rs. 974 crores (USD 176.34 million) were pushed for sale, of which Rs. 650 crores (approximately USD 125 million) were bought by counterparties that had placed buy orders at lower levels. Within seconds, stocks plunged as a large number of index shares hit the market.

BJ Dilip, the SEBI Deputy General Manager at MRD, who had issued the notice to NSE, said that the exchange could have prevented the fall but did not. Then too, the MRD came under Agarwal.

According to SEBI, the NSE failed to ensure that system-based control on trading limits and exposure taken by clients were in place at the time of granting computer-to-computer link permission to brokers and “ensure the effectiveness of system-based controls on a continuous basis.”

SEBI also pointed out that margin obligations exceeded the collateral deposited (at the exchange) by brokers like Inventure Growth, Prakash K Shah Shares and Securities, Labdhi Finance and Focus Shares and Securities, which were the counterparties to Emkay. Margins are the initial money collected from brokers as risk management.

SEBI said that NSE had failed to implement market-wide circuit breakers too. As per norms, exchanges have to implement circuit breakers at 10 per cent, 15 per cent, and 20 per cent of the index movement. It was observed that on October 5, the applicable 10 per cent circuit-breaker limit for Nifty was at 570 points (based on the specified parameters) as the index had opened at 5,815. The 10 per cent circuit breaker for Nifty was triggered at 9:50.58 AM and NSE’s trading system broadcasted a message ‘M/C:1’, requesting the members that there would be a trading halt as per the applicable circuit limit.

But by the time the market was halted at 9:51.50 AM, the index had even breached the 15 per cent circuit-breaker limit. It was the work of the *‘flash boys’ orders’*.

As per SEBI, in ‘*six seconds*’ of the total 52 seconds between the broadcast alert and actual shutting of the market, several orders already in NSE’s trading engine got executed. This further pushed the market down.

NSE refused to reverse the trades after the internal probe committee (comprising board members of the exchange) solely blamed Emkay and systems failure at the brokers’ end for the crash. But SEBI probe is a testimony as to how orders worth few millions can cause a stock market shutdown in India. That was 2012.

But a similar thing had happened in 2009 when HFT-driven Algo orders took the Nifty index order book by storm and shut down markets on ridiculously low volumes. The inside knowledge of working of the trading systems at NSE benefited a handful of traders at the cost of wider markets. If markets could be shut in 2012 on the back of orders worth just USD 176 million, the combined order flow in cash and derivatives in 2009, which shut markets was USD 300 million. Compared to 2012, the market-wide depth in 2009 too was shallow and trading volumes higher.

In this context, it is necessary to go back to what Ajay wrote to Susan in an email, just 12 weeks before markets hit upper circuits.

“... on Day 1 Anupam is a finance guy. He is not a programmer. So drive him appropriately. He can go into all your existing projects—but in a domain knowledge roll e.g. he can start working on trading strategies which can go into all algorithm trading work. But you have to swear everyone to silence on the fact that the data that we are getting out of NSE for VIX (volatility index) and LIX (liquidity risk index) is being used for algorithmic trading work — it would be a severe problem if this fact comes to light since NSE has not given anyone else this data.”

The investigation found that IFPL had no intention of doing any work towards volatility index, it was later done by an NSE subsidiary. Availability of confidential data with IFPL and market shutdown of 2009 thus needs to be studied in the view of the ‘*tell-tale*’ email by Ajay to Sunita, which sets the

perspective in place. SEBI has gone to say that Ajay schemed to gather confidential data from NSE.

“It is also evident from the records and statement of Ajay that the LIX (liquidity index) project was his brainchild and he was instrumental in getting the LIX project awarded to IFPL, in which he and his wife had ‘*vested interest*’. Ajay, IFPL and its directors pre-planned their scheme to obtain data from NSE for which the LIX project was itself conceived by Ajay to which NSE and its senior officials agreed,” SEBI said in an SCN it issued to Narain and Ramkrishna in 2019.

Evidence showing how data was going into the creation of an Algo trading software was abundant – both at NSE and MCX. In February 2009, Ajay had password access to NSE’s servers for the data extraction. Similar to what Susan and Anand had at MCX.

“Their association with NSE had apparently transcended beyond the institutional framework as he had personal relationships with most of the senior management personnel,” based on the probe, SEBI was compelled to say in point number 14 of its SCN to Narain and Ramkrishna.

When asked by SEBI if she knew Ajay, and the rationale for NSE to share data with him prior to 2012, when there was no such contract to do so, Ramkrishna said, “I may have met him infrequently. He was known to NSE as a researcher and academician. I’m not able to recollect any arrangement for sharing data with Ajay in his personal capacity. Research and Ravi Varanasi team may be able to provide the necessary details if any.”

Ajay was ingrained into the workings of NSE’s core operations since its inception, wrote the risk management codes for derivatives, created Nifty, held board seats and committee positions in crucial exchange subsidiaries and sister companies. Yet, Ramakrishna, NSE’s Joint MD since 2009, says she met Shah infrequently and knew him only as a researcher, but ended up awarding key contracts to his associates and friends.

On his part, Varanasi said data was provided to Ajay based on the clauses in the agreement. But who had seen that agreement and what were the clauses?

Was the COLO Grid at NSE Operational at NSE in 2009?

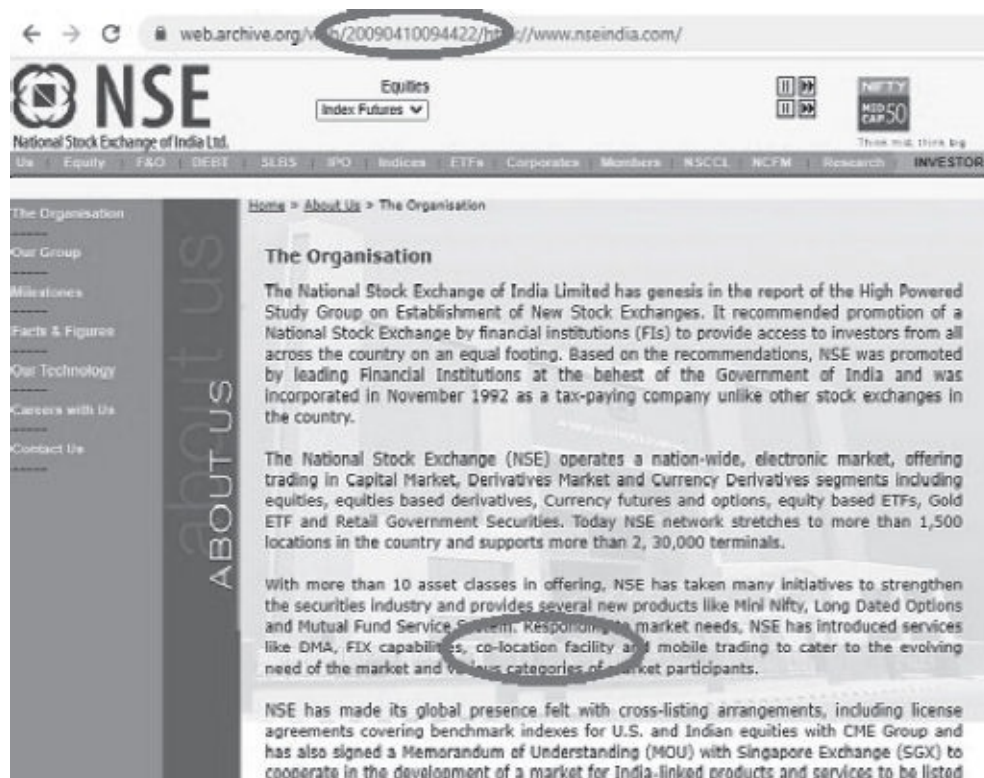
Officially, the NSE said that it became operational in 2010. But there is evidence to suggest that the COLO grid at the exchange was functional in 2009 and could have played a role in the market shutdown on the election results day that year.

Google has a '*Wayback Machine*', which is nothing but an internet archive library that stores billions of web pages that have been taken down by the operators or organisations but were created in the past. Since 1996, the *Wayback Machine* has been collecting and cataloguing websites, which to date exceeds 240 billion web pages or almost two petabytes of data. While Google may not archive every dot on a website but still whatever it records is just a snapshot of the web page and nothing is added or deleted from it.

This is why many investigators the world over trust the *Wayback Machine*, to dig out past information that was hidden by organisations to wash-off records. Similar such past record of NSE's website provides vital information about the COLO grid being operational at the exchange in 2009 too.⁴² The following is the screenshot of NSE's web page captured by the *Wayback Machine*, at 9:44.22 AM on April 10, 2009. (Figure No)

On the page, NSE boasted that it offered a COLO facility along with other tools. The circle on the top displays the date and time of the capture of NSE's webpage by the *Wayback Machine*, (2009/04/10 – 9:44.22) and the bigger circle below shows NSE's offering, which included COLO.

How can NSE's website claim to have delivered COLO trading system a year before the exchange officially went live with it?



In 2009, ahead of the mega event, which was the general election results that year, Ajay and Susan had the TBT data, tech tools of IFPL and COLO grid too at their disposal (if we go by the claims of NSE website itself), to which the wider markets were in pitch-darkness When the markets hit an upper circuit, perhaps all this was at play.

And if it indeed was, then it becomes a scandal of UNPARALLELED magnitude, since the then NSE management had not divulged to the markets or SEBI that its COLO grid was operational. The Bhave apathy factor again...

Was this the reason that Narain had offered to quit the NSE in 2009 itself?

“In 2009, I informed the board of directors my desire to retire. However, at the insistence of the board, I continued further and retired on March 31, 2013. After 2009, the day-to-day management rested with the then Joint MD Ms. Chitra Ramkrishna till I retired in March, 2013,” Narin told SEBI.

He said that between 2009 and 2013, he was travelling extensively overseas to build an overseas presence on the NSE and to explore the possibility of setting up stock exchanges in other European countries and Australia. He also became one of the board of directors of the World Federation of Exchanges and was involved in framing global policies on various issues of the securities market.

“After 2013, I was appointed as the Non-Executive director till June 2017, after which I had no involvement with NSE. Further, even as MD & CEO, it was impossible for me to be constantly involved with every detail of every department,” Narain said.

According to him, Ramkrishna was responsible for all that was happening inside NSE. But on her part, Ramkrishna put the blame on various departmental teams of NSE.

“I was in a ‘*top management*’ role with policy and strategic direction for the company. The key heads of different functions were reporting to me only post April, 2013, after I became MD & CEO. Further, I relied upon the judgement, information and advice of the functional head and had no reason to suspect the integrity and competency of these persons,” she said during an interrogation by SEBI.

How Did SEBI Nail Ramkrishna?

“I find the explanation of Ramkrishna as evasive and not acceptable,” SEBI’s Mohanty said in his report.

His reasoning was that from 2009 onwards till 2013, Ramkrishna was the Joint Managing Director and was looking after the day-to-day management of the exchange as Narain was preoccupied with his overseas assignments. Further, Narain has also stated in his submission that in 2009, Ramkrishna was elevated as Joint Managing Director so that her position was equivalent to him. She could start taking over all his functions.

Thus from the explanation of Narain, Mohanty gauged that Ramkrishna was practically in complete command and control of NSE, first as Joint MD until 2013 and then as MD and CEO from 2013 until she resigned in December 2016.

And so, any act of negligence, lack of due diligence or any actions or inactions on the part of NSE reflecting on its poor governance shall automatically reflect on the performance of Narain and Ramkrishna. As the executive head of the organisation, as the Key Management Personnel and as a Whole-Time Director on the Board, Ramkrishna is answerable for all the actions, performance or lack of performance by her functional heads, which could have a bearing on the governance of the exchange.

Ramkrishna cannot take shelter under the plea that her role was a top management role and she was not responsible for the functions and performance of her functional heads, Mohanty reasoned.

“Therefore, all the arguments advanced by Ramkrishna with a view to escape from her responsibility and accountability to maintain high standards of corporate governance, due diligence and adoption of fair, equitable and transparent policy towards the market participants in the interest of securities market, are not maintainable,” Mohanty said.

SEBI further said that Ramkrishna’s contention that she was not aware of any arrangement of sharing of data with Ajay or about the conflict of interest involved with regard to IFPL are elusive.

“It gives an impression as if she was not at all involved in day-to-day operations of the exchange which were left to the discretion of the respective functional heads. Such an explanation is nothing but an attempt to escape from her liability as the executive head of the exchange,” Mohanty said.

For all the grave fraud by design, SEBI’s punishment to Narain and Ramkrishna was that they cannot associate themselves with capital markets for a few years (five years or less). That was all that elite got from SEBI for

compromising the integrity of India's market and putting millions of investors and billions of dollars at risk. Not to mention the erosion of trust in the market.

There are several aspects of the scam that fall outside SEBI's preview and require a probe as per India's criminal procedure code, like rogue stockbrokers Harshad Mehta and Ketan Parekh who were tried for criminal charges and convicted. Considering how lightly the partners in crime at the NSE market manipulation were let off, we can only have pity on Mehta and Parekh who were the demigods of India's Dalal Street.

For a scandal that was far lower in magnitude than the data theft and COLO scam, Mehta died in prison and Parekh was banned for 17 years from markets for a case related to stock rigging by SEBI. Parekh's time in prison and CBI custody eventually made him a recluse. Special agencies recovered more than Rs. 20,000 crores of Mehta's wealth since a proper course of criminal investigation followed. Parekh returned nearly Rs. 396 crores to Madhavpura Mercantile Bank.

All this was possible as the probe moved out of SEBI's realm and professional crime detection and money laundering agencies were entrusted with the job. On the other hand, when the top executives left NSE; they took home millions in salaries and variable pay, not to mention the fringe-benefits.

"As far as NSE, Narain and Ramkrishna are concerned, there is no evidence found from the records to suggest that they were aware about the hidden agenda of Ajay. Therefore, I have no adverse finding against them in respect of their culpability. With regard to alleged misuse of data by Infotech, NSE has said that misuse of data by Infotech would be at best called a matter of contractual breach and it cannot be held liable for such misuse by Infotech. I agree with the contention," said Mohanty in his snap judgement. Whose tune was SEBI singing here, and why?

There was no estimate of how much Ajay, Susan, IFPL benefited from

data or his network. Were the NSE bosses and SEBI also in the dark about the fact that the place where IFPL was registered at 208 Sungrace, Raheja Vihar, Chandivali, Andheri (E), was actually the residential address of NSE's VP Suprabhat Lala?⁴³

Lala was not a small fry; he was NSE's head of trading and even of other departments. NSE never did its due diligence even when it gave the project of writing its derivative risk management code to IFPL? The question is, why did the two bosses never feel the need for it?

"The hubris of our people has hurt the institution," a senior NSE official told the press. (xli)

When Ajay was milking the data from NSE in 2009-2010, he was also a member of the Finance Ministry, and constituted a high profile working group for suggesting policy on '*foreign investments*'. The report gave vital policy insights with regard to FIIs, Foreign Direct Investment (FDI), Private Equity and Foreign Venture Capital Investments in the country. Its highlight was the study on P-Note policy, something which could potentially decide the market direction.

"One of the few economists known in the capital market who writes policy papers based on data analytics," Narain once said with regard to Ajay.

Surely, by virtue of his high patronage, Ajay had the code to NSE's data goldmine and vital knowledge of government policy that could shape the long-term direction of markets. The working group's report was submitted in July 2010 when NSE was also operating COLO farm. Besides, there was too much muck in the COLO trading. Missing emails, lost trails, contradictory statements and ominous silence of exchange employees mark the forensic audit that highlighted clear possibilities of NSE's total disregard for rules and running a national exchange without well-laid-down protocols only to favour and give preferential treatment to a few brokers.

THE 'HEART OF DARKNESS'

The Art of '*Deal-Making*'

Nobody understood the real game of NSE's top brass when the exchange purchased a 26 per cent stake in Omnesys Technologies Private Limited in 2008 and inducted Chitra Ramkrishna on its board as a shareholder director.

For the onlookers, it seemed like a move by the NSE to ward off competition from FTIL's *ODIN* that held a near monopoly in providing trading solutions to equity brokers. So the stake purchase in Omnesys Technologies was perceived as an ideal deal by NSE to compete in the segment with FTIL. But as has been the usual case in the story so far, there was more to it than met the eye in the deal between NSE and Omnesys.

Bangalore-based techie, Shrikant Pandit, had set up Omnesys as a stock market trading solutions company in 1997. When asked for his view on the deal with NSE, Pandit, also the CEO of Omnesys, told the media that his company had a confidentiality agreement with NSE, which prevented them from divulging anything about the deal.⁴⁴

The question is, why should a legitimate business partnership, entered into by a premier stock exchange be mired in such secrecy?

Matter of fact, adhering to standards of prudent disclosure norms,

features of the deal and relevant details should have been voluntarily released as public information. So why was Pandit tight-lipped about it. Was it because it involved a grave conflict of interest? The answer as we will understand is, yes.

It would be interesting to know that Omnesys Technologies had a sister concern company called Omnesys Tradenet (OTN) Private Limited, which was actually a '*stock broking company*' playing on the NSE platform.

Its balance sheet shows that OTN earned a brokerage of Rs. 6.01 crores in March 2008, Rs. 3.25 crores in March 2009 and Rs. 4.09 crores in March 2010 from the equity segment. Usually, brokerage fees can be as high as 0.5 per cent on the trading turnover of a client. To generate that amount of brokerage, which OTN did, the company must have churned hundreds and thousands of crores' worth of turnover in equity market trading on the NSE. Omnesys also had another sister company called OTN Commodities, which was a member of NSE-sponsored NCDEX among other exchanges.

Pandit and his family members were key managerial persons and shareholders in Omnesys Technologies as well as OTN and OTN Commodities. Apart from OTN and OTN Commodities, the key shareholders of Omnesys also had an interest in other stockbroking companies. Gana Yantrika Systems Private Limited, which was the holding company for Omnesys Technologies, was also the holding company for OTN and OTN Commodities.

After the acquisition, NSE's Chitra Ramkrishna was nominated on the board of Omnesys Technologies as a director. Perhaps it would not be far-fetched to say that there cannot be a conflict of interest more blatant than this in India's stock market history. The key insiders of India's largest stock exchange, privy to a lot of sensitive market information, were in material relation with brokers, trading software developers and data vendors.

Nowhere in the world has there been an example of a stock exchange becoming a shareholder of a company that had its group company running

equity and commodity broking business. Forget the tenets of corporate governance: accountability, fairness, transparency etc., it was just against the rules of *'fair play'*.

But then NSE, supposedly India's young, tech-savvy, most professionally run exchange, was setting new standards in the industry. Also with SEBI, ostensibly the super-cop of the financial market totally turning a blind eye, the head honchos at NSE was obviously free to dabble in whatever shenanigans they were up to.

However, to his credit, Bhave never appeared unaccountable on maintaining a strict vigil over other exchanges and even disciplined them, often. High ownership of foreign investors and domestic business houses was a threat to exchanges but the dubious dealings of stock exchange officials, insiders and their links to the web of private companies was aiding the stock market ecosystem. Even when SEBI had a chance to investigate all of it, post the whistleblower letter presenting an opportunity, the regulator goofed-up, this time under Ajay Tyagi and, of course, SK Mohanty, who was then looking after the MRD.

The nexus between NSE's management and brokers ran deeper.

"We draw attention to Note 19 of the notes to accounts, that the company (OTN) has sold its entire share broking business to M/S Way2Wealth, retaining its membership with NSE & BSE," said a note dated August 5, 2010 from KS Ranga and Co, the auditors for OTN.

It arouses further curiosity when you see that the stockbroker Way2Wealth Securities, in turn, became a client of Omnesys Technologies, which facilitated it to trade using NSE's COLO grid. Later, it came to be known that Way2Wealth was among the brokers that got *'preferential access'* to NSE's COLO farm. An SCN was also issued to Way2Wealth as its name cropped up prominently in investigations post the whistleblower exposé.

After the story thus far, it will be a little surprise to know that

Way2Wealth was also a client of IFPL and used its trading software Chanakya to trade on NSE. Still, SEBI has never probed the companies thoroughly, that facilitated preferential access to several brokers.

So why did Way2Wealth agree to buy OTN's broking business? What did it gain from the deal? What did Omnesys, Pandit, Ramkrishna, NSE all of them gain from the deal with Way2Wealth?

The broking outfit Way2Wealth was owned by the Coffee Day Holding group, which was promoted by a politically exposed person VG Siddhartha. He was the son-in-law of former Karnataka Chief Minister, SM Krishna, another former Congress party veteran, who later joined the BJP. Karnataka was also the home state of Pandit and Omnesys. Did any of the Omnesys shareholders have a political link?

Way2Wealth also had a subsidiary called AlphaGrep, which was exclusively into HFT and Algo trading, with offices in Mumbai, Bangalore, Singapore, Hong Kong, London, etc. Alphagrep is said to have got '*preferential access*' to NSE's COLO grid. Where was the money moving globally?

SEBI has silently buried all these aspects of NSE's dealings with brokers and never actually went into the shareholding of Omnesys or it has not even probed the company. The regulator simply declared in its final verdict that no '*fraud*' angle comes out of the COLO scam and everything that happened was just '*procedural lapse*'.

A very senior merchant banker who was my regular source of information, and who was keeping a very keen eye on SEBI's anticipated actions told me very disappointingly, "For far too long SEBI has lacked any sense of accountability."

That was about the broking business, but there are more dimensions to NSE's deal with Omnesys. For more than a decade, Omnesys Technologies did not see much pick-up in its business and was loss-making. But it

witnessed an 180 degree turn in its luck in 2008 after NSE picked up a stake in it.

Before the exchange came in as a shareholder, Omnesys had accumulated a loss of around Rs. 4 crores, but in the following year, i.e., for the financial year (FY) 2009-2010, the pre-tax profit of the company stood at Rs. 9.5 crores. In the following years, the pre-tax profit of the company rose to around Rs. 15 crores and higher. There was no looking back. In 2013-2014, NSE and Pandit-led shareholders took an exit from Omnesys and the company was sold to Thomson Reuters for a total value of around Rs. 240 crores.

A stupendous rise in valuation in a matter of a very short time. And the story does not end here. After its founder shareholders and NSE exited, suddenly, Omnesys was again staring at a downward spiral in its luck. It reported a loss of Rs. 12.6 crore by March 2014. (xli)

Omnesys was among the first trading solutions providers that could connect brokers to NSE's COLO facility. In 2008, the NSE put FT's software ODIN on '*watchlist*' and did not allow it access to its currency trading platform or get any other new client for any segment. But Omnesys got that access as soon as NSE bought a stake in it.

While the markets were focused on its currency segment war with rivals, the NSE started its DMA scheme in 2008 under which came the tech tools for trading and connecting to the exchange platform. Omnesys became a pioneer in the tech tools even as FTIL remained blocked by NSE in any of the new segments.

At a 2009 technology conference, James Shapiro, BSE's marketing head and the former NYSE executive, said that NSE had illegally blocked its Algo trading customers from accessing trades at BSE. The NSE was super quick to respond, it retorted by saying that it involved a security risk.⁴⁵

But what was that risk? It never came out with any explanation to that

effect. Effectively, this meant that Omnesys was among the very few large software firms that could give brokers access to NSE's DMA facility and COLO trading grid. In 2010, some of the publicly known names among brokers like Edelweiss, SMC, Kotak Securities, Motilal Oswal, Enam Securities and discount broking firm Zerodha were the key clients of Omnesys.⁴⁶

In fact, Zerodha has now become the largest discount brokerage and has Pandit of Omnesys as its board member and key managerial person. Zerodha also does proprietary and Algo trading.

Deutsche Bank, Citi, Morgan Stanley, Goldman Sachs and MF Global were among the top foreign broking firms that were trading at NSE's COLO from the beginning. In 2008, when NSE started DMA, top global players signing up for it included UBS, Morgan Stanley, JP Morgan and DSP Merrill Lynch. Domestic retail players had no access to it. Foreign players do business with local brokers to trade on NSE and BSE. In turn, some of the local brokers were using Omnesys.

“Omnesys was the market leader with its DMA product being highly popular on the institutional desk. Its Algo trading solution was also popular among the domestic proprietary trading firms. It is important to note here, that since NSE was the second largest shareholder of Omnesys and worked closely with them for several projects, there was a natural bonhomie between the COLO staff of NSE and Omnesys than any other independent software vendors (ISVs). Another major domestic ISV was Greeksoft,” the whistleblower informed SEBI, virtually pleading the regulator to probe all these links. Of course, SEBI did nothing more than just scratching the surface.

As seen earlier, NSE kept the stake purchase deal of Omnesys as a closely guarded secret, which otherwise was the duty of the exchange to reveal to the markets as to what it intended to do with its trading technology.⁴⁷ What then was the reason behind a sudden rise and fall in the profits of Omnesys?

The golden run of Omnesys, as per its financial results, coincided with the period when NSE was using a COLO system architecture that released TBT data, which was '*advanced knowledge*' of price and order information, sequentially. (xli) It meant that the first to connect to NSE's servers would receive the information first. Just around the time when NSE was to change its COLO architecture, Omnesys was sold off to Thomson Reuters. The company was once again a struggling enterprise and fell into losses after NSE changed its COLO infrastructure.

Among other things, the NSE promoted Omnesys also had one of its key clients as New Delhi-based proprietary trader OPG Securities, whom SEBI had named as one of the key beneficiaries in the COLO scam. At some point, OPG was also a client of Ajay's company IFPL.

Inside the COLO Farm

"NSE tick-by-tick (TBT) architecture was prone to abuse thereby compromising market fairness and integrity, in that it provided quicker order dissemination to those who managed to login early."

"NSE has not fully co-operated by not giving timely response, or not responding at all, and not deploying enough resources to answer all queries in a timely fashion."

Technical Advisory Committee (TAC), SEBI

"System architecture of TCP based TBT system was prone to manipulation."

Deloitte

"Speed of some of the TBT Servers of NSE were higher than the others."

"The members who consistently logged in first preferred the faster servers. The means by which these members received the information

regarding the faster server needs to be examined in detail.”

SEBI CFT

“Oral instructions of server allocations and other changes were often received from seniors.”

Abhishek Soni, COLO Employee, NSE

“Alleging fraud against the exchange, in this scenario, (is) tantamount to attributing intention or knowledge of which there is no proof.”

WTM, G Mahalingam, SEBI

Like the senior SEBI official SK Mohanty, Mahalingam was another WTM in the regulator’s office who was involved in issuing a final verdict in the COLO scam.

Before he joined SEBI in 2016, Mahalingam spent most of his life working in the banking sector. If one has to go by their diagnosis of the COLO scam, it can be summed up that both the SEBI officials lacked a keen sense of deductive skill. They could only see, but they did not observe that the tail was wagging the dog. It was a murky world, where latency traders were obsessed with TBT data and NSE gave it to them on a platter. Along came the tools that also gave these traders the speed of trade execution.

As discussed, SEBI was blind to the intricacies of the COLO-based trading, which was laced with numerous conflicts. But NSE officials further complicated the matter by choosing favourites even in the COLO farm. For years, privileged brokers got advanced market information and nobody had wind of it, until came the whistleblower.

“Smart guys had figured out that the way to game the system lay in being the first to connect to the (NSE) server and preferably the one that was the

fastest. A server could be fast due to lesser load or it could be that the hardware of one could be more powerful than the others,” the whistleblower said.

Oblivious to the high-tech grid, domestic retail investors and mutual funds were busy in their plain vanilla game. Little did they know that the knowledge of the buy or sell orders that they were punching, reached the operators in NSE’s COLO farm. This knowledge of the order book, which is TBT data, was fully exploited by the HFT Algos that were in control and it was all a hand-in-glove operation.

Reports after reports pointed out that NSE’s COLO data dissemination architecture was faulty. But instead of joining the dots, such as linking NSE’s defects in the COLO, which lacked essential safeguards, with the sharing of key data with conflicted researchers and brokers, SEBI broke down the colossal case of *‘preferential access’* into several isolated instances and concluded that there was only *‘procedural lapse’*, which cannot be termed as *‘fraud’*. The regulator forgot that most financial market scandals have emerged out of exploiting the loopholes. Too much autonomy to SEBI officials has made them their own judge and jury, who have the luxury to selectively decide on the due course of action when they see the face.

A look at NSE’s TBT data dissemination system prevalent in its COLO farm before 2014, reveals the plot.

The exchange used TCP/IP (Transmission Control Protocol/Internet Protocol) for delivering TBT data feed. TCP/IP is an internet communication language that allows one computer (in this case NSE’s server) to talk to another (broker’s server). This way NSE sent the data packets to the rightful broker server. NSE had two kinds of data to share: the usual stock price and market change data that goes to receivers including television news channels, wire service providers like Bloomberg and also brokers for display on their trading screens.

This feed sharing was limited to 2 Mbps based on NSE’s leased line

network capacity. The other data was the coveted TBT, which was also being shared through TCP/IP in COLO. TBT data was the real-time change in price, the quantity of the stock up to the last tick in the order book, which remained hidden to the wider markets but was revealed only to the favoured few in the COLO farm.

The biggest flaw of TCP/IP was that it sent out TBT data sequentially, meaning first come first serve basis. It thereby bestowed an unfair advantage on members who were ahead in the queue in receiving the information packets. By hook or crook, brokers attempted to login first to NSE's data dissemination server. Those who got the TBT data first, their Algos could react first. The allocation of ports and servers to trading members, crucial to their position in the queue, was left to be decided at the whims and fancy of NSE's lower-level COLO staff – but as an employee pointed out, the oral instructions always came from the top.

“If I was first to connect to the TBT server in the morning and subsequently 40 other guys followed, the exchange sent out a price packet, it would first send that to me and then to others in the order of connection. This cycle would be repeated for each and every price information sent, which makes me ahead of the other 40 guys throughout the day. The only redeeming assumption is, next day I would not be the first guy and so across days the advantage would even out. This is a crucial point to appreciate as it forms the crux of market manipulation,” the whistleblower revealed.

TCP/IP is connection-oriented – once a connection is established, data can be sent bi-directional. Unlike the NSE, global exchanges followed the UDP (user datagram protocol) model, in which multiple messages are sent as packets in chunks to all, irrespective of their position in the connection cue.

NSE eventually switched from TCP to UDP. But those who knew the nuts and bolts of the system earlier reaped the benefits of its flaws for years. The flaws were definitely not a mere procedural lapse as underlined by the probe reports.

Who Designed NSE's Defective COLO System?

The making and functioning of technology at any exchange is the exclusive domain of the chief technology officer (CTO). Ravi Apte was NSE's CTO between 2007 and 2013. Umesh Jain, who was to replace Apte, joined NSE in September 2012 as a senior VP. After joining NSE, he was neither made in-charge of technology-related functions at NSE nor did all technology-related employees report to him or Apte.

"I was surprised to find out that none of the staff belonging to the technology team reported to Ravi Apte. Only the business solutions group (development team) of NSE reported to Ravi Apte. All the staff belonging to the technology team reported to N. Murlidaran, the CEO of NSE Infotech Service Limited, the wholly owned subsidiary of NSE, who in turn reported to Chitra Ramkrishna, Joint Managing Director at that time. No documents/files were handed over to me by Ravi Apte and/or N. Murlidaran during the handing over process," Jain told SEBI in his deposition.⁴⁸

Jain, who was a new entrant at the NSE and not part of the old gang, saw that Ramkrishna played a key role even in technology-related matters through Murlidharan, who was the head of NSE's technology company NSE IT. When he joined NSE, software development and design was under Murlidharan, who reported to Ramkrishna, Jain said. Also, Apte had told SEBI that he had facilitated the transfer of data, related to equity orders and trades, to Ajay using Murlidharan's team and on the request of MD and DMD.

"Despite formal roles in the technology team, the technology related functions of NSE were scattered. There were other employees in NSE who were made in-charge of certain technology related functions and such people carried it out along with /as part of their own department/ roles and they reported directly to Chitra Ramkrishna and at no point of time to me," Jain said.

After Jain was elevated to the role of CTO, he argued for changing the TCP/IP based COLO architecture to Multicast TBT (MTBT) dissemination. Unlike TCP/IP, where the data packets are received on a first come first serve basis, the MTBT ensured that the packets reached all those who were connected through the COLO grid simultaneously.

“I observed that the governance and management framework was completely missing at NSE. There was no strategy planning, risk management, quality assurance or control. None of the customer management, architecture framework and process were in place,” said Jain.

But when Jain could see so many flaws in the COLO system, why were those working at NSE for a long time silent? Also, who designed the TCP/IP based architecture with inherent flaws that could be exploited?

“As regards selection of TCP/ IP over MTBT, I was not involved in the detailed design and selection. However, I opined in its favour,” Apte told SEBI.

In effect, it simply meant that *‘officially’*, the COLO farm’s architecture could have been selected only by Narain, Ramkrishna and Muralidharan. But Narain said that since 2009, after he had elevated Ramkrishna to an equal position, he kept away from the day-to-day functioning of the exchange. That left Ramkrishna and Murlidharan as directly responsible for the selection of the TCP/IP architecture, which of course came with its inherent flaws. Yet, when Narain was stepping down from his position as MD and CEO, he insisted and even made sure that the board of the exchange selected Ramkrishna for the top job.

SEBI believed what it wanted to, but the design of the COLO farm revealed its own tale. Every step seemed meticulously planned by those who had the nerve and the knowledge of the stock exchange operations. The forensic probe pointed out that undoubtedly, brokers got preferential access through the COLO farm. And even though the high-tech grid was built inside its own premises, the NSE had no control over it.

“NSE does not have regulatory jurisdiction in this matter and no penalty structure. The exchange and broker in COLO had a lessor-lessee relationship. The landlord (lessor) does not have jurisdiction on underutilization or non-utilization of leased assets,” the exchange told investigators. Brokers were free to operate as per their convenience when they rented the rack space inside NSE premises.

‘Officially’, the NSE started selling COLO rack space in 2010 and it was done in three phases. Each rack could handle 2 to 4 broker servers. The exchange sold a single rack for Rs. 22.5 lakh and half rack space at between Rs. 10 lakh to Rs. 16 lakh. A total of 110 full racks and 137 half racks were sold by the exchange. The rack rate was halved sometime in 2013. Still, the NSE easily earned more than Rs. 50 crores (approximately USD 9 to 10 million) by conservative estimates in just renting the rack space at its premises.

“The COLO facility helps in faster refresh of market data as well as execution of trades. This will benefit brokerages using Algo strategies,” Muralidharan admittedly told the media in a marketing pitch when NSE was selling the rack space.⁴⁹

An exchange that swore on transparency, never actually revealed the precise data with regard to who among the foreign and domestic entities rented the rack space.

Were any of the P-Note issuing FIIs, also in NSE’s COLO? Perhaps, no one knows. Although brokers kept repeating that Algo trading was not yielding them much profit, the NSE racks never remained empty.

The exchange provided a ‘*guideline handbook*’ that described processes and procedures for a broker to ‘*ensure discipline*’ to use the infrastructure. It talked about security, controlled access, formats to be used, escalation matrix, etc.

Deloitte, however, said there were no documents, policies and protocols

about the functioning of COLO, TBT system, connection or allocation of servers to brokers, etc. Most information was provided to the auditor orally and often information shared differed in various discussions. Deloitte also could not find any historical information due to the absence of protocols on data retention, email and other information for certain key employees of NSE. It was all indeed an '*Orwellian*' state of affair.

“In the absence of a specific policy and operating procedure it appears that NSE relied on broker undertakings,” Deloitte said.

The catch was that NSE's COLO area staff provided server and login time information to brokers secretly on many occasions. The staff even fixed the time at which NSE's TBT would be started in the absence of documented policies.

“For some strange operational reasons, NSE did not start its TBT servers at a fixed time every day. First, it could not physically start all TBT servers concurrently and unlike the market, which opens at the same time for all persons, the TBT connection was established before the market opened, typically one hour before, but the time was not fixed. This was the first of the many steps, which cemented the bond between NSE's data centre staff and OPG. Every day, he (OPG's promoter Sanjay Gupta) would be privy to the information as to which server would be started at what time so that he would be the first to connect and enjoy the advantage,” the whistleblower revealed in his extensive letter.

“Emails reviewed suggested certain members may have received advice from someone within the exchange that there was an advantage in receiving market feeds on early logins to TBT servers. We also found that in various instances, statements given to us by the NSE team conflicted with emails that we came across in the course of our review,” Deloitte made its revelations.

There were several emails found during the probe where brokers were inquiring about the advantages of early login with COLO staff much before

everyone else could come in.

“During our discussion with members of the NSE IT team, we were given to understand that the TBT application was manually started at around 7:30 a.m. on trading days using a script written on Epsilon server. A broker could only connect, once the application to the server was started by the PSM (Product Support Team) member,” Deloitte said.

The person responsible for starting the server manually should be able to say who asked him to follow a certain time and passed that information to the brokers.

Deloitte said there were no logs or records pertaining to Epsilon server to understand the process or sequence of applications that started the servers. Epsilon server is an internal networking and communication solution. Random checks by SEBI’s CFT team had revealed that the starting time of servers in COLO varied from 1 second to 10 minutes. Information on the exact time of starting of the servers was the key for brokers.

“COLO staff provided server and login time information to brokers. The staff even fixed the time at which NSE’s TBT would start,” the CFT report said.

Investigations also revealed that servers giving out TBT data at NSE *‘were not time-synchronised’*. The NSE submitted that they intend to have equipment that were time sync protocol compliant, in the next four years (CFT gave its report in 2015).

Additionally, NSE also said that all servers were deriving time sync individually through a common independent source. Due to the lack of synchronisation among the servers, the time logs provided by NSE had to be treated in isolation.

In simple words, NSE servers could have had a different time clock. What was 7:00 AM for one server could very well have been 7:02 AM for another. Only the insiders knew exactly which server was ahead. The brokers who

logged in early got the TBT data first throughout the day and could hit their trade ahead of others. It was this early login when seen with faulty TCP/IP infrastructure became a tool for manipulation.

If one considers the intricate planning behind such a complex plot, Harshad Mehta and Ketan Parekh really look like *Lilliputians*.

“As per your suggestion to connect to TBT servers as early as possible, we would connect to TBT servers by 7.30 am onwards and give you the feedback,” broker AB Financial Services wrote to NSE’s COLO support staff in an email.

“Early TBT login will be given preference in trading feeds, we have few queries regarding this process,” said another email from IKM Investors to the COLO team.

Jagdish Joshi, the head of NSE’s COLO responded on the same day to both the emails, refuting the theory of early login advantage and even said that TBT connected randomly. But the same Joshi remained silent when OPG and some other brokers were favoured.

But to a question from SEBI on what were the *advantages* of early *login*, Apte said, “Yes, there could be advantage of a few microseconds.”

“Over period of time others figured out the importance of this (early login) and did what all geeks do. They wrote a piece of code which would continuously ping to check if the TBT server was on and attempt to connect immediately when it got the first response from the TBT server,” the informer mentioned in his letter addressed to SEBI.

Specifically, about Omnesys, he further revealed that the tech company had launched a managed data centre service for its Algo clients. Under this, Omnesys claimed to provide some custom hardware and software as a result of which it guaranteed superior performance over other service providers.

“But the crux of the edge to Omnesys was its knowledge connecting

early to the faster server that would put the trader ahead in the queue. In fact, it became so popular that Omnesys started profit sharing for the gains made from its system. This was obstinately billed as man-hours for managed services so that it could vary every month. It was a very clean way to earn out of their knowledge. Everyone just believed it was the difference in the software of Omnesys, which gave them an edge. Whereas it was simply front-running,” the whistleblower said in his most shocking revelation.

As in other matters, why SEBI did not investigate Omnesys remains a mystery. Instead, the regulator made Sanjay Gupta and OPG the kingpin. There seems little doubt that the broker-cum-proprietary trader was complicit, but SEBI’s absurdity and a desk job at the probe shielded some of the other large players who could have benefited from the scam.

The SEBI constituted CFT team had also observed that NSE was using machines, which had inbuilt speed variation. It said that three different server models were deployed by the exchange for TBT data dissemination, which had varying clock speed and processor memory. Naturally, the question arises, who gave the brokers the idea of the fastest TBT machine?

“As per the server manuals, the memory speeds of (Hewlett Packard Enterprise ProLiant) Server Models DL-380 G5 and DL-380 G6 servers are different. The server manuals state that the RAM used in DL-380 G5 were DDR2 which had a speed of 667 Mhz whereas the RAM used in DL-380 G6 are DDR3, which have a Speed of 1,333 Mhz and in certain cases 1,067 Mhz. Hence, it prima-facie appears that the speed of (TBT) servers 23, 24 and 26 were higher than remaining servers,” CFT said.

The whistleblower information on early login as well as some TBT servers being faster than others was point-blank.

Plumbing, Patchwork and Secret Servers

“Implementation of ‘*Load-Balancer*’ is required on a priority basis,” said

NSE's COLO employee, Smrati Kaushik, in her email to most of her colleagues in the technology team highlighting the danger of running without adequate check and balances.

Load-balancer is nothing but a technology that re-distributes network traffic across multiple computer servers. For instance, when a busy website like Google is witnessing incremental or high traffic, its *load-balancer* will distribute incoming search requests evenly across its multiple servers that will execute the task and provide the resolution on your browser. Imagine the situation without an auto *load-balancer* that Google can face – it can slow down and even crash due to the extra burden or load on the systems. Such auto *load-balancer* was missing in NSE's COLO grid, which handled millions of trade messages per second. Was this procedural lapse?

Probes revealed that load-balancing was done manually by COLO support staff and brokers got a preference in this. The information of the server with the lowest load was passed on by the exchange staff to brokers, who connected to the fastest machines very often.

“This issue was not escalated to me. A dynamic *load-balancer* was not envisaged at that point,” said Murlidharan.

This is as ridiculous as a trained doctor saying that the matter regarding the requirement of oxygen cylinder in the operation theatre during surgery was not escalated to him and it was not envisaged. It's like saying that the doctor never felt it would be required? Likewise, Murlidharan was a trained technology doctor and he was saying that he was not sure if the critical safeguards were needed or not.

Murlidharan's spin though did not last long. According to Deloitte, agenda items and minutes of NSE's meetings showed that they took note of the fact that dynamic *load-balancers* were configured into the network but not implemented in the TBT architecture. The records of this can be found in the NSE tech committee's 28th meeting held on August 7, 2012. Yet SEBI

did not take any cognisance of this, the regulator in its wisdom, was not sure if this was a procedural lapse or not?

“The absence of *load-balancers* appears to have created advantages for certain brokers. We were informed that load-balancing was done manually,” Deloitte said in its report.

Shockingly, the auditor also found additional servers in the COLO farm, of which nobody in the NSE had any inkling of.

“After NSE confirmed completion of data restoration exercise from all TBT servers during the audit, we came across additional TBT servers, which had not been put through the restoration process. NSE team was unaware of them. We also found that in various instances, statements given to us by the NSE team conflicted with emails that we came across in the course of our review.” Deloitte’s report said.

It was like an additional currency printing press lying in a government mint, which was kept well oiled and covertly used but nobody was mindful about it. Ludicrous...! But true.

Who was using these hidden servers in NSE? If this was not a criminal act, what was? India’s largest exchange had been hijacked by traders, data thieves and wily executives, and as you would have guessed it by now, no one was taking note, definitely not the super-cop.

Similar to the ‘*load-balancer*’, there is another technical feature called ‘*randomiser*’, which indiscriminately picks a connection to begin dissemination of data, rather than starting with the first connection each time. But the ‘*randomiser*’ too was missing and NSE’s COLO was exposed to brokers connecting early. Though the NSE had developed ‘*randomiser*’ in 2011, it was not replicated on the broader TBT systems, the probe said.

The Mystery of the ‘*Secondary Server*’

As always, there is an interesting play between '*Primary servers*' and '*Secondary servers*', which also existed in the scheme of working of an exchange. *Secondary servers* are an emergency backup machine supposed to come into play only when the *primary* goes down. Temporarily, the load is shifted to the *secondary server* until the *primary* is back and running. Since this shift occurs in '*milliseconds*', it remains unnoticeable. But even when not in use, *secondary servers* keep running alongside the primary machines as they are needed to be ready to take the load immediately as and when required. Since *secondary servers* are light on load until in full use, they remain miles ahead in terms of speed when compared to the *primary servers*. A select few with patronage in the NSE COLO farm were enjoying the party via *secondary server* – front-running the markets by milliseconds.

“NSE told us a secondary server was in place since 2010, as a contingency measure for members, in case the primary server failed for any reason. TBT-LV9 was the secondary or fallback server until January 31, 2012, when TBT COLO-27 was introduced to replace it. We were given to understand that the secondary server was also an active server and there was no system, whereby the secondary server would start only when the primary server failed or was down. Brokers could log into secondary servers anytime,” Deloitte revealed.

When a broker took up a new TBT connection, NSE sent activation emails with information of both *primary* and *secondary servers* and even off-line arrangement to the brokers. Again, there was no documented policy that the exchange had regarding connecting to the *secondary server*. Also, there was no documented monitoring mechanism to determine if brokers connecting to *secondary servers* had any legitimate reason. So could it be a procedural lapse, well, SEBI needed to have taken note of this.

“We noted an email by Bhavya Gandhi who wrote to the COLO support team providing a list of 24 IPs of nine brokers, stating that they were connecting to the secondary servers. The email requested COLO support to

ask the brokers to connect to Primary Servers. This indicates that there was a mechanism to monitor connections,” Deloitte said.

Prior to 2013, there were several emails that indicated how senior systems analyst Gandhi kept warning brokers against connecting to the *secondary server* as it was against the rule. Gandhi also kept alerting the other COLO members about the dangers of brokers randomly logging into them, though it did not cut much ice with seniors. Often, brokers wrote emails to key COLO officials asking them to move their connections to the *secondary servers* at their own will.

Another COLO staff member, Sunny Sachdev told Jagdish Joshi that they had clear instructions to not move any clients to other TBT servers, without appropriate justification and approvals. He also requested him to seek evidence of performance degradation from clients. On an occasion, a broker called ‘*Open Futures*’ was deactivated from a particular server to accommodate OPG.

The auditor said it did not come across any correspondence to brokers in 2013, that discouraged them from connecting to the *secondary server*. Thus, it appeared to Deloitte, “That access to the secondary server was not possible without the knowledge of the NSE team.”

“OPG wrote to Jagdish Joshi, requesting to be moved to a different TBT server, citing performance degradation. (Ironically), the email was not marked to COLO support that was tasked with handling members’ queries. Joshi instructed the COLO team to move four of OPGs IPs to a different server. The COLO team resisted, told Joshi that there was neither proof of performance degradation nor approvals but he insisted that this was to be done to mitigate risk. Eventually, the team deactivated the IPS of another member to accommodate OPG. Server logs also indicated that after these IPS were migrated, OPG was successful in frequently establishing first 5 connections on (the fastest) server,” Deloitte said.

The auditor was not done with its revelations. Nearly two years after

submitting its first investigation report, Deloitte another startling disclosure. “NSE had provided incorrect and partial data. The same was even agreed by NSE officials during the meeting held on 16th May, 2018 and in submissions made by NSE on 22nd May, 2018.”

NSE officials told Deloitte that they had withheld crucial information earlier about how two of the servers in the COLO data centre were interchangeable and used as primary and secondary servers between January 2011 to February 2012. Based on this, Deloitte concluded, “There was no way to ascertain which member was connecting to the primary or secondary server.”

The entire system ran on the whims and fancy of NSE officials. Deloitte has repeatedly pointed in its report that the exchange lacked a clearly documented policy on every aspect in COLO. There were absolutely ‘no set procedures’ that should have been in place for such a vital installation. But strangely, SEBI still felt the COLO scam at NSE was just a procedural lapse.

According to the TAC, none of the circulars issued by NSE since 2009 with regard to TBT data sharing or COLO had any details about the backup server. It was a trading system designed to be exploited. Only a few insiders had the means and the ways around them. In that, NSE’s unregulated COLO trading facility, not only defeated the ‘principle of fair access’ it also challenged SEBI’s ability to be vigilant.

Deloitte submitted its report to NSE in December 2016. But days before this, SEBI’s Rajiv Agarwal, the WTM who was diligently following the trail and asking pertinent questions to the exchange, lost his job. In the meantime, NSE quietly discontinued its TCP/IP based infrastructure even before SEBI could examine Deloitte’s report.

Also, post the submission of the probe report, Abhishek Soni, the COLO employee who had revealed to Deloitte that ‘oral instructions for server allocations came from seniors’ was asked to quit.

Since Deloitte could not find any historical information due to absence of protocols on data retention, email and other information of key employees, the trail to NSE's top brass was cut short. It is unknown if personal emails records of employees were considered during the probe, as discreet conversations often happen there.

Deloitte disclosed that it could not gain access to emails or other documents of some crucial employees including Apte, Joshi, Mamtha Rangaprasad (Associate VP at NSE between April 2010 to April 2013) and Nimesh Vaghasiya (system analyst at NSE Infotech Services between July 2007 to September 2013).

All these key people worked at NSE at the most crucial time when TBT data was being stolen by Ajay and up to the time when developments of granting preferential access to brokers at NSE's COLO data centre was at its peak. The trail, which could have led to the top, was obviously cut. After Jagdish Joshi left NSE in 2014, instances of OPG connecting first to NSE's server fell drastically.

Deloitte remarked that operating systems in the computers used by several employees were installed fresh in 2016. These included Soni, Devi Prasad Singh (NSE network infrastructure and security expert since 1996 who is currently the VP and Head of IT Operations), Arvind Goyal (who has managed derivative trading and membership operations at NSE since 2004 and currently heads the membership department), Hari K (NSE employee since inception and who was responsible for memberships) and Viral Mody (another technology expert who has been with NSE for more than a decade and is currently the VP at the exchange).

Also, it is unclear whether COLO chief Joshi or other members of the IT team had discussed the server issues with senior officials like Suprabhat Lala who was in-charge of trading and even customer relations until 2011 (he was also an official with a serious conflict of interest with NSE since IFPL's trading software *Chanakya* was manufactured and marketed in his own

garage). Then there was Ravi Varanasi (in-charge of inspection and surveillance) and Hari K. Not to mention how all this remained out of the knowledge of Ramkrishna and Murlidharan – the lack of communication was surely a huge procedural lapse.

In January 2020, a news report in The Economic Times said that SEBI was conducting further probe with regard to leak of NSE's internal documents of policy decisions on technology, specifically COLO infrastructure, internal studies and minutes of the board meetings, etc. The leak involved the period when Ramkrishna ran the show.⁵⁰

The '*Dark Fibre*' Speedway

When you hit the enter button to the Google search engine, you get the results instantly. How? The data you sought, travels through your internet service provider's fibre optics cable, which transmits it using the '*pulse of light*' and hence it is called as a '*lit*' cable. The network of fibre optic cables connects the world through internet, audio and visuals. Over the lit fibre, the data travels at the speed of light.

But in the work of financial markets, '*lit cables*' are like commercial, passenger flights, a means of mass transport system. Simply put, a lit cable network is used by many brokers to connect to an exchange via the internet and send and receive stock trading data. They are heavy on load as the masses depend on them and their wear and tear, too, make them slower in data transmission by a fraction of seconds. A '*dark fibre line*' may function similar to the '*lit cables*', i.e., transfer data to and fro. But it is more like a '*private jet*' owned by the privileged and more efficient for travel since it does not cater to the masses. At NSE, the privileged few used the '*dark fibre*' lines to connect to the COLO farm.

Since, '*dark fibre*' cables were extremely light on load, as they were not a mass transport system of data, they ensured faster access to its user in connecting to NSE's trading grid and TBT data travelled quicker to them.

NSE contracted a company called Sampark Infotainment to lay the '*dark fibre*' lines and provide its private jet-like service to only two brokers – Way2Wealth Securities and GKN Capital, another entity owned by the promoters of OPG Securities. In the first place, allowing access through such connectivity was a clear case of '*fraud*' perpetrated on the whole market since the exchange never had any policy on the use of '*dark fibre*'. Did the wider markets even know that such a thing was in use? If not, then there was nothing at all in it to say it involved fair access. If some can be ahead by 20 microseconds, they are a 100 per cent ahead of others.

“Confirm that you have shifted Way2Wealth and GKN to Sampark to the new set-up this has to be done on a priority,” Nagendra Kumar, an NSE employee wrote to a colleague. Who had ordered him to do so? When confronted by SEBI why he wrote such an email, he said, “My boss Ravi wanted to know.” That was incriminating. Apparently, there are half a dozen people named Ravi who have worked in NSE. But who was THIS boss?

While examining the documents submitted by Sampark, it was discovered that the company was not licensed to provide data connectivity and was only licensed to establish and maintain assets such as fibre, right of way, duct space, etc. for licensed telecom service providers and direct user customers.

Investigations also found that Way2Wealth left no stone unturned to acquire latency as low as possible by avoiding switches/hops, by shortening distances of the cable paths or by positioning the cable path in such a manner that it always remains ahead of the other trading members in terms of accessing the market data feeds through its connectivity at NSE COLO and BSE COLO.

“COLO facility at NSE was symbolized by maladministration and misgovernance. From the way the NSE and its officials, Way2Wealth, its officials and GKN have conducted themselves with respect to the COLO connectivity, it appears that there was an implicit collusion amongst NSE to protect the interest of brokers and Sampark,” SEBI observed.

NSE did not inspect Way2Wealth and its points of connection taken from Sampark. But the same exchange officials did not permit Shaastra Securities to take connectivity from Microscan (a service provider similar to Sampark). NSE officials waived their policy of physical inspection of the office of the trading members at BSE prior to granting permission for connectivity to Way2Wealth.

“Later when some eyebrows were raised, NSE denied Sampark permission to provide connectivity but the officials suggested a way out to the company. Subsequently, Reliance Communications communicated to NSE over an email dated August 7, 2015 that Sampark had sold its infrastructure to it. Reliance was allowed by NSE to provide data connectivity,” the whistleblower said.

In its final order, SEBI just banned Sampark from associating with market companies for two years and took no action against the fraud it did of laying the dark fibre lines without being an authorised service provider.

Way2Wealth and GKN were fined just around Rs. 15 crores and Rs. 4.9 crores, respectively. But there has been no proper estimation of the illegal gains generated by them over the years. Incidentally, GKN is a company related to OPG.

There was another interesting revelation by the whistleblower of which SEBI took no note of.

“Dark fibre links are operating in a regulatory vacuum at MCX. A Mumbai-based fibre network firm rejected by the NSE (Sampark), was operating at MCX with impunity. The firm provides differential speed access to different clients based on revenue share and MCX may not check this on the alibi that it was not responsible to ensure fairness across firms since it does not provide COLO. A Mauritius entity controlled by a United States parent made profits of Rs. 20-30 crore between 2013-14 by trading in currency futures on NSE and BSE accessing feeds prohibited to non-bank participants,” the whistleblower said.

All this was never properly investigated by SEBI. The speed links at NSE and MCX were also used by one of the largest HFT and Algo traders who has a base in Mauritius, US and India. There was a Deloitte report against several brokers and it had the name of this broker too. But this whole thing has been conveniently buried.

“Forensic review of electronic stored information of 14 custodians was conducted to identify possible collusion between NSE employees, vendors, brokers. Our review was focused on ‘*Keyword*’ based processes, which were drafted based on whistleblower allegations as undertakings given to us of the process followed for the COLO TBT system. On running keywords most responsive documents were reviewed further,” Deloitte said in its report.

The trail was incomplete since it was based only on keyword searches of company emails. A full access and probe to private emails of all the suspects would have unearthed a real goldmine.

SOME GAMBLES ARE 'MORE EQUAL'

Preferential Treatment

In its nascent stages, the business of stock broking in India was dominated by the Baniyas, a community involved in mercantile capitalism or money lending. Subsequently, Gujaratis in Mumbai, Marwaris in Kolkata and the Hindi speaking stockbrokers in New Delhi emerged as the dominant players in the country's stock market. Over the years, as Kolkata-based brokers found themselves entangled in several market scams and every major trail led to the loan sharks in that city, their perception among the investors' fraternity vitiated. While brokers in Mumbai patronised the fine art of the city's shrewd corporate culture and a knack for PR, those in Delhi established a strong bonding with top bureaucrats and politicians cutting across the party line.

OPG Securities Pvt Ltd (OPG), founded by Om Prakash Gupta, was a run-of-the-mill New Delhi-based equity brokerage house that seemed uninterested in client-based commission business, which is commonly referred to as retail business. OPG prioritised proprietary trading, which in market parlance is direct trading of stocks, bonds, commodities or currencies for '*self-gains*' rather than handling business for clients. OPG belonged to the old-world charm of '*prop*' trading firms that employed scores of traders, who

could speedily and relentlessly punch computer keys to buy and sell stocks.

This breed of traders would try to profit from arbitrage on stock spreads, as small as a few paise. However, post-2008, the noisy keyboards in the prop trading firms were all replaced by an automated software as the NSE introduced HFT and Algo tools. OPG too transformed itself into a new-age, tech-driven '*prop trader*', as Sanjay Gupta, the next generation of OP Gupta, took over the reins. Usually, Sanjay came across as a non-distinct personality but he had the '*unique*' knack of making friends in the right places.

One KK Daga of Millennium Securities, a Kolkata broker, kept pleading his case with the NSE for allowing his firm to connect to the COLO grid but got no, or just lame responses from the exchange staff for weeks and months. Once in a while, the NSE staff would reply to his emails titled '*My Grievance*' and insist that an office inspection of his brokerage was required before allowing him to enter the COLO farm.

But then, for brokers like OPG and Way2Wealth, the NSE top brass was just a phone call or email away. Once when Sanjay wrote to Chitra Ramkrishnan informing her about some issues and that he intended to buy more 'full racks', at the server farm, she immediately put people to work. "Do the needful, ASAP and revert," she directed the exchange team.

The immediate response to OPG's requests seems to contrast with responsiveness received by other members. Barclays Capital raised some query on December 29, 2011 and followed up on January 30, 2012, since they had still not received any response.

COLO chief Jagdish Joshi then suggested a call and missed it twice. However, Barclays got its work done later. Matter of fact, insiders' claim, during the said period most of the queries other than from those in the so-called '*good books*' went unaddressed, some totally ignored and junked.

As per the whistleblower account, OPG had figured that the secret to success in NSE's COLO was in knowing the fastest server and its starting

time. It is what the NSE-owned tech platform provider Omnesys was delivering to all its clients under the guise of managed services. The TBT connection was established before the markets opened (typically one hour before) but the time was not fixed. Could it be that some such non-public knowledge of COLO operations gave the clients of Omnesys an edge?

Sanjay hired Omnesys staff and gradually also warmed up to a few key people in NSE's COLO data centre, who would let him know the time at which the servers would be started in the morning. This was the first of the many steps, which cemented the bond between NSE's COLO staff and OPG. Every day, Sanjay would be privy to the information on the starting time of the fastest server.

“He is very good at building informal relationships with people. During his interactions with the NSE data centre team, it became clear to Sanjay, where the advantage lay. He needed a person who knew the nuts and bolts of the system. He was working with Omnesys for his Algos, so he picked a person called ‘*Nagbhushan*,’ in charge of the NSE COLO installations, to work on his payroll,” the whistleblower revealed about Sanjay.

Investigations and visitor log details showed that Nagbhushan visited NSE's COLO farm several times even though he was not working for NSE or Omnesys.

Subsequently, OPG got more innovative and wrote a code in their software that would continuously ping at NSE's COLO servers since early in the morning, to check if they were operational. On receiving the first response from the servers, OPG was often placed ahead in the cue to login.

Deloitte's probe revealed that OPG maintained an Excel file with ‘*time slots*’ of when its software application would start to attempt a login into NSE's COLO servers every day, between 2013-2014. Prior to that, OPG used services from Omnesys, Ajay's IFPL and little known Mumbai-based firm Greeksoft Technologies.

“OPG would initiate the first attempt to connect to NSE’s TBT servers between 6:55 AM and 7:05 AM every day. OPG also tried connecting to NSE COLO on 38 non-trading days (including Saturday and Sunday) between 6:55 AM and 7:05 AM,” Deloitte said in its probe report.

Market trading started at 9:15 AM. What was the need to connect to NSE’s COLO farm more than two or three hours early?

In its initial response to SEBI, NSE had said, “There is nothing inherently wrong with trading members competing to login first. Consistent early login by members in a level playing field is neither per se unfair, nor does it amount to market abuse.”

There was no *‘level playing’* in the COLO farm. Investigation reports suggested it gave preferential access to certain brokers. Obviously, NSE was purely lying.

There were other email exchanges between OPG and NSE’s COLO staff, which show how the broker got a free pass and easy access, bulldozing all rules. Due to their patronage, their attitude had become so arrogant, when NSE warned the broker to get off the secondary server, it never acted promptly.

“Please give us time till (derivative) expiry as I don’t want to fiddle around with the setup. We will do the needful on Thursday evening,” Sanjay wrote to NSE COLO. Derivative contracts at NSE expire on Thursday and the heightened trading activity tapers down before again picking up closer to the next expiry.

“We want to connect to the secondary server for a few days for some testing. Please allow us to do so,” OPG wrote again in another email. “You are enabled,” replied NSE quite promptly.

Who had such exclusive and unrestricted privileges to connect first to the fastest server in NSE? If this was not granting preferential access, then what was?

Deloitte found warning tickets issued to several brokers against connecting to the secondary servers, which meant that the exchange knew the thin red line that existed there. Yet, strangely, the top brass of NSE remained blindfolded.

“Mr. Narain was with us a week before,” Aman Kokrady, Sanjay’s brother-in-law, wrote in an email to one of NSE’s COLO officials. In that email, Kokrady spoke about the exact operational details of the exchanges’ tech grid that he knew from his discussion with Narain. When SEBI confronted the NSE boss about his meeting with Kokrady and also showed him the email, Narain was evasive and simply said, “I don’t recall the meeting.”

It was just one of the several emails from OPG to the NSE that were unearthed during the investigation into the preferential access scam. In some other emails, OPG is also seen dictating terms to the NSE staff, who in turn appeared more than ready to oblige.

SEBI has completely ignored these points and blamed it all on ‘*procedural lapse*’ on part of the exchange to refrain from calling the COLO scam an organised ‘*fraud*’. If one broker could game the system, there were others too. Where is the audit of the software used by all the traders in COLO from the beginning? Why were Omnesys, its managed data service centre, and IFPL not audited?

It seemed that SEBI totally forgot about this, instead, it has drawn a bizarre conclusion.

“In my understanding, the information advantage accruing to the ‘first connect’ trading member may not continue throughout the day and depending upon the load factor, it may get diffused and diluted in the course of the trading day to a ‘probabilistic’ advantage,” SEBI’s Mahalingam observed in his order.

Why would brokers make diligent efforts to connect first? The SEBI

official was his own master.

The only inference that can be drawn from his observation is that he believed that *'information advantage could have been for a few seconds, minutes or hours but certainly not throughout the day and that was just fine'*. At some point in his order, the SEBI official even took the pain to go into details of the numbers of days that brokers actually played *'fair'* by not connecting first or staying away from the secondary servers. Such a foolish assessment was incredible and similar to going into the clauses of the *'fake'* agreement that Ajay had presented to SEBI.

OPG was using services of Omnesys and Ajay's IFPL for quite some time in more than five years that the broker traded on NSE and its COLO farm. Omnesys claimed it provided hardware and also wrote custom software for clients. What was the custom software? Was it just the one that gave an early advantage? A mere presence of a code in Omnesys custom software that could allow a broker to login early shows it was an intentional front-running operation that the exchange's top brass was also a party to.

Once Omnesys started profit-sharing contracts with brokers on a large scale, Sanjay felt OPG was losing its competitive advantage. He also realised that by using a mass-market product like Omnesys, the facts would eventually leak out somewhere. He went ahead and got an in-house software developed, to which no one else would have access, and therefore, no one would be any wiser of what he was doing – informed the whistleblower.

Even though SEBI directed OPG and Sanjay to refrain from disposing of, deleting or modifying records, emails, communications, IT logs, etc., the vile broker performed a factory reset of his phone before handing it over to Deloitte for forensic imaging. The forensic auditor said it was *'potential destruction of evidence'*. Still, Sanjay could not hide a crucial link that revealed his connections with Ajay and perhaps also the reason why the researcher was extracting data from NSE, clandestinely.

OPG had contracted another New Delhi firm *'Algotech'*, for providing it

with trading software. Deloitte came across an agreement between Sunita's (Ajay's) company Chanakya Trade Vistas (CTV) and Algotech for a software deal. That software developed by CTV was being used by OPG and the broker had even signed a deal to share 40 per cent profit with Algotech in 2010. There was no agreement for loss.

Interestingly, all the details of Algotech portfolio, including its financials and shareholding, have been wiped out from the government database. There is no trace left of the company for the common public. But government probe agencies and SEBI can still get all its details, provided their intent and conscience is clear. It is unknown why SEBI did not aggressively go behind Algotech and people linked to it.

To cut the long story short, OPG was also using the trading software developed by CTV. Ajay's sister-in-law Sunita and another person Krishna Dagli, who just seemed like a paid employee from the meagre salary he collected, were the directors in CTV – fronts for Ajay and Susan.

Ajay's other company IFPL was registered at senior NSE official Suprabhat Lala's residential address. Whose residential address was CTV registered at?

Algotech, through its directors Kuldeep Garweal and Pankaj Arora, was also linked to Universal Stock Brokers Private Limited. This broker first came into existence in 2004-2005, when Ajay's company IFPL had started dealing in Algo software. In 2010, Universal was the second among the top five brokers, who enjoyed an advantage of early connection to NSE's COLO servers. NYCE Securities and Derivatives, AB Financial and Religare were others in the early bird list.

For reasons best known to senior SEBI officials alone, they have completely omitted the link between Ajay, IFPL, Algotech, OPG and Universal Broking, in the course of their investigation and final order.

Investigations have found that IFPL had a '*profit-sharing*' agreement with

brokers, the SEBI order said and left it at that without probing or intending to look into it any further.

“Our analysis highlighted that OPG, Universal and NYCE appeared to be the first to connect to the specific servers more significantly than others. ‘Multi Cast TBT’ was introduced in April 2014 and Jagdish Joshi left NSE the next month, after which we have seen a decline in the number of first connections,” Deloitte said.

Not only did broker profits fall after NSE’s COLO chief left and the new trading infrastructure became operational, but Omnesys was also sold off by NSE and its other promoters.

SEBI and The Art of Suppression

There has been a massive under-reporting by SEBI of the illegitimate gains made by OPG Securities. In fact, the regulator has not even considered illegitimate gains by other brokers, who got preferential access.

SEBI estimated OPG to have earned Rs. 25 crores (approximately USD 3.5 million based on the conversion rate in 2018) from its COLO operations. This in itself was nothing but gross under-estimation. If that was not enough, SEBI in its final order against OPG asked the broker to return only Rs. 15.55 crores of the illegitimate gains. The question arises, why such a sloppy estimation?

For no apparent logic or reason, the Indian School of Business (ISB) was tasked by SEBI to calculate the illegitimate gains made by brokers from NSE’s COLO due to *‘procedural lapse’*. ISB’s report is full of strange theories. Simple calculations show how ISB’s estimates of illegitimate gains are dubious, ridiculously miscalculated and deserve to be trashed. Systemic empirical evidence consistent with this view is plentiful.

Even while both SEBI and ISB concluded that OPG earned only Rs. 25 crores through NSE’s COLO between 2010 and 2015, the broker on its own

has declared that it had earned Rs. 160 crores through share trading. These numbers are part of SEBI's CFT report and can be corroborated in OPG's tax filings each year. How could SEBI miss that?

The CFT report, which gives a segment-wise trading turnover of OPG, shows the broker churned '*prop trade*' volumes worth a whopping Rs. 26.64 lakh crores (approximately USD 484 billion based on a conversion price of Rs. 55 against the USD) in the equity derivatives segment between 2010 and 2015. Add to this, Rs. 1.5 lakh crores (USD 15 to 20 billion) in currency futures trading. Comparatively, in the cash market segment, OPG's turnover in these five years stood at only Rs. 46,264 crores (between USD 2 to 5 billion). All these statistics are missing in the ISB report, not only with regard to OPG but even on other brokers.

Tax filings by OPG show that in a single year in 2012, the broker paid Rs. 4.19 crores as transaction charges to the NSE, which in the previous year was Rs. 3.22 crores. OPG paid Rs. 12.40 crores as securities transaction tax (STT) in 2012 and Rs. 17.21 crores in 2011. To generate such taxes and exchange transaction charges, a broker has to generate trading turnover worth thousands of crore as both the levies are less than even 0.01 per cent of the total turnover.

In addition, OPG paid Rs. 90.78 lakhs as NSE's COLO charges in 2012 and Rs. 62.24 lakhs in 2011. In these two years, the broker also paid nearly Rs. 1.5 crores as charges for a leased line, which is for connecting from his office to NSE. More than 90 per cent of OPG's trading is HFT and Algo-driven.

For the two years before 2015, when the broker was enjoying '*preferential access*', its turnover witnessed a tenfold jump, as estimated by the CFT report, which further said that OPG's derivative to cash ratio was 98:2. Simply put, OPG's derivative trading turnover was 98 per cent in the five years when NSE's COLO grid was being misused and the maximum of its profits came from there.

Now, if one goes by ISB and SEBI's estimates that OPG made illegitimate gains of just Rs. 15.55 crores in five years, then it has to be believed that the broker earned Rs. 145 crores net profit up to 2014 (figures for 2015 are not available) from ethical business practices, as its regulatory filings say nearly all its profit was from share trading.

There are two key questions that arise if one goes by ISB and SEBI's 'naive' assumption: If the legitimate business practices gave such abundantly high profits to the broker, why would OPG waste so much time and energy in relentlessly trying to login first and use unethical means with NSE's COLO staff and the top brass?

If you assume that OPG made 90 per cent of its Rs. 160 crores worth of profits in share trading without the misuse of NSE's COLO grid, there was no motive for the broker to indulge in dishonest practices, which have been pointed out by various probe reports. The other question is, can Rs. 46,000 crores, which was just 2 per cent of OPG's total share trading activity in five years, give the broker 90 per cent of its profit?

On page number 22 of its report, ISB said, "OPG's profit advantage, when logging first to NSE servers, is not substantial if examined on an average daily basis. It appears that the profit advantage, considering the total profits, is due to more frequent first login, than not."

Only, the incumbent SEBI chairman Ajay Tyagi and his senior team members including Mahalingam and Mohanty, under whom the final verdict was delivered, understood what ISB meant in this statement. To any expert, who can notice the inconsistencies, the ISB report, by all means, was questionable and seemed to be without the application of mind. But even more puzzling was its mindless acceptance by SEBI. On the face of it, the report itself calls for a CVC probe.

Quite illogically, ISB said it had considered only the intraday trading positions of the brokers, when they connected first to NSE servers and were in an '*advantageous position*'. But the report nowhere gives bifurcation of

volumes generated by brokers when they were not in an advantageous position. A bifurcation of OPG's Rs. 26.64 lakh crores (USD 484 billion) worth trading volumes in equity derivatives and Rs. 1.5 lakh crores (USD 15 to 20 billion) churn in the currency segment is missing in the ISB report. Is it practically possible to arrive at such a bifurcation?

Entire profits are illegitimate if NSE's COLO infrastructure was defective and designed to give preferential access since it lacked the essential safeguards, checks and balances.

A few among the NSE's COLO staff kept pointing to the flaws in the infrastructure repeatedly but the top brass was not bothered. Otherwise, how did the profit of Omnesys fall and OPG too started surrendering its connections, after NSE changed its infrastructure from TCP/IP to Multi Caste TBT? Once again, we see that much has been left to probabilities and assumptions by SEBI.

On page number 16, ISB said, "Total average profit of the brokers during the unicast (NSE's old system between 2010 and 2015) is five times higher than multicast period (the new system that NSE silently introduced after Deloitte submitted its first report in December 2016). The difference is starker when we look at medians, where the profit during the unicast is about 15 times higher than multicast period."

The statement is evidence enough to show that brokers reaped higher gains from NSE's defective COLO system and their profits reduced when the systems changed. But ISB has attempted to give a spin to the data. "It may be attributed to more number of days during the unicast period than under multicast."

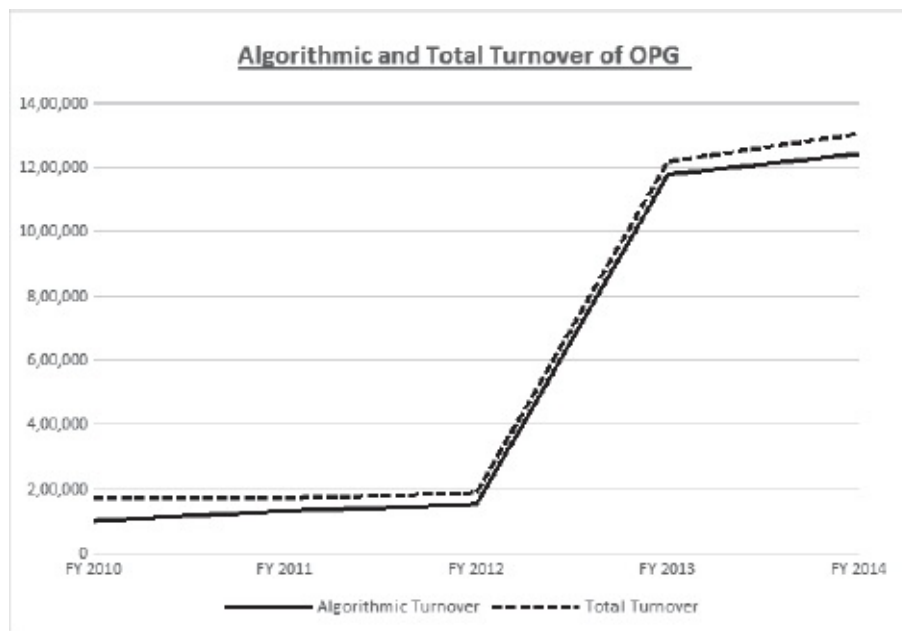
ISB has linked the lower profits of brokers under the new system to less number of days under review. But to overcome such a lacuna, the logical question arises, should it not have relied on the methodology of calculating average profits per day and see if the benefits were lower or the same?

From what ISB said, it is clear that average profits per day too went down under NSE's new trading infrastructure. ISB's report loses its ground of any further arguments. After all, share trading profits should be linked to mathematics and trading strategies or the ability of brokers to make the right bets. Or is it supposed to be linked to changes in a stock exchange order matching and data sharing infrastructure? Here, authors of the ISB report stood fully exposed in their attempt to set the agenda.

How come each and every broker in NSE's COLO saw a sharp decline in their profits after the change in its trading infrastructure, which was done after Deloitte pointed out the flaws? Nothing more is left to ponder over the ISB report as SEBI accepted all of ISB's observations without any second thought and let go of brokers with just a mediocre rap.

The chart below prepared by SEBI's CFT, when Rajiv Agarwal was the WTM, gives a precise picture of OPG's earnings from NSE's COLO. It is another piece of crucial evidence to dump the '*pliable*' ISB report.

Algorithmic and Total Turnover of OPG FY 2011-14 – Trend Analysis



It may be seen from the above graph that both the algorithmic and total

turnover for OPG has increased considerably since FY 2012 as compared to earlier Financial Years, CFT said.

Further, CFT noticed that the fastest servers were reserved for OPG. Around 2012, all first, and top-3 logins by OPG on NSE's dissemination servers named TBTCOLO23, TBTCOLO26 and TBTCOLO27 were from OPG's rack I9 and had the IP subnet as 10.230.54.XXX. This IP subnet was not allotted to any other member as per NSE's submission dated July 31, 2015. The IPs allotted to OPG had the subnets as 10.230.21.XXX, 10.230.54.XXX, 10.230.39.XXX and 10.230.81.XXX. The IPs belonging to these subnets had also not been allotted to any other member during 2012 as per NSE's submission to CFT.

How OPG and Other Leading Brokers Escaped Regulatory Action

Before SEBI, it is the duty of the stock exchanges to initiate action against brokers, who are found to have indulged in any wrongdoings or malpractices. Stock exchanges are called the '*first level regulators*'. NSE did not take any action against OPG.

I reported this story in September 2019 in *The Hindu Business Line* as to how the exchange withdrew its SCN against OPG on the grounds that SEBI had already passed an order against the company.⁵¹ The simple fact is, if the exchange took action against the broker, it would have had to take action against its own staff and also hold top brass guilty. It would have definitely stirred up a hornet's nest.

OPG told NSE that other brokers too connected to the secondary servers. It indicated that if NSE had initiated any action against OPG, it would have had to go behind other big names too. And there were plenty.

For shutting down its case against OPG, the NSE said that it found legal difficulties would arise in the disposal of the matter on merits as there was a SEBI investigation against other brokers too and its earlier order against

OPG was only in specific segments.

“The developments belatedly for first time indicated the ‘the prudent and legal difficulties’ that would arise in the event of a ‘de novo’ (starting from beginning) disposal of the matter on merits,” the NSE said and dropped the case against OPG.

OPG’s argument to NSE was that the exchange had no law or provision to deal with monitoring of connections to secondary servers and disciplinary proceedings can be initiated only if there is a legal provision.

“Whereas the warnings given by the exchange were not based on a legal provision.” It made front-running legal? Why were the provisions of front-running not invoked by the exchange? Also, it was a perfect crime and exchange had knowingly kept the loopholes open for the brokers to exploit.

NSE was pleading helplessness and sending a wrong message to those who were playing by the rules. Instead, the exchange was required to set an example by its action and let the broker appeal against it.

Still, OPG was just the tip of the iceberg. There are other large brokers who escaped any action from NSE as well as SEBI.

In 2018, Deloitte submitted its second probe report to SEBI. That report had two parts, wherein the second part of the report spoke about preferential access to brokers like OPG, Barclays and SMC.

The first part of the report was based on the analysis done against brokers including Crosseas Capital Services Private Limited, GDR Securities, KIFS Securities Private Limited, KIFS Trade Capital Private Limited, Estee Advisors Private Limited and Quadeye Securities Private Limited. It was related to “appointment of an unauthorised service provider for connectivity to NSE’s COLO.”

SEBI has simply buried this crucial part of Deloitte’s report. There is no mention of Deloitte’s findings against any of these other brokers and their

wrongdoings in SEBI's final order. Nor has SEBI conducted an interrogation or investigation of these brokers like it did for OPG.

There is enough evidence to say that OPG got preferential access, but somehow, SEBI seems to have made the broker a scapegoat or the poster-boy of the entire scam and let other larger players off the hook.

Few brokers, with regard to whom the Deloitte report has been suppressed, hold a presence in tax havens like Mauritius and are also affiliated to international trading venues including the Dubai Gold and Commodity Exchange and Singapore Stock Exchange (SGX). Their statutory filings show that their management is Indian but they have created a wide network of HFT trading firms across the world, where profits can be transferred. They have the company vehicles capable of round-tripping of money across various jurisdictions including the US and are also registered as sub-accounts of FIIs in India.

These brokers run collective investment schemes in Mauritius, where they can give and take money from any investors, without much questions being asked. Did such brokers use dark fibre trading links, Smapark, Omnesys or did they have anything to do with Ajay or his family members?

Omnesys was one service provider; there were a few foreign players too. Has SEBI investigated them all? This fact is unknown. The only thing that is clear so far is that the gains most brokers made from trading through NSE's COLO could all be dubious since they got the advantage of the defective infrastructure of the exchange. Because if OPG connected to the secondary servers and it was illegitimate, then even others who logged into the secondary servers or used the speed links made illegitimate gains.

Yet, SEBI has done nothing against them and instead it has attempted to suppress several aspects of the scam, in which it has been successful so far.

The findings of Deloitte of the many brokers mentioned in the first part of its 2018 report will remain under wraps, unless and until SEBI thinks it

wants to reveal it. Ideally, the investigators should not miss the woods for the trees, which shows that the entire COLO trading was illegal and none can be spared.

Oddly, SEBI provided Deloitte with call data record (CDR) data for the period after 2016 of brokers and some of the NSE employees for a probe.

Supposedly, the COLO scam and data theft took place at the NSE between 2008 and 2015. What will Deloitte do by studying the CDR records of the subjects post-2016?

On the face of it, it really comes out very lame from an institution with '*Statutory Powers*', which is supposed to be the regulator of the securities market of a country the size of India, which is undoubtedly the world's fastest-growing economy.

If SEBI thought it could not dig out the CDR details for past years, there are other government agencies who had the expertise in the job and could have been called in, but the regulator never took their help. What was worrying SEBI? Digital evidence is never extinct. Only retrieving it becomes little difficult as time passes.

The absurdity does not end there. Investigators could not question NSE's COLO chief Jagdish Joshi, a key link in the chain, as he had already quit his job. Also, it seemed that his computer system and emails were not available to Deloitte. Whatever the investigators gathered about him, which proved to be incriminating, was from the emails of his colleagues or OPG. How is the probe complete?

Personal email and CDR data of NSE's top brass, Ajay, Susan, IFPL, key employees and promoters of Omnesys and various such other key subjects have been ignored by SEBI, which never even tasked the forensic auditor to study them and get to the full ramification of the scam. Another glaring omission by SEBI was that an entity linked to an NSE promoter, which also traded in NSE's COLO farm and was found to have logged into the

secondary servers was not even probed.

Goldman Sachs (GS), the top US-based bank and FII, is among the marquee names in the list of NSE promoters and holds a 5 per cent stake in the bourse. In a single year in 2010, its Indian arm, GS (India) Securities Private Limited logged into secondary servers 69 times in NSE's COLO. But SEBI had restricted the scope of investigation of its TAC committee to a period between 2012 and 2015. Otherwise, even before OPG, did GS know the advantages of connecting to NSE's secondary server?

This also raises a slew of legitimate questions. Did GS face any problem in trading via the primary servers and hence connected to the secondary servers? Were any warnings issued to it? If not, why? What was the amount involved in its trading via the secondary servers? Did it indulge in any unethical activity? What were its trading volumes and profits from the defective COLO grid? On whose behalf was GS trading?

If OPG Securities is guilty of logging into a secondary server, why not a company belonging to NSE's promoter group, which too indulged in the same act by login into secondary servers 116 times in four years? Who provided trading software to GS?

The same question also arises for other brokers in the list, who were trading in the COLO farm, of which no assessment has been made by SEBI. Why is there no SCN against GS? Answers to all these questions are just missing, as SEBI has never considered a probe against the FII broker.

The chart below shows top 20 brokers who logged into NSE's Secondary Servers

A January 11, 2019 report in *The Mint* newspaper suggested that average turnover and profits of 62 brokerages were higher during 2010-14 when they were supposed to have got unfair access. The profits fell when NSE strengthened its infrastructure. The profits of GS India Securities fell too.⁵²

Newspaper reports have named Motilal Oswal Securities, SMC Global

Securities, Religare Securities, IKM Investors, Kotak Securities, Barclays among others having suffered a fall in their profits.

It came to light that OPG was far aggressive in breaking into NSE's trading systems, but the forensic audit reports suggested there were 62 brokerages who enjoyed '*preferential access*' at NSE between 2010 and 2015. Were the emails of those brokerage employees checked like OPGs? SEBI served SCNs mainly to three firms including OPG Securities, its associate firm GKN Securities and Way2Wealth Stock Brokers. All we can say with a '*pinch of salt*' is that the rest just got lucky.

Although the ISB report failed miserably in estimating the full extent of the illegitimate gains made by several brokers who enjoyed preferential access, nevertheless, it shared a glimpse of who all have benefited. GS, the report said, made intraday gains to the tune of Rs. 90.74 crores and Barclays Rs. 146 crores. Check out the table below.

Table 3: Total Intraday Profits of Deloitte Sample - All Servers [Profits in lakhs of Rs.]

Trading Member Name	Unicast	Multicast
	Total Intraday Profit	Total Intraday Profit
A.P.T. PORTFOLIO PVT. LTD.	19,075.75	6,593.38
AB FINANCIAL SERVICES	1,156.37	214.45
ADROIT FINANCIAL SERVICES PVT. LTD.	20,839.15	4,108.55
BARCLAYS SECURITIES (INDIA) PVT. LTD.	14,610.21	237.10
CITIGROUP GLOBAL MARKETS INDIA PVT. LTD.	10,754.06	-117.03
CNB FINWIZ PVT. LTD.	5,181.18	2,455.67
CPR CAPITAL SERVICES LTD.	5,343.79	806.65
DIVYA PORTFOLIO PVT. LTD.	7,949.43	115.88
GKN SECURITIES	711.73	472.30
IKM INVESTORS PVT. LTD.	-583.24	-264.41
NYCE SECURITIES & DERIVATIVES LTD.	-14.92	-35.35
OPEN FUTURES	-367.32	34.34
OPG SECURITIES PVT. LTD.	14,181.56	258.73
PACE STOCK BROKING SERVICES PVT. LTD.	17,671.75	1,712.70
SMC GLOBAL SECURITIES LTD.	14,931.09	721.50
UNIVERSAL STOCK BROKERS PVT. LTD.	420.14	12.95
WAY2WEALTH BROKERS PVT. LTD.	8,988.23	11,080.85
Average	8,285.24	1,671.07
Median	7,949.43	258.73

Table 4: Total Intraday Profits of Benchmark Sample - All Servers [Profits in lakhs of Rs.]

Trading Member Name	Unicast Total In- traday Profit	Multicast Intraday Profit
CRIMSON FINANCIAL SERVICES LTD.	8,260.40	4,255.70
EDELWEISS SECURITIES LTD.	1,275.96	-58.23
ESTEE ADVISORS PVT LTD	456.76	578.56
GOLDMAN SACHS (INDIA) SECURITIES PVT. LTD.	9,074.69	198.82
KOTAK SECURITIES LTD.	46,364.70	649.90
N.K. SECURITIES	23,426.35	830.16
NEWEDGE BROKER INDIA PVT. LTD.	12,623.64	291.01
PHILLIPCAPITAL (INDIA) PVT. LTD	222.06	-199.65
QUADEYE SECURITIES PVT. LTD.	39,144.30	5,778.50
RELIGARE SECURITIES LTD.	13,537.48	11,358.93
SG ASIA HOLDINGS INDIA PVT. LTD.	10,906.87	404.23
SHAASTRA SECURITIES TRADING PVT. LTD.	9,380.75	10,418.79
SHARE INDIA SECURITIES LTD.	1,716.34	1,740.45
Average	13,568.48	2,788.24
Median	9,380.75	649.90

The Deloitte report disclosed by SEBI showed that SMC Global Securities and Barclays Securities were repeatedly connecting to the secondary servers.

NSE's COLO team wrote to SMC Global on several occasions asking them not to connect to the secondary server. But the broker said it wanted to continue as the primary server was too slow. SMC Global connected to the secondary server on 389 days in the equity derivatives segment but it complained against the difficulties it faced only for 210 days. SMC Global was unable to provide evidence for the days they had connected to the secondary server without a corresponding complaint regarding the connectivity. Yet, NSE's DAC just let off the broker with just a '*dire warning*', the Deloitte report found.

It noted that before June 2012 there was no explicit communication about the circumstances in which the secondary server could be accessed. Although, it said that "this requirement was implicit in the servers having been categorised into primary and secondary server."

In the case of Barclays, no warnings were issued, nor was any other action initiated. During the probe, Barclays simply said that it was not aware of the

reasons as to why their application IT team was switching between *primary* and *secondary servers*. It said its IT team was transferred to an entity named Squarepoint Capital in 2015. That was the end of it. SEBI and NSE did not move any further against the FII broker.

SEBI fined SMC only Rs. 5 lakhs for violation of code of conduct rules of brokers – that was the meagre cost at which it gained an unfair advantage and earned millions?

By conservative estimates, NSE's equity derivative segment generated more than Rs. 2,00,000 lakh crore worth of trading volumes between 2010 and 2015, each day. In this, again conservatively, 50 per cent of volumes, almost half, came from the COLO traders. In the five years, even if one considered 1,000 trading days when all the brokers got preferential access to the market, the math on illegitimate gains would run into a mind-boggling thousands of crore rupees.

If OPG could churn Rs. 26.64 lakh crores (USD 484 billion) worth volumes in five years and earn 0.01 per cent profit (lowest possible estimate), one can only imagine how high the stakes could have been for some of the large players in the game.

Most of the HFT Algo players and FII have their base in Singapore, where the Nifty index is traded. A lot of their profits are also generated in a jurisdiction that is out of the reach of India's market regulator.

The Chidambaram '*Rahasaya*'

It gets utterly boring when you repeatedly come across SEBI's dull techniques of investigations, interrogation and arriving at a conclusion, more than often on absurd grounds. But there was something intriguing that the regulator asked Ajay during his interrogation.

“What is the nature of your association with Algotech, Alpha Grep, OPG, Lares, CTV, Way2Wealth, Advantage Strategic, Advantage

Corporation?”

Was SEBI just scratching the surface or was there more to this question? Nobody knows as the regulator itself has abandoned the links and chopped-off the trail by an incomplete and shoddy probe.

Could the question, in such crucial investigations, be without any context, or did SEBI have some clue of its link to the COLO scam? The question has deep ramifications since it involves a direct link to former FM Palaniappan Chidambaram and his son Karti Chidambaram.

Except Advantage Strategic Consultancy Private Limited (ASCPL), all the other companies mentioned by SEBI in its course of interrogation were directly involved in trading at NSE COLO farm. ASCPL's emergence in this is shrouded in secrecy.

ASCPL is a shell company that came under the scanner of India's ED in 2016 for receiving kickbacks in the deals, where the Finance Ministry under Chidambaram gave clearance for foreign investments in India. On certain occasions, ASCPL is also alleged to have paid certain expenses and travel bills of Chidambaram.

ASCPL directors Ravi Visvanathan and Padma Bhaskararaman told the ED that they held '*namesake*' positions and their role was limited to signing invoices and documents on the instructions of Karti and S Bhaskararaman, his chartered accountant and co-accused in the INX Media bribery case, in which Karti Chidambaram was arrested in February 2018.

Karti has repeatedly denied any business relationship with ASCPL, although upon questioning, he told the ED that he knew the directors very well and they were his friends.

ASCPL's director, Padma, is the wife of Karti's accountant Bhaskararaman. ASCPL has a subsidiary in Singapore called Advantage Strategic Consulting Singapore Pte Ltd. The ramification also stretches to Mauritius and the 2G telecom scam as ASCPL is alleged to have received

bribes from a Mauritius company in 2006, which was then controlled by a suspect in the 2G scam.

The brokers who were indulging in COLO trading in NSE and are supposed to have got preferential access, also have sister companies in Singapore. AlphaGrep, a Way2Wealth company, is also registered in Singapore and was incorporated in 2010, the same year when NSE started its COLO trading. AlphaGrep also has a Hong Kong subsidiary.

Way2Wealth Illuminati Pte Ltd is another company linked to the same group and registered in Singapore, which had around Rs. 140 crores (nearly USD 1.09 million) dues outstanding from related parties that it extended as unsecured loans and advances as on March 2015, this is revealed in its statutory filings.

Many of the broking entities in Singapore, owned by Indian shareholders, have their primary objective as trading in stock markets in India. But still, they have kept their money transactions outside the country. Some of the other trading companies in the Deloitte list that were buried by SEBI too have Singapore links.

Noticeably, a large number of HFT brokers have established a cross-country network of trading firms. They are registered on the Singapore Stock Exchange, Dubai Gold and Commodity Exchange (DGCX) and even in the US. These brokers can trade Indian stocks and assets on the overseas exchanges and show their profits outside.

It is a key reason that Nifty trading on the SGX has been highly popular. Authorities in India have no idea about those taking positions in Singapore, even if it could be a Way2Wealth or an OPG trading there and maybe even transferring their wealth. It is the same reason that brokers who trade gold and silver on the MCX are the same who trade it on the DGCX.

To the question that SEBI asked him on the various companies and ASCPL, Ajay simply said he had never heard about any of them. He even

denied the knowledge of the fact that his sister-in-law Sunita had spun another company called CTV. He said he was ignorant about IFPL's dealings with Way2Wealth, Lares Softech and some of its key directors and AlphaGrep.

There is evidence to show that these companies were clients of IFPL or Chanakya. Was Sunita even qualified to run an Algotech company than can code and write trading software? But SEBI believed Ajay on everything he said as if he was a torchbearer of truth.

Evidently, based on Ajay's statement, SEBI left the trail untouched and did not get into further details like the shareholding pattern or any of the financial information of companies connected to NSE's COLO. The incorporation dates of some of the key companies linked to NSE COLO have been changed and their old shareholding and financials, like that of Algotech, are nowhere available in public records. Even OPG had companies registered overseas? SEBI has never looked at this.

Crime and Punishment

For a super-cop of India's stock market that was created with a commitment to the cause of retail investors and ensure fair and equitable access to all the players, SEBI came close to Rodion Raskolnikov, the antagonist of Dostoyevsky's classic *Crime and Punishment*, as it rolled out its final verdict in NSE's COLO scam on April 30, 2019.

Raskolnikov is the central character of Dostoyevsky's novel written sometime at the end of 18th century Russia. The young law student harbours some perplexing hypothesis to 'justify' his crime after killing a corrupt pawnbroker and believes that extraordinary people have the right to transgress. Much in a similar way, SEBI believed there was no '*fraud*' involved in NSE's COLO trading firm, but it was a case of 'mere procedural lapse'.

There is '*compelling evidence*' to show that '*few*' derived benefits from

this '*systemic lapse*'. SEBI's logic to consider 'by-design systemic lapse as no fraud' is a crude, distorted and ridiculous representation of the scam, which implies that '*Murder by Decree*' is no crime.

SK Mohanty and G Mahalingam, the two senior SEBI officials, passed five separate orders in the entire scam. Instead of joining the dots, such as linking NSE's defective COLO systems, which lacked essential safeguards, with the sharing of key data with researchers (who were in conflict of interest) for software development and dubious activities of brokers, SEBI sliced and diced the colossal case of '*preferential access*' into several isolated instances. A link by link chain of events and story that is essential to be constructed in a scam was cut into pieces to arrive at varied conclusions.

Mahalingam dealt with the '*lapse*' in the COLO farm and access to OPG. Mohanty looked into brokers connecting via dark fibre as a separate case, even though it was just another side of the same coin, as it gave preference in connecting to a few.

Then separately, Mohanty dealt with sharing of data by NSE with researchers Ajay and Susan (who were in conflict of interest) and the role of their enterprise IFPL and members including Sunita and Suprabhat. He termed their offence as a mere failure of '*corporate governance*'.

Mahalingam concluded that procedural lapse cannot be termed as '*fraud*'.

"Alleging fraud against the exchange, in this scenario, (is) tantamount to attributing intention or knowledge of which there is no proof," he said and dropped the more charge of Prohibition of Fraudulent and Unfair Trading Practices (PFUTP) against the exchange as well as all its officials.

His justification for NO 'fraud' by exchange officials was due to, "absence of facts pointing towards collusion of employees with brokers, or proof of specific discrimination towards any specific broker... or accrual of monetary benefits/unjust enrichment to any employee or brokers."

Although instances of collusion are plenty, yet no proper assessment of

monetary gains had been done of any of the suspects. Could the gains not be in cash and kind? One can only speculate? What about the overseas vehicles that could give loans to anybody?

SEBI ignored probe reports that showed how NSE's architecture was prone to manipulation, market abuse, and that some brokers got preferential access and advantage over others. How come all of it happened without any collusion at any levels in NSE?

SEBI has pinned this to "flouting principles underlying the conduct of the business of a stock exchange." The fact that NSE ignored complaints of misuse of its architecture has been conveniently neglected.

"At the outset as the allegation of fraudulent and unfair trade practices levelled against NSE stands disproved, the same can no longer survive against the employees," Mahalingam said, in his order.

Naturally, if there is no '*fraud*' or 'collusion' by the exchange or its officials, how can any such charge stick against the broker(s) who merely 'used' the COLO farm? Still, Mahalingam held OPG guilty of 'fraud' and violating PFUTP norms.

"OPG Securities had made unlawful gains of Rs. 15.57 crores. It violated the PFUTP regulations," the SEBI order said. Is it not strange how a single broker can do this without any collusion?

What about those who provided the broker with trading software and facilitated it to access the NSE platform? What about the observation of probe reports, both public and the one hidden, against other brokers? There were 62 brokers who had gained '*unfair advantage*' by logging into the secondary servers but SEBI issued a SCN only to three.

Mahalingam's justification in imposing a penalty against the NSE lacked further logic.

SEBI asked NSE to '*disgorge*' nearly Rs. 1,000 crores (in unlawful gains).

It included Rs. 624 crores earned through COLO operations for four years with 12 per cent interest post-2015 up to 2019. Now, in not alleging a '*fraud*' by NSE in COLO operations, but still asking the exchange to pay such a huge amount, SEBI just played to the gallery and created a sensation so that people do not prod over the lacuna in its orders.

When a person or entity in the securities market makes a profit by '*fraudulent means*', a '*disgorgement*' order is issued to take the money from the wrongdoer and repay those gains to affected investors with interest. '*Disgorgement*' can be ordered off '*ill-gotten wealth*'. But SEBI has said that there was no '*fraud*' at NSE with regard to COLO.

Conveniently, the exchange can argue in court that its profit from COLO cannot be termed as '*ill-gotten wealth*' as there is no charge of '*fraud*' against it and hence no disgorgement of such proportion is required. Same goes for brokers. Since there is no '*fraud*' at NSE, what wrong did brokers do? Why should they agree to '*disgorge*' even a penny?

Similarly, SEBI asked NSE bosses Ravi Narain and Chitra Ramkrishnan to give away 25 per cent from their salary for two years and also banned them from associating from markets for five and three years respectively. Why, if they did not commit any fraud? Also, does it matter to the bosses, who anyway got a hefty golden handshake when they left NSE?

SEBI's method of calculating NSE's '*ill-gotten wealth*' that is marked for disgorgement too is questionable. By its logic, NSE earned Rs. 1,000 crores from co-location but all the accused, including brokers who got preferential access to the exchange systems, did not even make Rs. 100 crores. It all boils down to the '*Doctrine of Proportionality*', which is a principle that is prominently used as a ground for judicial review in cases of administrative action.

The '*Doctrine of Proportionality*' was adopted in a case involving Om Kumar v/s Union of India, in which a disciplinary authority had asked the SC of India to reconsider the quantum of punishment given to four civil

servants. The court refused to reconsider the quantum of punishment as '*no principle of law was violated*' nor was the punishment '*shockingly disproportionate*' to the mischiefs committed by the concerned persons.

This position of law was crystallised by the SC itself in later cases. In the current case, SEBI itself says that '*NSE has not committed any fraudulent activity in the COLO case*' nor does it show anybody who earned Rs. 1,000 crores or more from '*illegitimate*' trading at the exchange and thereby caused such a loss to their counterparty. Under such circumstances, what holds any court or appellate authority from putting SEBI's so-called '*disgorgement*' to the test of Doctrine of Proportionality and say that the decision arrived by SEBI is arbitrary and whimsical?

Was NSE the first case of a lapse in corporate governance in India that it has shocked everybody so much? In fact, the court may observe that SEBI's punishment is '*shockingly disproportionate*' to the offence or misconduct specified in SEBI's order as its charge of mere '*violation of code of conduct*' and '*corporate governances*' norms. SEBI's Rs. 1,000 crore disgorgement order does not measure up to the scale of NSE's wrongdoing. In that sense, SEBI's loosely knit orders in the COLO scam are nothing but giving an escape route to the exchange.

Come Mohanty's turn and he too does not disappoint those who are looking to find a usual pattern in SEBI's action. The NSE and its officials, who could not be charged with violation of PFUTP in one case of preferential access were subject to '*fraud*' and charged with violation of PFUTP (by attributing intention or knowledge) in another case involving dark fibre cable network.

Dark fibre and COLO are just two aspects of preferential access or connectivity to the exchange and involve the same set of brokers. In the dark fibre case, SEBI asked NSE to deposit Rs. 62 crores of estimated '*ill-gotten wealth*', but where is the estimate of the money, the exchange officials made or the brokers earned illegitimately? There is no monetary penalty Mohanty

wishes to impose on NSE officials, despite finding them indulging in '*fraud*'.

How did they commit fraud and enrich themselves? Did they take any bribes? Mohanty's order does not answer these questions. But there is an interesting inference he draws from the SC judgement, which Mahalingam has failed to refer to.

In the case involving SEBI & Others v/s Kanaiyalal Baldevbhai Patel and Others, the SC observed, "The definition of fraud which is an inclusive definition and therefore has to be understood to be broad and expansive, contemplates even an action or omission, as may be committed, even without any deceit if such act or omission has the effect of inducing another person to deal in securities. Certainly the definition expands beyond what can be normally understood to be a fraudulent act or a conduct amounting to fraud."

Thus, the definition includes even a mere expression or omission to act without having any intention to deceive or collude, if such an act or omission or expression or concealment leads to inducement of another person to deal in securities irrespective of whether there is any wrongful gain or avoidance of loss in dealing with such securities.

Using this, Mohanty says there is a fraud and does not impose any monetary penalty on NSE officials. But Mahalingam does not use this benchmark SC verdict and calls omission of duty by NSE and its officials as '*procedural lapse*' and yet imposed a penalty of Rs. 1,000 crores.

OPG has already argued that SEBI's action has been discriminatory against it. There were 62 brokers who logged into secondary servers, which is supposed to have given an unfair advantage, but SEBI issued a SCN only to three. With such inconsistencies, SEBI's COLO case may not stand water in the court.

If NSE's COLO case is rife with investigative and judicial misconduct, the entire blame for this lies with SEBI. Its officials have indulged in blatant

tampering, suppression of exculpatory evidence and the subornation of perjury. Instead of conducting an inquiry to establish fraud, SEBI dismissed the preferential access to data and servers as a code of conduct lapse.

The COLO case points to market rigging, through a '*deliberate procedural lapse*' in the trading systems at the NSE. But SEBI has viewed this merely as a '*violation of code of conduct and corporate governance*' norms – basically stupidity and dishonesty played an equal part.

History shows that the origin of any market scam can be traced to a '*procedural lapse*' at different levels. But a follow-up criminal probe based on the initial leads has often led to the unearthing of mega scandals. There are instances when the SEC, the US stock market regulator, had involved the Federal Bureau of Investigations (FBI) where a scandal or transactions involve wide ramifications.

SEBI's reluctance to involve criminal investigative agencies raises enough suspicion on its intent, especially since NSE's trading technology and COLO infrastructure lacked any regulatory scrutiny since 2008 and high ranking regulatory officials should be held guilty for this oversight

Curiously, the key mandate of forensic auditors against the NSE was just to check systemic lapse; the emphasis was NEVER on detecting fraud. How can SEBI be certain that there was no fraud when no investigation was conducted to determine the same? Was NSE COLO grid, which allowed brokers to front-run the markets through fraudulent means, a mere code of conduct violation? No, Mohanty had a different take in his order.

"The way COLO facility has been mismanaged and manipulated by certain trading members with the active connivance of an unauthorized service provider and the officials of NSE shows (NSE's) apathy towards the principle of transparency, fairness and equity mandated by SEBI on them," he said. Clearly, if one goes by the assumption of Mohanty and Mahalingam, dark fibre access was mismanaged and hence fraud, but COLO server-based trading was appropriate and only the brokers gamed it.

R Nandakumar (former senior VP for operations), Mayur Sindhwad (COO for trading), Sankarson Banerjee (CTO for projects), G. Shenoy (CTO for operations), Suprabhat Lala, Ravi Apte (former CTO), N. Muralidharan (former CTO) and Jagdish Joshi (former head for COLO) all were almost exonerated in the COLO case.

If Narain and Ramkrishna were held responsible for a lapse in their duty, how did other officials in-charge of respective departments, go scot-free? Is this an acceptance by SEBI that the rank and file was just following orders from the two powerful bosses?

In the '*dark fibre*' Ravi Varanasi along with Chitra was banned from markets for three years. All the NSE officials, against whom SEBI took action have appealed against it and got a stay from the SAT. Till date the hearing has yet to be concluded.

When it came to Ajay and Susan's data theft from NSE, which was used to strengthen their Algo software that was then used by brokers to play in the COLO, SEBI held researchers Ajay, his sister-in-law, Sunita, her partner, Krishna Dagli, and their company IFPL at fault for violating PFUTP rules in order to extract data from the exchange for '*commercial gains*'. But when it came to pronouncing punishment, they seemed to have gotten away very lightly. They just had to disassociate from the markets and related companies for two years – a Raskolnikov galore.

The complete trail of their alleged collusion with NSE bosses and brokers has been left unexplored. Suprabhat Lala, NSE's VP, regulations, was not the subject of a forensic probe. His email records were not scanned. He was barred from holding any position with any market infrastructure intermediaries for two years but was not held guilty of any conspiracy at the exchange.

Like bathing in the sacred Ganges can wash away your sins, SEBI's ban cleansed the researchers of their criminal acts of data theft and agreement

forgery. Susan, whose PhD degree is questionable, escaped any action by SEBI at all.

Were NSE's top bosses Narain and Ramkrishna aware of Susan's missing degree as data was being shared with the two since 1995? Why could SEBI not pin Susan for lying about her PhD degree? Despite saying that Ajay held too much influence on NSE and its management, SEBI has failed to make a case about how he enriched himself with that.

The key foundational principle of SEBI, when it came into being in 1992, was its commitment to retail investors, the common investing public. When the Harshad Mehta and Ketan Parekh stock rigging scandals hit India's markets, they did not prove as fatal to the '*cult of equity market investing in the country*' as the COLO scam at the NSE.

Way back in 2011, the International Organisation of Securities Commission (IOSCO), the global body of stock market regulatory of which SEBI is a signatory and an active member, had published its concerns on inequality.⁵³ The report clearly mentions that SEBI had informed IOSCO that it had set up a TAC in March 2010 to advise on issues relating to HFT, COLO trading and security issues related to internet-based trading. Also, SEBI is responsible to conduct an annual inspection of exchanges and their trading infrastructure.

"Co-location raises issues related to potential distortion of competition between market members, equal access to the market and the cost of such services. These issues could be especially relevant if access to co-location services is limited in the short-run by the physical limit of spare capacity. In particular, the fact that some participants may receive information on order book trading interest and executions sooner than others, and have their orders entered in the trading system more rapidly may raise questions regarding the fairness and integrity of the markets," the IOSCO paper said while highlighting the grave issues of market access involved in market access via COLO.

So, let us for argument sake presume that there was no knowledge about the manipulation of COLO grid. Despite that, it is amply clear that SEBI behaved irresponsibly with regard to NSE. What was the influence? In the background of this, whom does SEBI hold responsible for the failure in duty of its own officials in allowing NSE to start high-tech trading and COLO operations without requisite permissions?

There is compelling evidence of a mega scandal involving NSE's COLO grid with wide ramifications, and the travesty of justice. Assumption of SEBI and its probe lack deductive logic and **empirical observations**. Was such a scandal possible without political patronage or the backroom brokers in the loftiest corridors of power?

The same can be seen even in the case of the data scandal at MCX, where the forensic auditor report was a screaming evidence and involvement of the same ring involving Ajay and Susan? Who in SEBI is responsible for not probing the matter further and burying the scam? Ignorance has been SEBI's strength.

In May 2018, the CBI registered a first information report (FIR) against OPG Securities, Sanjay, his brother-in-law Kokrady, Ajay Shah and other '*unnamed officials*' of the NSE and SEBI. The CBI is probing the case under the Indian Penal Code (IPC) section 120 B (criminal conspiracy), 204 (destruction of evidence), Prevention of Corruption Act 1988 and IT Act 2000, section 66 that make the unauthorised access to computer resources and breaking into digital systems a punishable offence. However, the need is to involve an agency that can probe possible money laundering angles too as the money trail has been left totally untouched.

In 2019, the agency told the Delhi HC that it would investigate the '*wider angle*' and '*wider conspiracy*' involved in the NSE COLO scam and that its scope of the investigation was not limited to the names mentioned in its regular case and could be expanded to persons yet to be named. However, even two years after registering the case, the CBI so far has not achieved any

meaningful breakthrough.

In October 2019, the Madras HC has issued notices to the SEBI, CBI, ED and three other agencies in connection with the COLO scam. Will the CBI investigation be also going the SEBI way?

Naturally, brokers, bureaucrats, regulatory officials and exchange executives, “*The Market Mafia*”, have spread their propaganda thick and fast as per which if the scam was investigated further, the markets would crash severely. Rather naively, the incumbent government seems to have taken the bait. It has ignored the fact that nowhere in the world has the stock markets ever crashed due to investigations into some of the largest financial scandals, as it does not affect the corporate profits, which ultimately drive stock prices.

If anything, the COLO scandal has surely left the Indian markets in a more vulnerable state, as it has eroded confidence in SEBI and its regulatory prowess. The need is to review the requirement for the COLO trading firm, the dark force, within the exchange premise. SEBI in its current form is a supreme example of official incompetence and obstinacy, where high ranking officials are thriving on collusion and cronyism since their appointment hinges on the ‘*bureaucratic brotherhood*’.

The only way to fix the ‘*conflict*’ that runs deep in SEBI’s working is to split the organisation into two – one to form the rules and regulate the market and another to investigate and adjudicate the scams triggered by inefficiency in those regulations. Both the verticals should have separate chairs. It will, for a long time, scuttle the rise of another Bhavé and his ‘*brotherhood*’.

THE WHIFF OF MONEY

Manna from Mauritius and Cayman

The Grand Hyatt Plaza is Mumbai's swanky five-star hotel in equal vicinity to the international airport to its North and BKC in the South. It is strategically located, at least for the fund managers and investment bankers who want to make that last quick sortie to the business district BKC and escape the rush hour load to take the next flight back to their home destination.

But it's not all for which money managers stay here. Few suits in the hotel act as the registered address for a few funds houses and companies. In them, a certain F-9C suite in the hotel seems to be sought after. Nearly a dozen entities share this address: F-9C suite, Grand Hyatt Plaza, Santacruz (East), Mumbai MH 400055, as their common registered addresses.

Entities registered at the F-9C suite have billions of USD flowing to it from the mystic blue-sea islands of Mauritius and Cayman, the backyard of the world's filthy rich. The cash flowing from these *'tax havens'* is all *'white money'* ready to be moved anywhere in the world. Another similar address that attracts much *'Manna from Heavens'* is Rocklines House, Ground Floor, 9/2, Museum Road, Bengaluru, Karnataka.

When a Limited Liability Partnership (LLP) company in Mumbai or Bengaluru gets Manna from Mauritius and Cayman, it most often morphs into Private Equity (PE) investment and can be further deployed or just held ideal until those holding it chalk-out a roadmap for its use. SEBI has even accorded some such PE firms a licence under Category-II fund, which means high-risk funds that cannot issue P-Notes or derivatives.

SA Dave, aged over 80 years, who was the first boss of SEBI and currently a chairman in Ajay's family-owned company CMIE, along with Susan, primarily a professor and researcher, are '*designated partners*' in PE schemes of a fund called the '*True North*'.

Director Details				
DIN	Director Name	Designation	Appointment Date	
00001456	SURESH NARSAPPA TALWAR	Director	18 September 2006	View other directorships
00001480	SURENDRA AMBALAL DAVE	Director	18 September 2006	View other directorships
00472794	SUSAN THOMAS	Director	13 March 2007	View other directorships
Other Companies Associated with				
Company	Designation	Appointment Date		
TRUE NORTH FUND V LLP	Nominee-Body Corp Partner	19 January 2017		
NORTHERN ARC CAPITAL LIMITED	Director	11 October 2011		
TRUE NORTH FUND TRUSTEESHIP PRIVATE LIMITED	Director	13 March 2007		
FINCARE SMALL FINANCE BANK LIMITED	Director	02 September 2014		
00001480	SURENDRA AMBALAL DAVE	Director	18 September 2006	View other directorships
Other Companies Associated with				
Company	Designation	Appointment Date		
TRUE NORTH FUND V LLP	Body Corporate OP Nominee	07 August 2017		
CENTRE FOR MONITORING INDIAN ECONOMY PRIVATE LIMITED	Director	01 May 1995		
TRUE NORTH FUND V LLP	Body Corporate OP Nominee	20 December 2015		
DECCAN CEMENT LIMITED	Director	12 August 2011		
TRUE NORTH FUND TRUSTEESHIP ENTITY PRIVATE LIMITED	Director	24 January 2018		
PHOENIX TOWNSHIP LIMITED	Director	30 June 1996		
TRUE NORTH FUND TRUSTEESHIP PRIVATE LIMITED	Director	17 July 2000		

As per its annual returns up to March 31, 2019, the True North Fund V

LLP, just a single scheme under the umbrella of True North, had received a total partner contribution of Rs. 33,64,25,75,591 (Rs. 3,364 thousand crores or USD 442 million at the current exchange rate of 76) out of the total obligation of Rs. 44,32,40,00,000 (Rs. 4,432 cores – more than USD 500 million) that was committed to it.

Who Contributed This Money?

A meticulous search for the document trail shows that as of March 2019 a contribution of Rs. 28,42,64,38,445 (Rs. 2,842 crores or USD 375 million) into True North Fund V LLP came from '*Indium V Mauritius Holdings*'. Another Rs. 3,16,18,42,590 (Rs. 316 crores or USD 41.60 million) was received by the scheme from Cayman Island fund '*QS FF EM India*'.

Both these sources, just mail-box companies, have been painted '*white*' to the best of the ability of top-notch legal shops ensconced in the paradise islands. Yet, the trail reveals itself.

The below snapshots reveal partner and registration of True North Scheme Fund V LLP. It is Form-11 filing of MCA and shows who contributed money to the scheme.

The below snapshot is a document of True North Fund V LLP registration details from MCA.

State of bodies corporate as partners

Type of body corporate	Companies incorporated outside India/registered in Sikkim	Pre-Fill
Corporate identity number (CIN) or Foreign company registration number (FCRN) or limited liability partnership identification number (LLPIN) or any other identification number	126686 C1/GBL	
Name of the body corporate	INDIUM V MAURITIUS HOLDINGS LIMITED	
Full address of the registered office or principal place of business in India	Mauritius 201 STERLING TOWER 14 POGRIERE STREET PORT LOUIS	
Country where registered	INDIA	
Obligation of contribution (in Rs.)	38,07,00,00,000	Contribution received and accounted for (in Rs.) 28,42,64,38,445
Name and particulars of person signing on behalf of body corporate as nominee		
Category	Partner	
DPIN/ Income-tax PAN/ Passport number	1466397A	
Name	ASHRAF RAMTOOLA	
Father's Name	ELIAS RAMTOOLA	
Permanent Residential Address	4 BIS CHURCH STREET, PORT LOUIS, MAURITIUS Mauritius	
Present residential address	4 BIS CHURCH STREET, PORT LOUIS, MAURITIUS	
Nationality	MAURITIUS	
Date of Appointment	28/12/2016	(DDMMYYYY)
Date of Cessation		

LLP FORM NO. 11
[Pursuant to rule 25(1) of Limited Liability Partnership Rules, 2009]

Annual Return of Limited Liability Partnership (LLP)

Note - All fields marked in are to be mandatorily filled.

1. (a) Annual return made upto 31st day of March of 2019 Year

(b) Start date of financial year for which annual return is being filed 01/04/2018 (DD/MM/YYYY)

2. Limited Liability Partnership identification number (LLPIN) AAI-0542 Pre-filled

3. Name of the Limited Liability Partnership TRUE NORTH FUND V LLP

4. (a) Address of the registered office of the LLP
ROCKLINES HOUSE, GROUND FLOOR, 9/2, MUESUM ROAD
BANGALORE
Karnataka
560001
India

(b) e-mail ID compliance@truenorth.co.in

5. Other address if declared under section 13(2) for service of documents
ROCKLINES HOUSE, GROUND FLOOR, 9/2, MUESUM ROAD
BANGALORE
Karnataka
560001
India

6. Business Classification Service

7. Principal business activities of the LLP
LLP will operate as a category II Alternate Investment Fund under Securities Exchange Board of India (Alternate Investment Fund) Regulations, 2012 and seek registration as such with the SEBI.

8. Details as on 31st March of the period for which annual return is being filed

(a) Total number of designated partners 2 Total number of partners

(c) Total obligation of contribution of partners of the LLP (Rs.) 44,32,40,00,000

(d) Total contribution received by all partners of the LLP (Rs.) 50,84,25,75,501

Note: 'Contribution received' to be entered in corresponding Form 8 should be same as the value entered in field 8(d) above.

A Mauritius resident, Ashraf Ramtoola, having his permanent house address as '4 BIS Church Street, Port Louis' is the nominee for 'Indum V Mauritius Holdings Limited'.

Nearly 18 months after the 'Panama Paper Leaks', a German newspaper *Sueddeutsche Zeitung* broke the story on 'Paradise Paper Leaks'.⁵⁴ Akin to Panama leaks, the Paradise Papers too were a massive trove of secret financial data revealed by the International Consortium of Investigative Journalists (ICIJ).

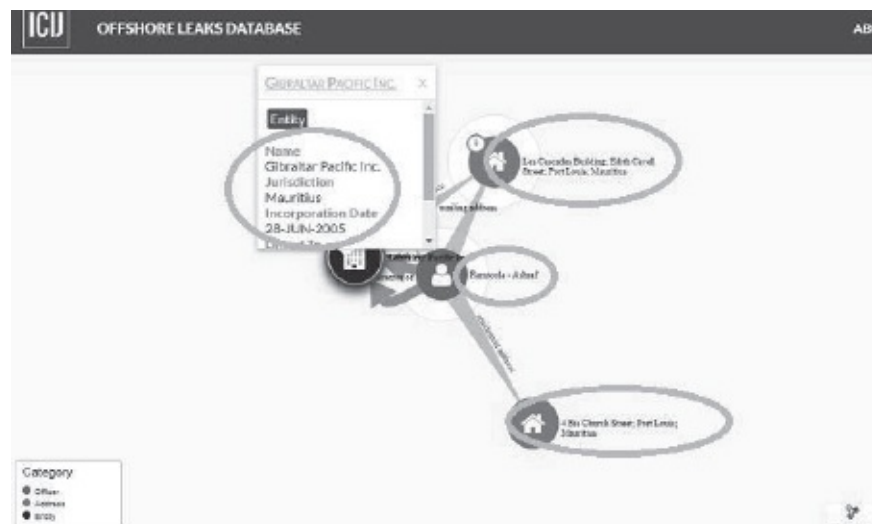
It contained 13.4 million documents from leading off-shore finance firms Appleby in Bermuda and Asiatic Trust in Singapore. Both these firms had a

global network of lawyers, accountants and bankers who offered their services for setting up off-shore vehicles that diverted cash as '*white money*' to near-zero tax rates in 19 tax havens. Among their clients were global elite, many of them politicians, who were also put on court trial in their respective country and had to step down from their position.

India too had its share of names that came out in the media with links to Paradise Papers. They included tainted political-cum-corporate lobbyist, Niira Radia, absconding and fugitive liquor baron, Vijay Mallya, film star, Amitabh Bachchan, gun-running convict Sanjay Dutt's wife, Manyata, BJP's Jayant Sinha and Ravindra Kishor Sinha.⁵⁵

But those who were smart enough to borrow pseudo names or recruit a front were not identified by the media and India's investigative agencies too did not pay attention in deciphering it. A trail still exists for those who want to pick it up.

ICIJ in its database has linked Ramtoola with his residential addresses in Mauritius to '*Paradise Paper*' company Gibraltar Pacific Inc, a client of Appleby that was busted in the leak.



The above pic has four circles. They display the person's name (Ramtoola) who was busted in Paradise Paper Leaks as a client of Appleby,

his two addressed in Mauritius and connection to Gibraltar Pacific.

Who is Ramtoola?

His name first cropped up in SEBI investigations with regard to dealings of Mavi Investment Fund Limited that played heavily in Indian stock markets between 2008 and 2012. SEBI, which was investigating the fund for insider trading, said that Mavi Investment was a registered sub-account of FII Warburg Bank.

Ramtoola was a company secretary of Mavi Investment and GE Capital Mauritius Investment Company (not to be mistaken with global giant General Electric). He was also a director on board of International Management Mauritius (IMM). His LinkedIn profile lists him as an accountant at IIM.⁵⁶

SEBI found that IMM was the company secretary of Mavi Investment, since its inception on June 5, 2005, and apart from acting as company secretary it was also its fund administrator and bank signatory. Ramtoola was earlier a director of Mavi Investment from June 5, 2005 to February 24, 2006. Post that he remained the authorised signatory for Mavi Investment and SEBI observed that “he was a man of confidence to sign for the company and had to be aware of the activities of the company,” which was trading heavily in Indian markets then.

These observations were made by SEBI adjudicating officer, Barnali Mukherjee, with regard to investigations into an insider trading case. In that case, SEBI let off Ramtoola and Mavi Investment for *‘lack of proper evidence’*. The SEBI adjudication report is available on its website.⁵⁷

But fate caught up a few years later and Mavi Investment was banned by SEBI for its role in the Global Depositary Receipt (GDR) scam that was unearthed in 2011.

Six companies including Cals Refineries, Asahi Infrastructure, IKF

Technologies, Avon Corp, K Sera Sera (later renamed KSS Limited), CAT Tech and Maars Software issued GDRs in London. These were subscribed by Mavi Investment, India Focus Cardinal Fund, KII Limited (Sub-Account), Sophia Growth (a share Class of Somerset India Fund, Sub-Account), European American Investment Bank Ag (FII), Basmati Securities Pvt Ltd, Oudh Finance & Investment, Alka India, SV Enterprises and JMP Securities.

When Indian entities sold GDR, '*white money*' parked in Mauritius came into these companies in a legitimate way. Who were the ultimate owners of these FIIs and sub-accounts? SEBI could not find out. Later, these FIIs quickly converted their GDRs into ordinary shares and dumped them on Indian retail investors.^{58,59} A 100-page report that gave details about SEBI investigation into the GDR is available on its website.⁶⁰

SEBI had charged Mavi Investments for colluding with merchant banking firm Pan Asia Advisors Limited (now renamed as Global Finance and Capital Limited) by subscribing to issues marketed by it, converting the GDRs into shares within a short time and then selling them back to entities controlled by Pan Asia and promoters of the companies. Dubai-based NRI, Arun Panchariya, then MD and chief promoter of Pan Asia held 100 per cent equity in the company. Coincidentally, Panchariya was also the main promoter of K Sera Sera.

In a loss of face then too, SEBI let off Mavi Investment ahead of other FIIs in 2013 under '*benefit of the doubt*'. But a tough nut as it was, Mavi Investment was again caught up in the net, this time by India's CBI in relation to the country's most illustrious money laundering racket involving the 2G scam.⁶¹

The CBI managed to track down the spectrum scam money trail passing through 22 different Indian and Mauritian companies to a single building in Mauritius. 10 of the 12 Mauritius-based companies under the CBI scanner including Mavi Investment Fund, Delphi Investment, Capital Global, Black Lion, Inditel Holding, Deccan Asian Infrastructure, Aidtel Holding, Kaif

Investment, Electro Investment Ltd, Palab Investment had one thing in common – their address at Les Cascades Building, Edith Cavell Street, Port Louis, Mauritius. (lviii)

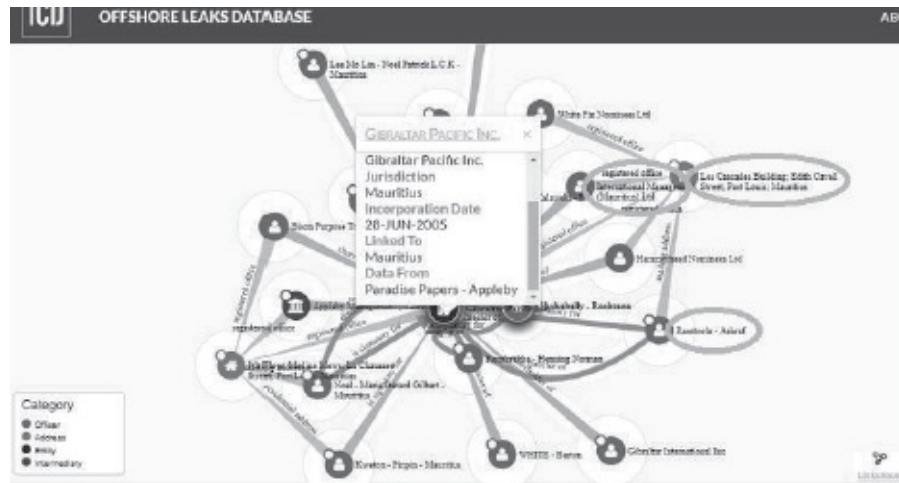
The money trail in the 2G scam led the CBI to discover an amount of USD 31 million with Mavi Investments. Like many companies in Mauritius, the official owner held only 1 per cent in Mavi Investment and the CBI never got to know who invested USD 4 million from Mavi Investment to buy Reliance Telecom's 9.9 per cent stake in Swan Telecom.⁶² Mavi Investment also held shares in DB Realty Limited whose promoter Sahid Usman Balwa was kept in jail custody in 2G scam for months before being released on bail.⁶³ DB Realty had its part-owners as the Patel brothers — Abdulla, Adil and Nabil – the three sons of late Mumbai underworld don Yusuf Patel.

Mavi Investment shares common links in terms of company director in Mauritius with Passport Capital LLC. The fund Passport Capital LLC (Mauritius) too has had its share of skirmishes with SEBI and its sub-account was even banned for front-running.⁶⁴ Both Mavi Investment and Passport LLC share common registration address – Les Cascades Building, Edith Cavell Street, Port Louis, Mauritius.

An officer named Noel Marie Gerard Gilbert in Mauritius, who is also a director of Mavi and Passport Capital LLC is connected to 363 entities as per ICIJ database on off-shore leaks.⁶⁵ Then, there are Pakistani and Chinese linked funds and corporations registered at the same building in Mauritius. Some of the Mauritius firms that cater to these Pakistani and Chinese funds and corporations also serve Indian-linked entities.

The ICIJ database shows that Ramtoola's Gibraltar Pacific named in the '*Paradise Papers*' like Mavi Investments, where he was a company secretary and designated signatory, is registered at – Les Cascades Building, Edith Cavell Street, Port Louis, Mauritius.⁶⁶ Ramtoola is a signatory and director in Gibraltar Pacific, which in turn is connected to 18 other entities.^{67,68} All of these were linked to '*Paradise Paper Leaks*'.

The above snapshot shows that Ramtoola is a signatory and director at Gibraltar and his residential address matches with that of Indium Mauritius Holdings that has contributed to True North. Below Ramtoola's Gibraltar is connected to 18 other entities.



Ramtoola's link has now come a full circle as investments in True North. His Indium VI Mauritius Holdings, which is registered as a partner in True North Fund VI LLP a SEBI-recognised Category-II fund, is registered at Edith Cavell Street, Port Louis, Mauritius where all the 2G scam companies and Mavi Investment was registered.

Below image is an MCA filing of True North Fund VI LLP that shows its partner, address and contribution.

11. Details of bodies corporate as partners

1	Type of body corporate	Companies incorporated outside India/registered in Sikkim	Pre-
	Corporate identity number (CIN) or Foreign company registration number (FCRN) or Limited liability partnership identification number (LLPIN) or Foreign limited liability partnership identification number (LLPIN) or any other identification number	C151704 C1/GBL	
	Name of the body corporate	Indium VI Mauritius Holdings Limited	
	Full address of the registered office or principal place of business in India	SGG Fund Services (Mauritius) Ltd Edith Cavel Street Port Louis Mauritius	
	Country where registered	INDIA	
	Obligation of contribution (in Rs.)	22,55,50,00,000	Contribution received and accounted for (in Rs.) 0
	Name and particulars of person signing on behalf of body corporate as nominee		
	Category	Partner	
	DPIN/ Income-tax PAN/ Passport number	001466597	
	Name	Ashraf Ramtoola	
	Father's Name	Elias Haji Allimamode Ramtoola	
	Permanent Residential Address	4 Bis Church Street, Port Louis, Port Louis Mauritius	
	Present residential address	4 Bis Church Street, Port Louis, Port Louis	

Sometimes he is a '*designated signatory*' in the bank accounts of a Mauritius company that keeps pouring funds into India, in some other instance, an '*accountant*', or a company secretary and now a '*partner*' in a fund contributing millions of dollars in PE True North.

He got in trouble with SEBI multiple times for manipulation and moving money but the regulator could not gather further evidence and he was let off very lightly. His company and fund fell into CBI net in 2G money laundering but the agency hit a wall and could not lift the veil over his firms in Mauritius. His LinkedIn profile lists him as an accountant and a senior manager.⁶⁹ The trail spins around like a merry-go-round.

In November 2010, the New Delhi Income Tax Tribunal had passed an order against SMR Investments Limited, Mauritius, in which it said that the

investment company did not have a *Permanent Establishment* certificate. The tax department had then found that Ramtoola was a former director and signatory of SMR Investments Limited and an Indian called Suresh Rajpal held 99 per cent shares in the company. Rest 1 per cent share was held by Ms. Mavis Tse Rajpal. The judgement is still available. (Ixviii) Suresh Rajpal did not furnish his passport to the tax department. But it made clear that Ramtoola was just a rubber-stamp.

As a matter of fact, the tribunal also made an observation which said, “The purpose of setting up a company in Mauritius with 99 percent shareholding was merely to escape capital gains liability in India since there is no tax on capital gains in Mauritius.”

In Mauritius, the structure of a company secretary in many cases is not an individual but a registered firm that services many other companies. The employees of that company secretary firm or service provider can be designated partner, director or an authorised signatory.

Further evidence about this can be examined in another example. India’s most respected and diversified business house, Godrej Industries has a subsidiary called Godrej East Africa Holdings (Mauritius) Limited. In its corporate filings, Godrej East Africa states Ramtoola as one of its directors along with Rooksana Shahabally, Sevin Chendriah, Jatinkumar Onkar Brahmecha and Naveen Gupta.

In the same filing, a company SGG Corporate Services (Mauritius) based out at 33, Edith Cavell Street Port Louis, 11324, has been shown as ‘*administrator and secretary*’ to Godrej East Africa up to March 21, 2019 (it is also a service provider to Indium VI Mauritius Holdings, which is a partner in True North). After March, the filing of Godrej East Africa says that IQEQ Corporate Services, Mauritius became its ‘*administrator and secretary*’. The same filing on page no. 41 and point no. 17 has stated that Ramtoola is an employee of IQEQ Corporate Services along with all others. The report is available on the web.⁷⁰

There are other filings with the UK regulators and SEC that show Ramtooala as a ‘*secretary*’.

In tax havens, there are several financial services firms that provide end-to-end solutions from incorporating to managing a company or fund and give it a name, face and legitimacy. The money is held in a custodian bank, where there are signatories, which could be another front of ultimate beneficences. The use of money is decided by the beneficiaries to whom it belongs and the structures merely execute their mandate.

In those terms, Mauritius rules are very convenient for hidden beneficiary owners. They say a company can hold its board meeting, which can be chaired and minutes are recorded via telephone. There is no full public access to accounts of the company, which can be represented by two or more Mauritius residents. The shares issued may be redeemable, non-voting or confer preferential, special or limited rights to income, capital or voting as specified in the bye-laws (constitution) of the company incorporated there.

As per laws prevailing in Mauritius the names of key beneficial owners of corporations and funds are kept highly confidential. Further, Mauritius will not make any disclosures to any court, tribunal, and committee of enquiry or other authority in Mauritius or elsewhere except on a court order made. That too only if the court is ‘*satisfied that the confidential information is bona fide*’ required for the purpose of any enquiry or trial into, or relating to the trafficking of narcotics, drugs, arms trafficking or money laundering.

The powers of Mauritius regulatory authority the Financial Services Commission (FSC) to transmit information have been extended to include disclosure to foreign supervisory agencies – applies – with respect to financial services providers only and not to private holdings/special purpose vehicles. Mauritius has tax treaties with 34 countries, which exempt entities coming from the island nation from tax. The ‘*maximum*’ tax rate in Mauritius is 3 per cent.

The Financial Action Task Force (on Money Laundering) (FATF), a

global body policing back-money and terrorist financing jurisdictions has put Mauritius in the grey list this year for serious deficiencies in its anti-money laundering laws. It falls in the list of countries including the Bahamas, Barbados, Botswana, Cambodia, Ghana, Jamaica, Mongolia, Myanmar, Nicaragua, Panama and Zimbabwe. Countries that were already on the list are Afghanistan, Iraq, Vanuatu, Pakistan, Syria, Yemen, Uganda, Trinidad and Tobago, Iran and North Korea.

Investigations carried out by ICIJ between 2016 and 2017, based on a cache of 200,000 confidential records from the Mauritius office of the Bermuda-based off-shore law firm Conyers Dill & Pearman, reveals how a sophisticated financial system based on the island is designed to divert tax revenue from poor nations. The files date from the early 1990s to 2017. ICIJ says that in Mauritius there could be some legitimate corporations too.

But Ramtoola's past record with regard to his links to 2G scam companies and market rigging operations should have been enough for the government and SEBI to raise a red flag when he returned as a front via huge PE investments.

"What Mauritius is providing is not a gateway but a getaway car for unscrupulous corporations dodging their tax obligations," said Alvin Mosioma, executive director of the non-profit Tax Justice Network Africa.⁷¹

The leaked records – including emails, contracts, meeting notes and audio recordings – provide a glimpse inside a busy off-shore law firm working with global accounting and advisory firms for some of the world's largest corporations and some very wealthy individuals. They've all found their way to an island built on helping the rich avoid paying taxes to nations as far-flung as the US, Thailand, and Oman, ICIJ says in its investigations.

ICIJ further says that where there are tax shelters, "There are Big Four accounting firms. KPMG, PricewaterhouseCoopers (PwC), Deloitte and EY all have offices in Mauritius."

The book *Thin Dividing Line: India, Mauritius and Global Illicit Financial Flows* by Paranjay Guha Thakurta and Shinzani, provides literature on Mauritius route for routing cash to India.

As per a July 2019 story in *The Economic Times*, investments in India via companies registered in Mauritius have shrunk since the government renegotiated the terms of the bilateral double tax avoidance treaty in 2016. But the tax haven still continues to be a hub for capital flows to companies in India. According to tax records accessed by the ICIJ, “As many as a fourth of all companies named in the leaked documents had India as their country of activity,” *The Economic Times* said in its report.⁷²

There have been numerous reports to show that P-Note investments in India largely came from Mauritius. After the clamp-down on P-Notes, entities have changed their form and manner of routing money into India.

Mauritius says it has made a high-level political commitment to get itself out of the list and it has said that repeatedly over the years whenever fingers were raised at it. But the European Union was contemplating black-listing it. Even if it manages to wriggle itself out of the ‘grey list’ through political jugglery and high patronage of big money bags and the global elites holding interests there, will Mauritius make the names of ultimate beneficiaries of its funds and corporations public for transparency?

Although going by its record, it would be easier said than done, let’s say; counting sand grains on Grand Baie (a popular Mauritius beach) could be a much easier task.

Is Ramtoola, who contributed Rs. 2,842 crores in True North, a front to laundering money for politicians, businessmen, senior company executives, bureaucrats, cops, brokers, bankers or researchers from India?

Otherwise, why would a person have such a complex and vague structure if he is so rich and interested in just investing? Why did the previous government let the trail go cold and the current government, too, has not

investigated Ramtoola links in relation to 2G? As a consequence, Ramtoola has reappeared as a PE investor, albeit as a front.

Indium V Mauritius Holdings, through which Ramtoola has contributed Rs. 2,842 crores to True North V, holds more than Rs. 8,344 crores (over USD 1,098 billion) in its kitty and has 30 beneficial owners.⁷³ The details of these 30 beneficial owners can only be shared by the government of Mauritius. But needless to say, these entities again will be registered in various tax havens, and those countries like in the 2G case may not give the data even to CBI.

Law in Mauritius defines the term '*Beneficial Owner*' or '*Ultimate Beneficial Owner*' as a natural person who holds by himself or his nominee a share or interest in a share which entitles him to exercise not less than 20 per cent of the aggregate voting power in a meeting of shareholders/partners or council members.

Fund Name INDIUM V (MAURITIUS) HOLDINGS LIMITED	General Partners, Managers, Trustees, Directors GLOBAL OPPORTUNITY ADVISORS (MAURITIUS) LIMITED
Fund Type Venture Capital Fund	
Fund Location Mauritius	Administrators (1) IQ EQ FUND SERVICES (MAURITIUS) LTD Mauritius Percentage of Assets Independently Valued: 100
Fund GAV \$1,098,000,000	
Number of Beneficial Owners 30	
General Partners, Managers, Trustees, Directors GLOBAL OPPORTUNITY ADVISORS (MAURITIUS) LIMITED	Auditors (1) DELOITTE MAURITIUS Mauritius Not Registered with PCAOB
Administrators (1) IQ EQ FUND SERVICES (MAURITIUS) LTD Mauritius Percentage of Assets Independently Valued: 100	Custodians (1) KOTAK MAHINDRA BANK LIMITED India

The snapshot above taken from Mauritius financial services database gives details of a number of beneficial owners of Indium V Holdings, its Gross Asset Value and the custodian bank. The Global Opportunity Advisor you

see in this is an advisor to Indium, a service provider. Belows of Indium IV also linked to India.

Fund Name INDIUM IV (MAURITIUS) HOLDINGS LIMITED	
Fund Type Venture Capital Fund	
Fund Location Mauritius	
Fund GAV \$392,000,000	
Number of Beneficial Owners 51	
General Partners, Managers, Trustees, Directors GLOBAL OPPORTUNITY ADVISORS (MAURITIUS) LIMITED	Administrators (1) IQ EQ FUND SERVICES (MAURITIUS) LIMITED Mauritius Percentage of Assets Independently Valued: 100
Administrators (1) IQ EQ FUND SERVICES (MAURITIUS) LIMITED Mauritius Percentage of Assets Independently Valued: 100	Auditors (1) DELOITTE MAURITIUS Mauritius Not Registered with PCAOB.
	Custodians (1) HDFC BANK LTD India

Indium has another scheme *Indium VI*, which has an obligation to contribute Rs. 2,255 crores to True North VI. Likewise, there is Indium III, Indium IV, Indium PCC – Cell Beta. The data can be found online. (lxxii) INDIUM IV (MAURITIUS) HOLDINGS LIMITED has Rs. 2,792 crores (US \$ 392,000,000) as Gross Asset Value. Obviously, the stakes are very high indeed.

Cayman Link – Barack Obama’s Revelations

A five-storey building called Uglan House in the Cayman Islands (An autonomous British Overseas Territory situated in the western Caribbean Sea) is home to one of the largest cross-border international law firms called Maples and Calder (Now renamed as Maples Group). The firm is part of the global off-shore magic ring spread across tax havens.

This apart, the Uglan House is home to more than 19,000 companies – on paper. Interestingly, entities registered in Cayman could be more than the country’s population of just around 64,000.

One corporation is registered in a measly less than 3 sq.ft. space in the inconspicuous office. In 2009, during his first Presidential campaign, Obama referred to Uglan House as, “Either the largest building in the world or the biggest tax scam.”⁷⁴

True North Fund V LLP has one of its partners as QS FF EM India Limited, registered at: Uglan House, KYL – 1104 Grand Cayman Islands. That fund has contributed Rs. 3,16,18,42,590 (Rs. 316 crores or USD 41.60 million) to True North Fund V, which is registered in Bengaluru, a metropolis in South India. QS FF EM is the second partner of True North V after Ramtoola’s Indium V Mauritius Holdings.

Why would a simple PE True North, which wants to invest in Indian companies, require a complex structure and money pouring in from a maze of entities across two of the world’s most infamous tax havens?

After Mauritius, Cayman was another large base of P-Note investors in India. As per a 2016 report, nearly 41 per cent of India’s P-Note holding entities had end-beneficial owners located in Cayman while only 11 per cent were left in Mauritius.⁷⁵

In this light, structures like Indium Mauritius and QS FF EM Mauritius, serve no other purpose than rerouting money via mail-box companies of wealthy investors who want to remain in the shadows.

2	Type of body corporate	Companies incorporated outside India/registered in Sikkim	Pre-filled
	Corporate identity number (CIN) or Foreign company registration number (FCRN) or Limited liability partnership identification number (LLPIN) or Foreign limited liability partnership identification number		NA
	Name of the body corporate	Q6 FF EM INDIA LIMITED	
	Full address of the registered office or principal place of business in India	Cayman Islands UGLAND HOUSE KYL - 1104 GRAND CAYMAN CAYMAN ISLANDS	
	Country where registered	INDIA	
	Obligation of contribution (in Rs.)	3,17,30,00,000	Contribution received and accounted for (in Rs.) 3,16,16,42,590
	Name and particulars of person signing on behalf of body corporate as nominee		
	Category	Partner	
	DPIN/ Income-tax PAN/ Passport number	12AV04612	
	Name	JOHANN DUMAS	
	Father's Name	Dumas	
	Permanent Residential Address	16 RUE DES COQUELICORS, 57570 BASSE RENTGEN, FRANCE	
		France	
	Present residential address	16 RUE DES COQUELICORS, 57570 BASSE RENTGEN, FRANCE	
	Nationality	FRANCE	

Indian Faces of True North

IVF was renamed as True North in 2017. Currently, its scheme True North V LLP has an amount of Rs. 3,364 crores (approximately USD 420 million) in its kitty. In this, the two mail-box companies in Mauritius and Cayman investors have contributed 99.90 per cent funds as of March 2019. While only 0.10 per cent contribution has come mainly from True North Trusteeship.

It has contributed Rs. 200 crore (USD 26.61 million) in the scheme and has IGIDR researcher Thomas as a nominated partner and board member along with CMIE's Dave and Suresh Talwar who is director of a host of other companies in India and Mauritius.

In 2019, Ashok Wadhwa, the owner of Mumbai-based investment bank-cum-equity brokerage and research house Ambit, held 49,990 shares or 99.9 per cent stake in True North Fund Trusteeship as a co-sponsor. He held the same 99.9 per cent stake when the company was called IVF. The idea of LLP is that the liability of every partner is limited to their contribution.

Type of body corporate	Company		Pre-Fill
Corporate identity number (CIN) or Foreign company registration number (FCRN) or limited liability partnership identification number (LLPIN) or any other identification number	U74140KA2006PTC040274		
Name of the body corporate	TRUE NORTH TRUSTEESHIP PRIVATE LIMITED		
Full address of the registered office or principal place of business in India	GROUND FLOOR, ROCKLINES HOUSE 9/2, MUSEUM ROAD, BANGALORE Karnataka 560001		
Country where registered	INDIA		
Obligation of contribution (in INR)	3,02,10,00,000	Contribution received and accounted for (in INR)	2,00,61,65,742
Name and particulars of person signing on behalf of body corporate as nominee			
Category	Partner		
DPPI/ Income-tax PAN/ Passport number	ACVPT2275J		
Name	SUSAN THOMAS		
Father's Name	VERGHESE		
Permanent Residential Address	U.G.I.D.R.E-7 Gen Arun Kumar Vaidya Mang Goregaon East Mumbai Maharashtra 400055		
Present residential address	U.G.I.D.R.E-7 Gen Arun Kumar Vaidya Mang Goregaon East Mumbai		
Nationality	INDIA		
Date of Appointment	15/01/2017	(DD/MM/YYYY)	
Date of Cessation		(DD/MM/YYYY)	
Previous Designation		Date of change in designation	(DD/MM/YYYY)
Previous Name, if any			
Whether resident in India	☒ Yes ☐ No		
Number of limited liability partnership(s) in which he/she is a partner	1		
Number of Company(s) in which he/she is a director			

Wadhwa was earlier the co-founder of tax consulting firms RSM and Co. and managing partner of Arthur Andersen. He is a law graduate and a chartered accountant. His firm Ambit is ensconced at a multi-storey plush commercial building in Parel, another hub for financial companies and corporate offices in Mumbai, yet Wadhwa has registered the PE fund at a hotel suite and a small office in Bengaluru.

Apart from Dave and Susan who are partners and key managerial persons with True North, Ajay is an advisor to the fund. After the fund picked up 100 per cent stake in 2014 in National Bulk Handling Corporation (NHBC), a commodity warehousing company, Ajay was put on its board. Another

board member of NHBC is Vijay Kelkar.

Why Should SEBI Have Investigated Ajay and Susan's Links With True North?

True North Fund V LLP **has incurred** a personal expense Rs. 61 crores. Administrative expense Rs. 26 crore. Total loss Rs. 85 crore. What part of this loss was incurred on people in the company and their expenses?

In the 2019 order, SEBI said, Ajay had derived undue benefit from his position at NSE. “Mr Shah held a position of high esteem and influence in the minds of the senior management of NSE, as evident from the statement of Ravi Narain, former MD and CEO of NSE. Mr Shah was capable of influencing the decision making of NSE management, which is exemplified by the fact that he and his wife suggested to NSE and NSE undertook a (liquidity index) project, for which the Infotech Financials was awarded the contract,” SEBI said.

Ajay has been long associated and actively engaged with NSE and several subsidiaries of the exchange. In fact, Ajay and Susan were the only two academicians with deep access into the NSE. They received trading data from the NSE, first, in their personal capacity and, later, as academics associated with IGIDR.

In case of senior NSE executive Lala, SEBI observed, “Suprabhat Lala being a senior official of NSE ought to have displayed professional integrity by disclosing to the senior management his connection with Infotech Financials through his wife i.e. Sunita Thomas, which had a strong cause to give rise to a conflict of interest to him as well as to Ajay Shah. Moreover, by sharing internal information and data about NSE and the minutes of technical advisory committee (TAC) with his wife (Sunita Thomas) who was a beneficiary party of the (liquidity index) project, Suprabhat Lala has committed breach of his duties to collude with other notices for the aforesaid fraudulent acts,” the order says.

All these statements by SEBI show that the regulator believed Ajay and Susan derived undue advantage from their position and access to NSE. In that sense, should or will the next logical step for the regulator not be to investigate the extent of benefit that Ajay and Susan derived from their advantageous position?

After all, it has been through their association with NSE in the early 1990s that they got access to capital markets and brokers. If they could use the data they stole from NSE for Algo software development, can they not use the same for other commercial purposes too?

If Susan is a designated partner and key managerial person in a company since 2010, which later contributed Rs. 200 crores to a PE fund, does the regulator not need to know who actually this money belonged to?

The order by SK Mohanty, WTM of SEBI, said, “Ajay Shah and other noticees have conclusively worked to fulfil their commercial goals by fraudulently using the data that was obtained by them from NSE to develop algo trading software. Keeping in line with the regular business of Infotech Financials Pvt Ltd, i.e. providing Algo trading software to traders in securities, the algorithmic trading software so developed from liquidity index (LIX) data was meant to be sold to the traders or brokers in the market to induce them to trade in securities with better trading results, on the strength of capability of such algo trading software prepared out of such exclusive data not ordinarily available to other market participants.”

“In my view,” Mr. Mohanty from SEBI says, “There is no difficulty in sharing of data by Mr Shah with an outside entity for commercial purpose as long as the agreement spells out clearly its objectives and provides for the commercial terms and conditions in the agreement in compliance with the stated data sharing policy of the exchange. However, when a data sharing agreement purely meant for research purposes assumes the colour of a commercial agreement thereby providing privileged access to confidential and exclusive trade data of the exchange to a select few persons who,

clandestinely exploit the said data for their commercial gains, it leads to serious issues leading to compromise on market integrity.”

In the case of True North, it is the association of Ajay and Susan that raises conflict of interest questions, especially since they were at the same time involved with crucial government and SEBI committees and also got vital data from NSE. Ajay and Susan’s proximity to Wadhwa, whose Ambit Institutional Equities also served FIIs, and their foothold in NSE and later at MCX is full of conflict that SEBI has not looked into. Susan was shown as a key managerial person in the filings of IVF even in 2010.

SEBI in its COLO scam investigation has also passed strictures against Ajay, hence it is the duty of the regulator to look deep into his dealings that could unravel the full aspects and ramifications of the scam he is involved in. Five years have passed since SEBI was first alerted by a whistleblower about the COLO scam in 2015.

Till date – the probe remains half-baked. Well, is anyone listening?

GLOSSARY

ACE – Ahmedabad Commodity Exchange

Algo – Algorithm

ASCPL – Advantage Strategic Consultancy Private Limited

BKC – Bandra Kurla Complex

BOLT – BSE On-Line Trading

BSE – Bombay Stock Exchange

BJP – Bhairtya Janta Party

CBI – Central Bureau of Investigation

CCI – Competition Commission of India

CDR – Call Data Record

CDSL – Central Depository Services Limited

CEO – Chief Executive Officer

CFT – Cross Functional Team

CMIE – Centre for Monitoring Indian Economy

CME – Chicago Mercantile Exchange

COLO – Co-location

COO – Chief Operating Officer
CTC – Computer to Computer
CTO – Chief Technology Officer
CTT – Commodity Transaction Tax
CTV – Chanakya Trade Vistas
CVC – Central Vigilance Commission
Deloitte – Deloitte Touche Tohmatsu
Demat – Dematerialised
DMA – Direct Market Access
DSE – Delhi Stock Exchange
ED – Enforcement Directorate
ED – Executive Director
EOW – Economic Offence Wing
E&Y – Ernst & Young
Fairgrowth – Fairgrowth Financial Services Ltd
FATF – Financial Action Task Force
FBI – Federal Bureau of Investigations
FDI – Foreign Direct Investment
FII – Foreign Institutional Investor
FIR – First Information Report
FM – Finance Minister
FMC – Forwards Market Commission

F&O – Futures and Options

FSC – Financial Services Commission

FTL – Financial Technologies (India) Limited

GDR – Global Depository Receipt

GS – Goldman Sachs

HC – High Court

HFT – High Frequency Trading

IAS – Indian Administrative Service

ICIJ – International Consortium of Investigative Journalists

ICJ – International Court of Justice

IFPL – Infotech Financials Private Limited

IGIDR – Indira Gandhi Institute of Developmental Research

IIHS – The Indian Institute for Human Settlements

IMM – International Management Mauritius

INC – Indian National Congress

ISVs – Independent Software Vendors

IVF – India Value Fund

IDBI – Industrial Development Bank of India

IIT – Indian Institute of Technology

IPC – Indian Penal Code

IPOs – Initial Public Offers

IPS – Indian Police Service

IRS – Indian Revenue Services

ISB – Indian School of Business

ISCO – International Organisation of Securities Commission

KYC – Know Your Customer

LaMag – Los Angeles Magazine

LIC – Life Insurance Corporation of India

LLP – Limited Liability Partnership

LME – London Metal Exchange

LSE – London Stock Exchange

MCA – Ministry of Corporate Affairs

MCX – Multi Commodity Exchange

MCX SX – MCX Stock Exchange

MD – Managing Director

MIMPS – Manner of Increasing & Maintaining Public Shareholding in Recognised Exchanges

MRD – Market Regulations Department

MTBT – Multicast TBT

NHBC – National Bulk Handling Corporation

NCDEX – National Commodities and Derivatives Exchange

NIPFP – National Institute of Public Finances and Policy

NSDL – National Services Depository Limited

NSE – National Stock Exchange

NSEL – National Spot Exchange Ltd

NYSE – New York Stock Exchange

ODIs – Off-Shore Derivative Instruments

OPG – OPG Securities Pvt Ltd

OTN – Omnesys Tradenet

PAN – Permanent Account Number

PE – Private Equity

PFUTP – Prohibition of Fraudulent and Unfair Trading Practices

PM – Prime Minister

PMO – PM's Office

P-Notes – Participatory Notes

PR – Public Relations

PRISM – Parallel Risk Management Software System

PSM – Product Support Team

PwC – PricewaterhouseCoopers

RBI – Reserve Bank of India

SAT – Securities Appellate Tribunal

SBX – Singapore Stock Exchange

SC –Supreme Court

SCN – Show-Cause Notice

SEBI – Securities and Exchange Board of India

SEC – Securities Exchange Commission

SFO – Serious Fraud Office

STT – Securities Transaction Tax

TAC – Technology Advisory Committee

TBT – tick-by-tick

TCP/IP – Transmission Control Protocol/Internet Protocol

TCS – Tata Consultancy Services

UDP – User Datagram Protocol

UPA – United Progressive Alliance

USC – University of Southern California

USD – United States Dollar

VP – Vice President

VSAT –Very Small Aperture Terminals

WTM – SEBI Whole Time Member

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Palak Shah has been a journalist in Mumbai for nearly 15-years. He has worked for most premier pink papers including *The Economic Times*, *Business Standard* and *The Financial Express*. Currently, he works with *The Hindu Business Line*. He was drawn to crime reporting at the age of 19 but a few years in the field told him that the fabric of crime had changed and the organised gangs, as Mumbai had witnessed during the eighties, no longer existed. It was business and markets that dominated the scenario. His passion to unravel the intricacies of the ‘white money’ economy led Palak to the world of finance and regulations.